

نظام نينجا لإدارة المصانع

Ninja Factory ERP — Complete System Analysis & Process Documentation

التحليل الشامل للنظام ودليل العمليات ومخططات قواعد البيانات

30 جزءاً تحليلياً | 16 قسمًا تشغيلياً | 30 دوراً ومستخدمًا | 94+ شاشة عمل
10 مخططات ERD تفصيلية | 25+ مسار ودورة عمل | 90+ قاعدة عمل حاكمة

الإصدار المؤسسي: Enterprise v4.0 | سبتمبر 2026

مصمم لمعمارية منشأة صناعية متكاملة وموحدة لمصانع البوليمر والبلاستيك

نظام نينجا لإدارة المصانع — التحليل الشامل للنظام (Ninja Factory ERP)



المشروع: نظام نينجا المتكامل لإدارة المصانع والمؤسسات الصناعية (Ninja Factory ERP)
قطاع الصناعة: تصنيع وتدوير البوليمر والبلاستيك (حبيبات HDPE و PP، الجالونات الصناعية، الأغذية والمقايض)
التقنية: تطبيق صفحة واحدة (Vanilla JS SPA) + منصة سوبابيز (PostgreSQL + Auth + Realtime + Storage)
اللغات: ثنائي اللغة لكل مستخدم — العربية (RTL) والإنجليزية (LTR)
قاعدة البيانات: PostgreSQL عبر سوبابيز — 170+ جدول، 16 قسمًا تشغيليًا، 94+ شاشة
الأمان: تحكم مركزي في الصلاحيات مبني على الأدوار (RBAC) مع أمان مستوى السطر (RLS)

الجزء 1 — التحليل الشامل للنظام

1.1 نظرة عامة على النظام

Ninja Factory ERP is a comprehensive, AI-powered enterprise resource planning system designed for a **polymer manufacturing factory**. The factory produces recycled HDPE/PP plastic granules, industrial jerrycans, and plastic caps/closures.

The system manages the **complete business lifecycle** from raw material procurement through production, quality control, sales, delivery, and financial accounting — alongside full HR, fleet management, and clinic/nursing operations.

1.2 قائمة الـ 30 ممثلًا وجهة فاعلة المعرفة في المنظومة

| الرقم | الجهة الفاعلة (EN) | نوع الدور | المسمى بالعربية | |-----|-----|-----|---|
بشري / مسؤول أعلى | صاحب المصنع | 2 | HR Manager | مستخدم بشري / مسؤول أعلى | مدير الموارد البشرية |
3 | Hall Manager / Department Head | مستخدم بشري | مدير الصالة | 4 | Employee | مستخدم بشري | موظف
5 | Accounts Manager / CFO | مستخدم بشري | المدير المالي | 6 | Chief Accountant | مستخدم بشري | كبير المحاسبين
7 | Accountant | مستخدم بشري | محاسب | 8 | Procurement Manager | مستخدم بشري | مدير المشتريات
9 | Procurement Specialist | مستخدم بشري | أخصائي مشتريات | 10 | Warehouse Manager | مستخدم بشري | مدير المخازن
11 | Production Manager | مستخدم بشري | مدير الإنتاج | 12 | Quality Manager / QC Inspector | مستخدم بشري | مدير الجودة
13 | Maintenance Manager / Technician | مستخدم بشري | مدير الصيانة | 14 | Engineering Manager / Engineer | مستخدم بشري | مدير الهندسي
15 | Technical

Office | مستخدم بشري | المكتب الفني | 16 | Sales Manager | مستخدم بشري | مدير المبيعات | 17 | Planning
Manager | مستخدم بشري | مدير التخطيط | 18 | Logistics Manager | مستخدم بشري | مدير اللوجستيات | 19 | IT
Admin / IT Support | مستخدم بشري / مسؤول أعلى | مدير تكنولوجيا المعلومات | 20 | Driver (Internal)
مستخدم بشري | سائق داخلي | 21 | Driver (External) | مستخدم خارجي | سائق خارجي | 22 | Spare Parts
Inspector | مستخدم بشري | مفتش قطع الغيار | 23 | Nurse / Nursing Management | مستخدم بشري | ممرض /
إدارة التمريض | 24 | Supplier (External Portal) | مستخدم خارجي | مورد خارجي | 25 | Customer / Guest
مستخدم خارجي | عميل / زائر | 26 | AI Engine | نظام داخلي | محرك الذكاء الاصطناعي | 27 | Notification
Service | نظام داخلي | خدمة الإشعارات | 28 | Supabase Auth | نظام خارجي | نظام المصادقة | 29 | Supabase
Realtime | نظام خارجي | البث المباشر | 30 | Brevo Email Service | خدمة طرف ثالث | خدمة البريد الإلكتروني |

1.3 جميع أدوار المستخدمين في قاعدة البيانات

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hr, employee, owner, hr manager, manager, hall manager,  
qc inspector, quality manager, warehouse manager,  
maintenance manager, technician, procurement manager,  
procurement specialist, purchasing manager, engineering manager,  
engineer, technical office, sales manager, planning manager,  
accountant, chief accountant, cfo, it support, it manager,  
spare parts inspector, supplier_external, nursing management, nurse
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1.4 جميع الأقسام الـ 16 في المصنع

#	القسم (بالإنجليزية)	القسم (بالعربية)
1	Production	الإنتاج
2	Warehouse	المخازن
3	Administration	الإدارة
4	Packaging	التعبئة والتغليف
5	Maintenance	الصيانة
6	Procurement	المشتريات
7	Finance	المالية
8	IT	تكنولوجيا المعلومات
9	Quality	الجودة
10	Engineering	الهندسة
11	Planning	التخطيط
12	Health & Safety	الصحة والسلامة
13	Secretariat	السكرتارية
14	Sales	المبيعات

#	القسم (بالإنجليزية)	القسم (بالعربية)
15	Logistics	اللوجستيات
16	HR	الموارد البشرية

1.5 جميع الشاشات والصفحات (+94 شاشة)

Employee Self-Service Screens

, my-salary , my-leaves , scan-checkout , scan-checkin , my-attendance , hr-personal , dashboard
my-expenses , my-missions , my-delays , my-medical , my-loans , my-overtime

HR Management Screens

, all-missions , all-delays , overtime , shifts , absence-leave , leaves , attendance , employees
, recruitment , hr-adjustments , payroll-funding , payroll , asset-assignment , employee-warnings
, medical-requests , complaints , expenses , loans , uniforms , performance , documents , hr-ats
, team-adjustments , friday-work , hr-qr-generator , performance-reviews , training , offboarding
nursing-medical-approvals

Communication & Collaboration

task-management , calendar , shift-swap , org-directory , internal-chat , announcements

Administration & Reporting

, activity-log-page , login-history , audit-log , kpi-dashboard , reports , ai-mind
, notification-settings , screen-permissions , global-search , approval-workflows , document-management
system-settings

Finance Screens

driver-payments , chart-of-accounts , financial-reports , petty-cash

Supply Chain Screens

supplier-performance , supplier-portal , erp-suppliers , purchase-requests , inventory

Sales & Products

customer-requests , erp-products , erp-sales

Operations Screens

, logistics , spare-parts , production , quality , maintenance , fleet , engineering , nursing-page
legal-affairs , equipment

1.6 ملخص الميزات بحسب الوحدات

الوحدة الوظيفية	عدد الميزات	الميزات التشغيلية والجوهرية
Authentication	12	Login, Logout, Sessions, RBAC, RLS, QR Auth, Multi-company
HR & People	+35	Employees CRUD, Attendance, Leaves, Absence tracking, Shifts, Overtime, Missions, QR check-in/out, Complaints, Disciplinary, Offboarding, Warnings, Performance Reviews, Training, Asset Assignment, Friday Work Requests, Recruitment/ATS
Payroll	+15	Payroll calculation, Salary adjustments, Loans, Medical requests, Late deductions, Expenses
Finance	+25	Treasury, Banks, GL, Journal Entries, Invoices, Payments, Checks, Custodies, Cost Centers, Fixed Assets, Taxes, Financial Closing, Chart of Accounts, Budgets
Procurement	+12	Purchase Requests, Purchase Orders, Quotations, Supplier Management, Supplier Performance (AI), Petty Cash
Inventory	+8	Items, Stock Movements, Transactions, Low Stock Alerts, Categories
Production	+10	Production Orders, BOM, Stages, Batches, Consumption, Traceability
Quality	+5	QC Inspections, Pass/Fail, Defect Tracking, Batch Quality
Sales & CRM	+12	Clients, Sales Orders, Products, Public Catalog, Customer Registration, Quote Requests, Customer Feedback
Fleet & Logistics	+18	Vehicles, Internal/External Drivers, Trips, Incidents, Fleet Maintenance, Fuel, Odometer, Cars, Car Trips
Engineering	+6	Projects, Drawings, Technical Office, Workflow Approvals
Maintenance	+10	Machines, Maintenance Logs, Companies, Visits, Payments, Equipment, Equipment Rentals
Nursing	+8	Clinic Visits, Work Injuries, Rest Permits, Pharmacy, Medication Dispense
Legal	+3	Legal Issues, Case Tracking, Assignment
IT & Admin	+20	IT Tickets, Notifications, Announcements, AI Reports, Global Search, KPIs, Tasks, Calendar, Chat, Documents, Webhooks, Email Templates, Approval Workflows

1.7 تصنيف البيانات التي يدخلها المستخدمون

- Employee personal data, attendance check-in/out, leave requests, overtime records
- Mission/errand details, complaints, expense claims, loan applications, medical requests
- Payroll adjustments, salary modifications
- Inventory items, stock movements, purchase requests
- Sales orders, client information, product catalog entries
- Journal entries, invoices, payments, checks
- Engineering project details, technical drawings

- Fleet vehicle data, trip logs, fuel logs, odometer readings
- Clinic visit records, injury reports, medication dispensing
- IT support tickets, legal case details
- KPI definitions, task assignments, calendar events, chat messages

1.8 تصنيف البيانات التي يولدها النظام تلقائياً

- Payroll net salary calculations (base + overtime + bonuses - penalties - deductions)
- Auto-generated document numbers (PO-, SO-, INV-, JE-, RCV-, PAY-)
- Late deduction calculations based on delay minutes
- AI scores for recruitment CVs (ATS)
- AI supplier recommendations and performance scoring
- Attendance status (present/absent) based on check-in/out
- Low stock alerts based on min_quantity thresholds
- Fleet alerts (license expiry, insurance expiry, maintenance due)
- QR codes for attendance
- Audit logs and activity logs
- AI-generated reports (monthly/semiannual/annual)
- Search index entries
- Notification records
- Distance calculations (odometer_end - odometer_start)
- Financial closing reports

All Relationships Between Data 1.9

The system has extensive FK relationships connecting:

- users** → ALL employee-related tables (attendance, leaves, payroll, loans, etc.)
- inventory_items** → inventory_transactions, purchase_requests, bom_items, production_consumption
- products** → product_categories, product_images, product_documents, bom, production_batches, sales_orders
- suppliers** → supplier_orders, supplier_transactions, supplier_performance, equipment, purchase_orders
- clients** → sales_orders, finance_invoices, quote_requests
- machines** → maintenance_logs
- fleet_vehicles** → fleet_trips, fleet_incidents, fleet_maintenance
- fleet_internal_drivers** → fleet_trips, fleet_vehicles
- companies** → branches → users (multi-company)
- clinic_visits** → clinic_dispense_logs
- clinic_medications** → clinic_dispense_logs
- approval_workflows** → approval_steps → approval_requests → approval_actions

All External Services 1.10

Purpose	Service
User authentication, JWT tokens	Supabase Auth

Purpose	Service
Database storage	Supabase PostgreSQL
Live notifications, permission changes	Supabase Realtime
File uploads (documents, images, CVs)	Supabase Storage
Email delivery service	Brevo (Sendinblue)
Chatbot, Analytics, Fraud Detection, Recommendations	AI Engine (Custom)

الجزء 2 — الجهات الفاعلة والمستخدمون (ملفات تفصيلية لـ 30 دوراً)

Actor 1: Owner (صاحب المصنع)

الخاصية	التفاصيل والبيان
Type	Human / Super Administrator
الوصف	Factory owner with complete system authority. Final decision maker for all critical approvals
Responsibilities	Final approval for leaves, loans, medical disbursements, payroll funding, major purchases, hiring
الصلاحيات المطلوبة	FULL ACCESS to all 94+ screens with all 8 actions (view, create, edit, delete, approve, reject, export, print)
Accessible Sections	ALL 15 modules
Key Actions	Approve/Reject leave requests (final stage), Approve loans, Approve medical disbursements, Release payroll funds, View all dashboards and reports, Configure system settings

Actor 2: HR Manager (مدير الموارد البشرية)

الخاصية	التفاصيل والبيان
Type	مستخدم بشري / مسؤول أعلى
الوصف	Manages all HR operations including employees, attendance, leaves, payroll, recruitment
Responsibilities	Employee lifecycle management, Attendance monitoring, Leave workflow (2nd approval), Payroll processing, Recruitment, Training, Performance reviews, Disciplinary actions
الصلاحيات المطلوبة	Full access to HR, Payroll, Recruitment, Attendance, Training, Performance modules. View access to Nursing

الخاصية	التفاصيل والبيان
Accessible Sections	HR, Payroll, Communication, Reports, Nursing (view)
Key Actions	CRUD on employees, Process attendance, Approve/Reject leaves (HR stage), Calculate payroll, Manage loans, Process offboarding, Manage recruitment, Issue warnings, Manage training

Actor 3: Hall Manager / Department Head (مدير الصالة)

الخاصية	التفاصيل والبيان
Type	مستخدم بشري
الوصف	Manages a specific department/production hall. First-level approver for employee requests
Responsibilities	Team management, First-level leave approval, Overtime approval, Mission approval, Absence monitoring
الصلاحيات المطلوبة	View/Approve for team's leaves, attendance, overtime. Department-specific views
Accessible Sections	Dashboard, Team management, HR self-service
Key Actions	Approve/Reject leave requests (manager stage), Approve overtime, Approve missions, View team attendance, Submit Friday work requests

Actor 4: Employee (موظف)

الخاصية	التفاصيل والبيان
Type	مستخدم بشري
الوصف	Regular factory worker with self-service access to personal HR data
Responsibilities	Check-in/out attendance, Submit leave requests, Submit expense claims, View own salary/payroll
الصلاحيات المطلوبة	View own data only. Create requests (leave, medical, expense, complaint, mission). View announcements and chat
Accessible Sections	Dashboard, My Info (self-service), Communication, Workplace
Screens	dashboard, hr-personal, my-attendance, scan-checkin, scan-checkout, my-leaves, my-salary, my-overtime, my-loans, my-medical, my-delays, my-missions, my-expenses, complaints, announcements, internal-chat, shift-swap, calendar, task-management

Actor 5: CFO / Chief Accountant / Accountant

الخاصية	التفاصيل والبيان
Type	مستخدم بشري
الوصف	Finance team managing all accounting operations
Responsibilities	General ledger, Treasury management, Invoice processing, Check management, Journal entries, Financial closing, Tax filing, Budget management
الصلاحيات المطلوبة	Full access to Finance module. View access to Payroll, Procurement, Sales
Key Actions	Create/Approve journal entries, Process payments, Manage checks, Financial closing, Tax calculations, Budget allocation, Custody management

Actor 6: Procurement Manager / Specialist

الخاصية	التفاصيل والبيان
Type	مستخدم بشري
الوصف	Manages purchasing, supplier relationships, and procurement workflows
Responsibilities	Purchase request processing, Supplier management, Purchase order creation, Quotation management, Petty cash
الصلاحيات المطلوبة	Full access to Procurement, Suppliers. View access to Inventory, Finance
Key Actions	Approve purchase requests, Create purchase orders, Manage suppliers, Request quotations, Track supplier performance

Actor 7: Warehouse Manager (مدير المخازن)

الخاصية	التفاصيل والبيان
Type	مستخدم بشري
الوصف	Manages inventory, stock levels, and material dispensing
Responsibilities	Inventory management, Stock movement tracking, Material dispensing, Low stock monitoring, Purchase request initiation
الصلاحيات المطلوبة	Full access to Inventory module. View access to Procurement, Production
Key Actions	Add/Update inventory items, Process stock movements (in/out/transfer), Dispense materials to production, Monitor low stock alerts

Actor 8: Production Manager (مدير الإنتاج)

الخاصية	التفاصيل والبيان
Type	مستخدم بشري
الوصف	Manages manufacturing operations, production orders, and batch tracking
Responsibilities	Production order management, BOM management, Stage tracking, Material consumption, Batch management
الصلاحيات المطلوبة	Full access to Production module. View access to Inventory, Quality
Key Actions	Create production orders, Manage BOM, Track manufacturing stages, Record material consumption, Create batches

Actor 9: Quality Manager / QC Inspector

الخاصية	التفاصيل والبيان
Type	مستخدم بشري
الوصف	Quality control for incoming materials and finished products
Responsibilities	QC inspections, Pass/Fail decisions, Defect tracking, Batch quality verification
الصلاحيات المطلوبة	Full access to Quality module. View access to Production, Procurement
Key Actions	Create QC inspections, Pass/Fail materials, Inspect batches, Record defects

Actor 10: Maintenance Manager / Technician

الخاصية	التفاصيل والبيان
Type	مستخدم بشري
الوصف	Manages machine maintenance, facilities, and external maintenance companies
Responsibilities	Machine status monitoring, Maintenance scheduling, External company management, Visit tracking, Payment processing
الصلاحيات المطلوبة	Full access to Maintenance module. View access to Inventory (spare parts)
Key Actions	Log maintenance activities, Update machine status, Manage maintenance companies, Track visits, Process maintenance payments

Actor 11: Engineering Manager / Engineer

الخاصية	التفاصيل والبيان
Type	مستخدم بشري
الوصف	Manages engineering projects and technical drawings
Responsibilities	Project management, Drawing management, Technical specifications, Workflow approvals
الصلاحيات المطلوبة	Full access to Engineering module
Key Actions	Create/manage projects, Upload/approve drawings, Update project progress

Actor 12: Sales Manager (مدير المبيعات)

الخاصية	التفاصيل والبيان
Type	مستخدم بشري
الوصف	Manages sales operations, clients, and product catalog
Responsibilities	Client management, Sales order processing, Product catalog management, Customer registration, Quote handling
الصلاحيات المطلوبة	Full access to Sales, Products, Customer Requests
Key Actions	Create sales orders, Manage clients, Manage product catalog, Process customer registrations, Handle quote requests

Actor 13: Logistics Manager

الخاصية	التفاصيل والبيان
Type	مستخدم بشري
الوصف	Manages fleet, drivers, trips, and transportation
Responsibilities	Fleet management, Driver management, Trip scheduling, Incident tracking
الصلاحيات المطلوبة	Full access to Fleet & Logistics module
Key Actions	Manage vehicles, Assign drivers, Schedule trips, Track incidents, Monitor fuel/odometer

Actor 14: IT Admin / IT Support

الخاصية	التفاصيل والبيان
Type	Human / System Administrator

الخاصية	التفاصيل والبيان
الوصف	System administration, IT support, and permission management
Responsibilities	System settings, Permission management, IT ticket resolution, User management
الصلاحيات المطلوبة	Full access to Admin, Permissions, System Settings, Audit logs
Key Actions	Manage screen permissions, Configure system settings, Resolve IT tickets, View audit/login history

Actor 15-16: Driver (Internal / External)

External Driver	Internal Driver	Property
مستخدم خارجي	مستخدم بشري	Type
External transport company driver	Factory employee driver	الوصف
Provide transport services	Drive company vehicles, Record trips	Responsibilities
Complete deliveries	Record odometer, Log trips, Report incidents	Key Actions

Actor 17: Spare Parts Inspector

الخاصية	التفاصيل والبيان
Type	مستخدم بشري
الوصف	Inspects spare parts quality, lifecycle tracking
Responsibilities	Quality inspection of spare parts, Lifecycle assessment, Repair recommendations
Key Actions	Inspect parts, Record damage category, Assess repairability, Recommend disposal/repair

Actor 18: Nurse / Nursing Management

الخاصية	التفاصيل والبيان
Type	مستخدم بشري
الوصف	Factory clinic staff managing employee health
Responsibilities	Clinic visits, Injury recording, Rest permits, Medication dispensing, Pharmacy management
الصلاحيات المطلوبة	Full access to Nursing module. View access to Medical Requests
Key Actions	Record clinic visits, Document injuries, Issue rest permits, Dispense medications, Manage pharmacy stock

Actor 19: Supplier (External Portal)

الخاصية	التفاصيل والبيان
Type	مستخدم خارجي
الوصف	External supplier with portal access
Responsibilities	View orders, Track payments, Update delivery status
الصلاحيات المطلوبة	Supplier portal only — view own orders and transactions

Actor 20: Customer / Guest

الخاصية	التفاصيل والبيان
Type	مستخدم خارجي
الوصف	Public visitor or potential customer
Responsibilities	Browse product catalog, Submit registration requests, Request quotes
الصلاحيات المطلوبة	Public catalog view, Registration form, Quote request form

Actors 21-24: System Actors

Actor	Type	Purpose
AI Engine	نظام داخلي	Chatbot, Analytics, Fraud Detection, CV Scoring, Supplier Recommendations
Notification Service	نظام داخلي	Push notifications, Email triggers, Real-time alerts
Supabase Auth	نظام خارجي	JWT authentication, Session management
Brevo Email	Third-party	Email delivery for notifications and reports

الجزء 3 — الأقسام والوحدات التشغيلية للمصنع



Module 1: Authentication & Security

الخاصية	التفاصيل والبيان
Purpose	Secure user authentication, authorization, session management, and audit trailing
Problem Solved	Ensures only authorized users access specific features; tracks all system activities

الخاصية	التفاصيل والبيان
Users	ALL users (login), IT Admin (configuration), Owner (full access)
Data Managed	Users, Sessions, Permissions, Login history, Audit logs, Activity logs
Connected To	ALL other modules (every module checks permissions)
الجدول المعنية	users, login_history, activity_log, audit_log, user_sessions, permissions, role_permissions, screen_permissions, permission_templates, companies, branches, used_qr_codes

Module 2: HR & People Management

الخاصية	التفاصيل والبيان
Purpose	Complete employee lifecycle management from hiring to offboarding
Problem Solved	Centralized HR operations — attendance tracking, leave management, disciplinary actions, performance evaluation
Users	HR Manager, Hall Manager, Employee, Owner
Data Managed	Employee records, Attendance, Leaves, Overtime, Missions, Complaints, Disciplinary actions, Performance reviews, Training, Asset assignments, Warnings, Absence records, ATS/Recruitment
Connected To	Payroll (salary data), Finance (expense processing), Nursing (medical), IT (tickets)
الجدول المعنية	users, attendance, leave_requests, overtime, missions, complaints, disciplinary_actions, offboarding, hr_absence_requests, hr_absence_records, hr_violations_warnings, hr_absence_audit, hr_absence_settings, performance_reviews, training_courses, training_enrollments, asset_assignments, employee_warnings, friday_work_requests, ats_applications, late_deductions

Module 3: Payroll & Compensation

الخاصية	التفاصيل والبيان
Purpose	Calculate and process employee salaries, manage loans and financial benefits
Problem Solved	Automated salary calculation integrating attendance, overtime, deductions, and loan installments
Users	HR Manager, Owner, Employee (view own), Accountant
Data Managed	Payroll records, Loans, Medical requests, Expenses, Late deductions, Salary adjustments
Connected To	HR (attendance/overtime data), Finance (payment processing)
الجدول المعنية	payroll, loans, medical_requests, expenses, late_deductions, salary_adjustments

Module 4: Finance & Accounting

الخاصية	التفاصيل والبيان
Purpose	Complete financial management including GL, treasury, invoicing, and tax management
Problem Solved	Double-entry accounting, cash flow tracking, check lifecycle, financial reporting
Users	CFO, Chief Accountant, Accountant, Owner
Data Managed	Safes, Bank accounts, Invoices, Payments, Checks, Custodies, Journal entries, Cost centers, Fixed assets, Taxes, Closings, Budgets
Connected To	Payroll (salary payments), Procurement (purchase invoices), Sales (sales invoices), Maintenance (payments)
الجدول المعنية	finance_safes, finance_bank_accounts, finance_clients, finance_suppliers, finance_invoices, finance_collections, finance_checks, finance_custodies, finance_journal_entries, finance_journal_lines, finance_cost_centers, finance_fixed_assets, finance_expense_categories, finance_taxes, finance_closings, finance_payments, general_ledger, chart_of_accounts, currencies, exchange_rates, department_budgets, doc_sequences

Module 5: Procurement & Purchasing

الخاصية	التفاصيل والبيان
Purpose	Manage the complete purchasing lifecycle from request to delivery
Problem Solved	Structured procurement workflow with approvals, supplier management, and spending tracking
Users	Procurement Manager, Procurement Specialist, Warehouse Manager, Owner
Data Managed	Purchase requests, Purchase orders, Suppliers, Supplier orders, Supplier transactions, Supplier performance
Connected To	Inventory (stock replenishment), Finance (invoicing), Quality (incoming inspection)
الجدول المعنية	purchase_requests, purchase_orders, suppliers, supplier_orders, supplier_transactions, supplier_performance

Module 6: Inventory & Warehouse

الخاصية	التفاصيل والبيان
Purpose	Track all inventory items, stock levels, and material movements
Problem Solved	Real-time stock visibility, automated low-stock alerts, movement traceability
Users	Warehouse Manager, Production Manager, Procurement Manager
Data Managed	Inventory items, Stock movements, Transactions

الخاصية	التفاصيل والبيان
Connected To	Procurement (incoming stock), Production (material consumption), Sales (finished goods)
الجدول المعنية	inventory_items, inventory_transactions, stock_movements



Module 7: Production & Manufacturing

الخاصية	التفاصيل والبيان
Purpose	Manage manufacturing operations from BOM to finished batch
Problem Solved	Production planning, stage tracking, material consumption, batch traceability
Users	Production Manager, Planning Manager, Quality Manager
Data Managed	Production orders, BOM, Stage logs, Consumption records, Batches
Connected To	Inventory (materials), Quality (QC inspections), Sales (finished products)
الجدول المعنية	production_orders, bom, bom_items, production_stage_templates, production_stage_logs, production_consumption, production_batches



Module 8: Quality Control

الخاصية	التفاصيل والبيان
Purpose	Ensure product and material quality through inspections
Problem Solved	Quality gate for incoming materials and finished products
Users	Quality Manager, QC Inspector
Data Managed	QC inspections (incoming materials + finished products)
Connected To	Procurement (incoming material), Production (batch quality), Inventory (accepted/rejected)
الجدول المعنية	qc_inspections



Module 9: Sales & CRM

الخاصية	التفاصيل والبيان
Purpose	Manage sales operations, client relationships, and public product catalog
Problem Solved	End-to-end sales cycle, customer self-service catalog, quote management
Users	Sales Manager, Customer/Guest

الخاصية	التفاصيل والبيان
Data Managed	Clients, Sales orders, Products, Categories, Images, Documents, Customer registrations, Quotes, Feedback
Connected To	Finance (invoicing), Production (production orders), Fleet (delivery), Inventory (stock)
الجدول المعنية	clients, sales_orders, products, product_categories, product_images, product_documents, customer_registration_requests, quote_requests, customer_feedback

Module 10: Fleet & Logistics

الخاصية	التفاصيل والبيان
Purpose	Manage company fleet, drivers, trips, and transportation
Problem Solved	Vehicle tracking, trip management, cost tracking, maintenance scheduling
Users	Logistics Manager, Driver (Internal/External), Owner
Data Managed	Vehicles, Drivers, Trips, Incidents, Fleet maintenance, Fuel, Odometer, Cars
Connected To	Sales (delivery), HR (driver employee data), Finance (driver payments)
الجدول المعنية	fleet_internal_drivers, fleet_external_drivers, fleet_vehicles, fleet_trips, fleet_incidents, fleet_maintenance, cars, car_trips, car_maintenance_logs, logistics_drivers, logistics_movements

Module 11: Engineering

الخاصية	التفاصيل والبيان
Purpose	Manage engineering projects, technical drawings, and approvals
Problem Solved	Technical project tracking, drawing version control, multi-department approval
Users	Engineering Manager, Engineer, Technical Office
Data Managed	Projects, Drawings, Workflow approvals
Connected To	Maintenance (equipment), Production (production lines)
الجدول المعنية	engineering_projects, engineering_drawings

Module 12: Maintenance & Facilities

الخاصية	التفاصيل والبيان
Purpose	Machine maintenance, external company management, equipment tracking

الخاصية	التفاصيل والبيان
Problem Solved	Preventive maintenance scheduling, repair tracking, equipment rental management
Users	Maintenance Manager, Technician, Owner
Data Managed	Machines, Maintenance logs, Companies, Visits, Payments, Equipment, Rentals
Connected To	Inventory (spare parts), Finance (payments), Engineering (equipment)
الجدول المعنية	machines, maintenance_logs, maintenance_requests, maintenance_companies, maintenance_visits, maintenance_payments, equipment_categories, equipment, equipment_rentals



Module 13: Nursing & Factory Clinic

الخاصية	التفاصيل والبيان
Purpose	Factory clinic operations — employee health, injuries, medication
Problem Solved	Occupational health tracking, injury documentation, medication inventory
Users	Nurse, Nursing Management, HR Manager
Data Managed	Clinic visits, Work injuries, Rest permits, Medications, Dispense logs
Connected To	HR (employee data, leave integration), Payroll (medical requests)
الجدول المعنية	clinic_visits, clinic_injuries, clinic_rest_permits, clinic_medications, clinic_dispense_logs



Module 14: Legal & Compliance

الخاصية	التفاصيل والبيان
Purpose	Track legal issues, cases, and compliance matters
Problem Solved	Centralized legal case management
Users	Owner, Legal Advisor (assigned_to)
Data Managed	Legal issues
Connected To	HR (employee-related cases)
الجدول المعنية	legal_issues



Module 15: IT & Administration

الخاصية	التفاصيل والبيان
Purpose	IT support, system administration, collaboration tools, AI analytics
Problem Solved	IT ticket management, system configuration, internal communication, AI-powered insights
Users	IT Admin, All users (notifications, chat, tasks)
Data Managed	IT tickets, Notifications, Announcements, AI reports, KPIs, Tasks, Calendar, Chat, Documents, Approval workflows, System settings
Connected To	ALL modules (notifications, documents, approvals span entire system)
الجدول المعنية	it_tickets, notifications, announcements, notification_preferences, system_settings, ai_reports, kpi_definitions, kpi_values, saved_reports, tasks, calendar_events, chat_channels, chat_messages, documents, document_categories, digital_signatures, webhooks, email_templates, import_logs, backup_logs, search_index, approval_workflows, approval_steps, approval_requests, approval_actions

Cross-Module Dependency Map

flowchart TD

```
AUTH[🔒 Authentication & Security] --> HR[👤 HR & People]
AUTH --> FIN[💰 Finance]
AUTH --> ALL[All Modules]
```

```
HR --> PAY[💰 Payroll]
HR --> NURSE[🏥 Nursing]
HR --> FLEET[🚚 Fleet - Drivers]
```

```
PAY --> FIN
```

```
PROC[🛒 Procurement] --> INV[📦 Inventory]
PROC --> FIN
PROC --> QC[🔍 Quality]
```

```
INV --> PROD[🏭 Production]
PROD --> QC
PROD --> SALES[📊 Sales]
```

```
SALES --> FIN
SALES --> FLEET
```

```
MAINT[🔧 Maintenance] --> INV
MAINT --> FIN
```

```
ENG[⚙️ Engineering] --> PROD
ENG --> MAINT
```

```
FLEET --> FIN
```

```
IT[💻 IT & Admin] --> ALL
LEGAL[⚖️ Legal] --> HR
```

الجزء 4 — القسم 1: الأمان، التحقق، وإدارة الصلاحيات

4.1 الغرض من الوحدة وأهميتها التشغيلية

The Authentication & Security module is the **foundation of the entire ERP system**. It controls

- **Who** can access the system (Authentication)
- **What** they can see and do (Authorization / RBAC)
- **When** and **how** they accessed it (Audit Trail)
- **Multi-company** isolation (Company/Branch scoping)

- Why it exists:** Without this module, sensitive factory data (salaries, financial records, production secrets) would be unprotected.
- What problem it solves:** Centralized Role-Based Access Control ensuring that a Production worker cannot see payroll data, and only authorized managers can approve requests.
- Which users interact:** ALL users must authenticate. IT Admin and Owner configure permissions.
- Data managed:** User accounts, sessions, permissions, login history, activity logs, audit trails, QR codes.

4.2 الجهات الفاعلة في هذه الوحدة

Processes Received	Processes Initiated	Data Access	المحظورات والقيود التشغيلية	الصلاحيات والإجراءات المصرح بها	Actor
Audit alerts	Permission changes, User creation	All tables	N/A (unrestricted)	Full system access, configure permissions, view all audit logs	Owner
Permission notifications	User creation, Role assignment	users, login_history, activity_log	Change system-level settings	Create/Edit employees, assign roles, view login history	HR Manager
IT tickets	Permission configuration, System settings	All security tables	Approve HR/Finance requests	Manage screen_permissions, system_settings, view audit logs	IT Admin
Notifications	Login, QR check-in	Own user record only	View other users' data, change permissions	Login, Logout, Change own password, QR check-in/out	Employee
Auth requests	Token validation	user_sessions	N/A (system)	Validate JWT tokens, Manage sessions	Supabase Auth
Chatbot requests	Analytics queries	Filtered by AIPermissionLayer	Access data outside user's permission	Query data within permission scope	AI Engine

4.3 الميزات التفصيلية والوظائف

Feature 1: User Login

الخاصية	التفاصيل والبيان
الوصف	User authenticates with username + password to receive session token
الجهة الفاعلة	Any User

الخاصية	التفاصيل والبيان
المدخلات	username, password
المخرجات	Session token, User profile, Permissions list
الجداول المعنية	users, login_history, user_sessions
عمليات CRUD	R (users), C (login_history, user_sessions)
قواعد العمل	BR-AUTH-01: Username and password must match. BR-AUTH-02: User status must be 'active'. BR-AUTH-03: Failed login must be recorded with reason
الصلاحيات المطلوبة	Public (no permission needed to access login page)
الأخطاء المحتملة	Invalid credentials, Account suspended, Account inactive, Account locked

Feature 2: User Logout

الخاصية	التفاصيل والبيان
الوصف	Terminate active session and clear client tokens
الجهة الفاعلة	Authenticated User
المدخلات	Session token
المخرجات	Session invalidated
الجداول المعنية	user_sessions
عمليات CRUD	U (user_sessions.is_active = false)
قواعد العمل	BR-AUTH-04: Session must be deactivated on logout

Feature 3: Session Management

الخاصية	التفاصيل والبيان
الوصف	Track active sessions with token hashing and expiry
الجهة الفاعلة	النظام (تلقائياً)
المدخلات	User login event
المخرجات	Active session record
الجداول المعنية	user_sessions
عمليات CRUD	C (on login), R (on validation), U (on refresh), D (on expiry/logout)
قواعد العمل	BR-AUTH-05: Tokens stored as hashes only. BR-AUTH-06: Sessions expire after configured time. BR-AUTH-07: Refresh tokens extend session

Feature 4: Screen-Level Permission Check (canView)

الخاصية	التفاصيل والبيان
الوصف	Check if user's role grants view access to a specific screen
الجهة الفاعلة	النظام (PermissionGuard)
المدخلات	user.role, screenId
المخرجات	Boolean (allowed/denied)
الجدول المعنوية	screen_permissions, permission_templates
عمليات CRUD	R
قواعد العمل	BR-AUTH-08: Owner role always has full access. BR-AUTH-09: Permission check runs before every page render. BR-AUTH-10: Denied users see "Access Denied" page

Feature 5: Action-Level Permission Check (canAction)

الخاصية	التفاصيل والبيان
الوصف	Check if user can perform specific action (create/edit/delete/approve/reject/export/print) on a screen
الجهة الفاعلة	النظام (PermissionGuard)
المدخلات	user.role, screenId, action
المخرجات	Boolean (allowed/denied)
الجدول المعنوية	screen_permissions, permission_templates
عمليات CRUD	R
قواعد العمل	BR-AUTH-11: Buttons disabled if action not permitted. BR-AUTH-12: 8 standard actions: view, create, edit, delete, approve, reject, export, print

Feature 6: Permission Management

الخاصية	التفاصيل والبيان
الوصف	Admin UI to assign/revoke permissions per role per screen per action
الجهة الفاعلة	Owner, IT Admin
المدخلات	Role, Screen, Action, Granted (boolean)
المخرجات	Updated permission record
الجدول المعنوية	screen_permissions, permission_templates

الخاصية	التفاصيل والبيان
عمليات CRUD	C, R, U, D
قواعد العمل	BR-AUTH-13: Changes broadcast via Supabase Realtime. BR-AUTH-14: Bulk actions available (Grant All HR / Finance / ALL). BR-AUTH-15: Permission templates can be copied from role to role
الأخطاء المحتملة	Duplicate permission entry, Invalid screen/action

Feature 7: Permission Templates

الخاصية	التفاصيل والبيان
الوصف	Pre-defined permission sets per role for quick assignment
الجهة الفاعلة	IT Admin, System (on user creation)
المدخلات	Role name
المخرجات	Auto-applied permission set
الجدول المعنية	permission_templates
عمليات CRUD	C, R, U, D
قواعد العمل	BR-AUTH-16: New users auto-receive template permissions based on their role

Feature 8: Audit Logging

الخاصية	التفاصيل والبيان
الوصف	Record significant system actions for compliance and investigation
الجهة الفاعلة	النظام (تلقائياً)
المدخلات	Action event (user_name, action, details)
المخرجات	Immutable audit record
الجدول المعنية	audit_log
عمليات CRUD	C (insert only — never update/delete)
قواعد العمل	BR-AUTH-17: Audit logs are append-only. BR-AUTH-18: Logs include IP address

Feature 9: Activity Logging

الخاصية	التفاصيل والبيان
الوصف	Detailed activity tracking with old/new value diffs per entity
الجهة الفاعلة	النظام (تلقائياً)

الخاصية	التفاصيل والبيان
المدخلات	Module, Action, Entity changes
المخرجات	Activity record with JSONB diffs
الجدول المعنية	activity_log
عمليات CRUD	C
قواعد العمل	BR-AUTH-19: old_values and new_values stored as JSONB for change tracking

Feature 10: Login History

الخاصية	التفاصيل والبيان
الوصف	Track all login attempts (success and failure)
الجهة الفاعلة	النظام (تلقائياً)
المدخلات	Login attempt event
المخرجات	Login history record
الجدول المعنية	login_history
عمليات CRUD	C, R
قواعد العمل	BR-AUTH-20: Both successful and failed attempts recorded. BR-AUTH-21: IP address and user agent captured

Feature 11: QR Code Authentication (Check-in/Out)

الخاصية	التفاصيل والبيان
الوصف	One-time QR codes for attendance check-in and check-out
الجهة الفاعلة	Employee (scanner), HR (generator)
المدخلات	QR code hash
المخرجات	Attendance record created/updated
الجدول المعنية	used_qr_codes, attendance
عمليات CRUD	C (used_qr_codes), C/U (attendance)
قواعد العمل	BR-AUTH-22: QR codes are single-use (hash stored after use). BR-AUTH-23: Reused QR codes are rejected
الأخطاء المحتملة	QR already used, Invalid QR format, Expired QR

Feature 12: Multi-Company Support

الخاصية	التفاصيل والبيان
الوصف	Company and branch hierarchy for multi-entity operations

الخاصية	التفاصيل والبيان
الجهة الفاعلة	Owner, IT Admin
المدخلات	Company/Branch data
المخرجات	Scoped data access
الجدول المعنية	companies, branches, users (company_id, branch_id)
عمليات CRUD	C, R, U, D
قواعد العمل	BR-AUTH-24: Users belong to a company and branch. BR-AUTH-25: Data scoped by company_id

PART 5 & 6 — PROCESSES (Start Actor → End Actor)

Process AUTH-P01: User Login

الخاصية	التفاصيل والبيان
Process ID	AUTH-P01
Start Actor	Employee / Any User
End Actor	Employee / Any User (receives session)
المحفز	User navigates to login page and submits credentials
Preconditions	User account exists in users table
Main Flow	User enters username + password → 2. System queries users WHERE username AND password_hash .1 match → 3. System checks user.status = 'active' → 4. System creates user_sessions record → 5. System inserts login_history (status='success') → 6. System loads permissions from screen_permissions / permission_templates → 7. User redirected to Dashboard
Alternative Flow	3a. If credentials invalid → Insert login_history (status='failed') → Show error. 3b. If status='suspended' → Insert login_history (status='locked') → Show suspended message
المدخلات	username: TEXT, password: TEXT
المخرجات	JWT session token, user profile object, permission array
الجدول المعنية	users, user_sessions, login_history, screen_permissions, permission_templates
DB Operations	SELECT users, INSERT user_sessions, INSERT login_history, SELECT screen_permissions
قواعد العمل	BR-AUTH-01, BR-AUTH-02, BR-AUTH-03, BR-AUTH-20, BR-AUTH-21
الصلاحيات المطلوبة	None (public endpoint)
الأخطاء المحتملة	Invalid credentials, Account suspended, Account inactive, Database connection error
Notifications	None on success. Security alert on multiple failures (future)

Process AUTH-P02: Permission Check

الخاصية	التفاصيل والبيان
Process ID	AUTH-P02
Start Actor	النظام (PermissionGuard)
End Actor	System (UI renders or denies)
المحفز	User navigates to any screen or clicks any action button
Main Flow	PermissionGuard.canView(screenId) called → 2. Check user.role → 3. Query screen_permissions for (user_id, 1 "screen_id, action='view') → 4. If granted → Render page → 5. If denied → Show "Access Denied
الجدول المعنوية	screen_permissions, permission_templates
DB Operations	SELECT screen_permissions

Process AUTH-P03: QR Code Check-In

الخاصية	التفاصيل والبيان
Process ID	AUTH-P03
Start Actor	Employee
End Actor	Employee (check-in confirmed)
المحفز	Employee scans QR code at gate
Main Flow	Employee scans QR → 2. System extracts qr_hash → 3. System checks used_qr_codes for hash → 4. If NOT .1 used → Insert into used_qr_codes → 5. Create/Update attendance record with check_in timestamp → 6. Show success
Alternative Flow	4a. If QR already used → Reject with "QR Already Used" error
الجدول المعنوية	used_qr_codes, attendance
الأخطاء المحتملة	QR already used, Invalid QR, Employee not found

Process AUTH-P04: Permission Assignment

الخاصية	التفاصيل والبيان
Process ID	AUTH-P04
Start Actor	IT Admin / Owner
End Actor	Target User (permissions updated in real-time)
المحفز	Admin opens Permission Management screen and modifies permissions

الخاصية	التفاصيل والبيان
Main Flow	Admin selects user role → 2. Permission grid loads (modules × screens × actions) → 3. Admin toggles .1 permissions → 4. System UPSERTS screen_permissions → 5. Supabase Realtime broadcasts change → 6. Target user's UI updates immediately
الجداول المعنية	screen_permissions, permission_templates
Notifications	Real-time permission update via Supabase Realtime

Process AUTH-P05: Audit Trail Recording

الخاصية	التفاصيل والبيان
Process ID	AUTH-P05
Start Actor	Any User (performing an action)
End Actor	System (audit record stored)
المحفز	Any significant action (CRUD operations, approvals, status changes)
Main Flow	User performs action → 2. System captures: user_name, action, entity_type, old_values, new_values, IP → 3. .1 System inserts into activity_log → 4. If critical action, also insert into audit_log
الجداول المعنية	activity_log, audit_log

الجزء 8 — المخططات الانسيابية لمسارات العمل

Login Process Flow

```
flowchart TD
    A[👤 User] --> B[Enter Username + Password]
    B --> C{Validate Credentials}
    C -->|Valid| D{Check User Status}
    C -->|Invalid| E[Record Failed Login]
    E --> F[Show Error Message]
    F --> A
    D -->|Active| G[Create Session]
    D -->|Suspended/Inactive| H[Record Locked Login]
    H --> I[Show Suspended Message]
    G --> J[(user_sessions DB)]
    G --> K[(login_history DB)]
    J --> L[Load Permissions]
    L --> M[(screen_permissions DB)]
    M --> N[Redirect to Dashboard]
    N --> O[👤 User - Authenticated]
```


Permission Check Flow

flowchart TD

```

A[👤 User Navigates to Screen] --> B[PermissionGuard.canView]
B --> C{Check user.role}
C -->|Owner| D[✅ Full Access]
C -->|Other| E{Query screen_permissions}
E -->|granted=true| F[✅ Render Page]
E -->|granted=false/missing| G[❌ Access Denied]
F --> H[Check Action Permissions]
H --> I{canAction per button}
I -->|granted| J[Enable Button]
I -->|denied| K[Disable Button]

```

QR Check-In Flow

flowchart TD

```

A[👤 Employee] --> B[Scan QR Code]
B --> C[Extract QR Hash]
C --> D{Check used_qr_codes}
D -->|Not Used| E[Insert used_qr_codes]
D -->|Already Used| F[❌ Reject - QR Reused]
E --> G[Create/Update Attendance Record]
G --> H[(attendance DB)]
H --> I[✅ Check-In Confirmed]
I --> J[👤 Employee]

```

PART 10 & 11 — RELATIONSHIPS & CARDINALITY

Business Meaning	FK Location	Optionality	Cardinality	Entity B	Relationship	Entity A
One user can have zero or many login records	login_history.user_id	Optional (user may never fail login)	N:1	login_history	places	users
One user can have zero or many sessions	user_sessions.user_id	Optional	N:1	user_sessions	has	users
One user can generate zero or many activity logs	activity_log.user_id	Optional	N:1	activity_log	generates	users

Business Meaning	FK Location	Optionality	Cardinality	Entity B	Relationship	Entity A
One user can generate zero or many audit logs	audit_log.user_id	Optional	N:1	audit_log	generates	users
One user can have zero or many screen permissions	screen_permissions.user_id	Optional	N:1	screen_permissions	has	users
Many users belong to one company	users.company_id	Optional	N:1	companies	belongs to	users
Many users belong to one branch	users.branch_id	Optional	N:1	branches	belongs to	users
One company has zero or many branches	branches.company_id	Optional	N:1	branches	has	companies
One user can use zero or many QR codes	used_qr_codes.used_by	Optional	N:1	used_qr_codes	uses	users

Crow's Foot Notation

```

USER ||--o{ LOGIN_HISTORY : "generates"
USER ||--o{ USER_SESSIONS : "has"
USER ||--o{ ACTIVITY_LOG : "generates"
USER ||--o{ AUDIT_LOG : "generates"
USER ||--o{ SCREEN_PERMISSIONS : "has"
USER }o--|| COMPANIES : "belongs to"
USER }o--|| BRANCHES : "belongs to"
COMPANIES ||--o{ BRANCHES : "has"
USER ||--o{ USED_QR_CODES : "uses"

```

Exactly One (mandatory) = || •

Zero or Many (optional many) = o{ •

Zero or One (optional one) = }o •

الجزء 12 — الاختيارية والإلزامية في العلاقات

Explanation	Optional Side	Mandatory Side	Relationship
A login record requires a user, but a user might not have logged in yet	user MAY have zero login records	login_history MUST have a user	users → login_history
A session requires a user, but not all users have active sessions	user MAY have zero sessions	user_sessions MUST have a user	users → user_sessions
A branch must have a parent company	company MAY have zero branches	branches MUST belong to a company	companies → branches
Multi-company is optional		company is optional for user	users → companies

الجزء 13 — العلاقات متعدد إلى متعدد (M:N)

role_permissions (Junction Table)

The relationship between **roles** and **permissions** is M:N: - One role can have many permissions - One permission can be granted to many roles

role_permissions :Junction Table

Description	Type	Attribute
Primary key	UUID PK	id
Role name (FK concept)	TEXT	role
FK → permissions(id)	UUID FK	permission_id
Whether permission is granted	BOOLEAN	granted

Why junction is required: A role like "hr manager" needs dozens of permissions, and a permission like "view employees" is shared across multiple roles

permission_templates (Junction Table)

:The relationship between **roles** and **screens+actions** is M:N

Description	Type	Attribute
Primary key	UUID PK	id
Role name	TEXT	role
Screen identifier	TEXT	screen_id
Action type	TEXT	action

Description	Type	Attribute
Permission granted	BOOLEAN	granted
(role, screen_id, action)		UNIQUE

PART 14 & 15 — SECTION ERD

```

erDiagram
    USERS ||--o{ LOGIN_HISTORY : "generates"
    USERS ||--o{ USER_SESSIONS : "has"
    USERS ||--o{ ACTIVITY_LOG : "generates"
    USERS ||--o{ AUDIT_LOG : "generates"
    USERS ||--o{ SCREEN_PERMISSIONS : "has"
    USERS ||--o{ USED_QR_CODES : "uses"
    USERS }o--|| COMPANIES : "belongs_to"
    USERS }o--|| BRANCHES : "belongs_to"
    COMPANIES ||--o{ BRANCHES : "has"
    PERMISSIONS ||--o{ ROLE_PERMISSIONS : "granted_via"
    PERMISSION_TEMPLATES }o--|| USERS : "applied_to"

```

```

USERS {
    uuid id PK
    text employee_id UK
    text full_name
    text email UK
    text username UK
    text password_hash
    text role
    text department
    text status
    uuid company_id FK
    uuid branch_id FK
    text preferred_language
    timestampz created_at
}

```

```

LOGIN_HISTORY {
    uuid id PK
    uuid user_id FK
    text user_name
    text ip_address
    text user_agent
    text login_status
    text failure_reason
    timestampz created_at
}

```

```

USER_SESSIONS {
    uuid id PK
    uuid user_id FK
    text token_hash
    text refresh_token_hash
    timestampz expires_at
    boolean is_active
    text ip_address
}

```

```

ACTIVITY_LOG {
    uuid id PK
    uuid user_id FK
    text module
    text action
    text entity_type
}

```

```
    uuid entity_id
    jsonb old_values
    jsonb new_values
    text ip_address
}

AUDIT_LOG {
    uuid id PK
    text action
    text user_name
    uuid user_id FK
    text details
    text ip
    timestampz timestamp
}

SCREEN_PERMISSIONS {
    uuid id PK
    uuid user_id FK
    text screen_id
    text action
    boolean granted
    text module
}

PERMISSION_TEMPLATES {
    uuid id PK
    text role
    text screen_id
    text action
    boolean granted
    text module
}

PERMISSIONS {
    uuid id PK
    text module
    text action
    text description_en
    text description_ar
}

ROLE_PERMISSIONS {
    uuid id PK
    text role
    uuid permission_id FK
    boolean granted
}

COMPANIES {
    uuid id PK
    text name
    text name_ar
    text tax_id
    boolean is_active
}

BRANCHES {
    uuid id PK
    uuid company_id FK
```

```

    text name
    text name_ar
    boolean is_active
}

USED_QR_CODES {
    text qr_hash PK
    uuid used_by FK
    timestampz used_at
    text action_type
}

```

PART 16 — CRUD ANALYSIS

Tables	Actor	D	U	R	C	Feature
login_history, user_sessions, users	Any User			✓	✓	User Login
user_sessions	Any User		✓			User Logout
login_history	IT Admin, Owner			✓		View Login History
activity_log	IT Admin, Owner			✓		View Activity Log
audit_log	IT Admin, Owner			✓		View Audit Log
screen_permissions	IT Admin, Owner	✓	✓	✓	✓	Manage Permissions
permission_templates	IT Admin	✓	✓	✓	✓	Permission Templates
used_qr_codes, attendance	Employee		✓	✓	✓	QR Check-In
(QR generated in memory)	HR Manager				✓	QR Generation
companies	Owner	✓	✓	✓	✓	Company Management
branches	Owner	✓	✓	✓	✓	Branch Management
users, permission_templates	HR Manager, Owner				✓	User Creation

الجزء 17 — مصفوفة الصلاحيات التفصيلية للقسم

Owner	IT Admin	HR Manager	Hall Manager	Employee	Feature
✓	✓	✓	✓	✓	Login/Logout
✓	✓	✓	✓	✓	View Own Profile
✓	✓	✓	X	X	View Login History
✓	✓	X	X	X	View Activity Log

Owner	IT Admin	HR Manager	Hall Manager	Employee	Feature
✓	✓	✓	X	X	View Audit Log
✓	✓	X	X	X	Manage Permissions
✓	✓	X	X	X	Permission Templates
✓	✓	✓	✓	✓	QR Check-In/Out
✓	✓	✓	X	X	QR Generation
✓	X	X	X	X	Company Management
✓	✓	X	X	X	System Settings

الجزء 18 — قواعد العمل الحاكمة (Business Rules)

Enforcement	Rule	Rule ID
Application logic	Username and password must match for login	BR-AUTH-01
Application logic + DB CHECK	User status must be 'active' to login	BR-AUTH-02
Application logic	Failed login attempts must be recorded	BR-AUTH-03
Application logic	Session deactivated on logout	BR-AUTH-04
Application logic	Tokens stored as hashes only	BR-AUTH-05
Application logic	Sessions expire after configured time	BR-AUTH-06
PermissionGuard	Owner role has full access (not bypassing, through permission system)	BR-AUTH-07
()PermissionGuard.canView	Permission check runs before every page render	BR-AUTH-08
PermissionGuard	Denied users see "Access Denied" page	BR-AUTH-09
DB seed data	standard actions: view, create, edit, delete, approve, reject, export, print 8	BR-AUTH-10
used_qr_codes PK constraint	QR codes are single-use	BR-AUTH-11
Supabase Realtime	Permission changes broadcast via Realtime	BR-AUTH-12
Application logic on user creation	New users auto-receive role template permissions	BR-AUTH-13
Application logic	Audit logs are append-only (insert only)	BR-AUTH-14
Application logic	Multi-company data scoped by company_id	BR-AUTH-15

الجزء 21 — سجلات المراجعة والتدقيق والتتبع الأمني

Retention	Purpose	Table	Log Type
Indefinite	Track all login attempts (success/failure)	login_history	Login History
Indefinite	Track entity-level changes with old/new JSONB diffs	activity_log	Activity Log
Indefinite	Track significant system actions	audit_log	Audit Log
Indefinite	Track active/expired sessions	user_sessions	Session History
Indefinite	Track QR code usage to prevent reuse	used_qr_codes	QR Usage Log

الجزء 22 — إشعارات وتنبيهات القسم

Notification Type	Receiver	Sender	Event
Real-time (Supabase Realtime)	Affected User	System	Permission Changed
Alert (future)	User / IT Admin	System	Login from New IP
Security Alert (future)	IT Admin	System	Multiple Failed Logins
Notification	HR Manager	System	New User Created
Notification	IT Admin	HR Manager	Account Suspended

الجزء 23 — مسارات معالجة الأخطاء والاستثناءات

Login Failure Flow

```
flowchart TD
    A[👤 User] --> B[Submit Credentials]
    B --> C{Validate}
    C -->|Invalid Password| D[login_history: status=failed]
    C -->|User Not Found| E[login_history: status=failed]
    C -->|Account Suspended| F[login_history: status=locked]
    D --> G[Show: Invalid Credentials]
    E --> G
    F --> H[Show: Account Suspended]
    G --> A
    H --> I[Contact HR]
```

QR Reuse Error Flow

```
flowchart TD
  A[👤 Employee] --> B[Scan QR]
  B --> C{Check used_qr_codes}
  C -->|Hash Exists| D[❌ QR Already Used]
  D --> E[Show Error Message]
  E --> F[Employee Must Get New QR]
  C -->|Hash Not Found| G[✅ Process Check-In]
```

Recovery	Database Effect	Impact	Error
User retries or resets password	login_history record created	Login blocked	Invalid credentials
HR must reactivate account	login_history record created	Login blocked	Account suspended
Generate new QR code	No attendance record created	Check-in rejected	QR already used
User must re-login	Session marked inactive	Access denied	Session expired
Admin must grant permission	No data change	Page/Action blocked	Permission denied
Retry or contact IT	No records	System unavailable	Database connection error

الجزء 4 — القسم 2: الموارد البشرية وشؤون الموظفين 👤

4.1 الغرض من الوحدة وأهميتها التشغيلية

The HR & People Management module is the **largest and most complex module** in the system. It manages the complete employee lifecycle from recruitment through offboarding, including daily .operations like attendance tracking, leave management, and performance evaluation

- Why it exists:** Centralizes all HR operations for a polymer manufacturing factory with 16 departments and hundreds of employees across multiple shifts
- What problem it solves:** Replaces paper-based attendance sheets, manual leave tracking, and disconnected HR processes with an integrated digital system
- Which users interact:** HR Manager, Hall Managers, Employees (self-service), Owner (final approvals), Nurse (medical integration)
- What data it manages:** 20+ tables covering employee records, attendance, leaves, absence tracking, overtime, missions, complaints, disciplinary actions, offboarding, performance reviews, training, asset assignments, warnings, Friday work requests, and recruitment/ATS

4.2 الجهات الفاعلة في هذه الوحدة

المحظورات والقيود التشغيلية	الصلاحيات والإجراءات المصرح بها	Actor
View other employees' data, Approve requests, Edit attendance	View own data, Submit leave/mission/expense/complaint requests, QR check-in/out, View own salary	Employee
Approve own leaves, Process payroll, Access other departments	Approve/Reject team leaves (1st level), View team attendance, Submit Friday work requests	Hall Manager
Final approve leaves (Owner), Release payroll funds	Full CRUD on employees, Process attendance, Approve leaves (2nd level), Manage recruitment, Issue warnings, Process offboarding, Manage training	HR Manager
N/A (full access)	Final leave approval (3rd level), Approve loans, View all HR data	Owner
Edit employee records, Approve leaves	View employee data for clinic visits, Issue rest permits	Nurse

4.3 الميزات التفصيلية والوظائف (Detailed)

Feature Group A: Employee Management

Business Rules	CRUD	Tables	Output	Input	Actor	Description	Feature
BR-HR-01: employee_id must be unique. BR-HR-02: Email and username must be unique	C	users, permission_templates	Employee record + auto-permissions	Personal data, role, department, salary	HR Manager	Add new employee record	Create Employee
BR-HR-03: Employees see only own data	R	users	Employee profile	employee_id	HR Manager, Hall Manager (team only)	View employee details	View Employee
BR-HR-04: Salary changes logged	U	users, activity_log	Updated record + activity_log	Updated fields	HR Manager	Edit employee information	Update Employee
BR-HR-05: Triggers offboarding	U	users	Status changed	employee_id, reason	HR Manager	Set employee status to inactive	Deactivate Employee

Business Rules	CRUD	Tables	Output	Input	Actor	Description	Feature
	R	users	Employee cards grouped by department	-	All authenticated	Browse organization structure	View Org Directory

Feature Group B: Attendance Management

CRUD	Tables	Output	Input	Actor	Description	Feature
C	attendance, used_qr_codes	Attendance record with check_in time	QR hash	Employee	Employee scans QR at gate entry	QR Check-In
U	attendance, used_qr_codes	Attendance updated with check_out, working_hours calculated	QR hash	Employee	Employee scans QR at gate exit	QR Check-Out
C/U	attendance	Attendance record	employee, date, check_in, check_out	HR Manager	HR enters attendance manually	Manual Attendance
R	attendance	Attendance list with stats	Date range, department filter	HR (all), Employee (own)	View attendance records	View Attendance
R	attendance	PDF/Excel report	Date range	HR Manager	Export attendance summary	Attendance Report
U/C	attendance, late_deductions	delay_minutes field, possible late_deduction	check_in time vs shift start	System	System calculates delay_minutes	Late Detection
-	in-memory) (generation	QR code image	-	HR Manager	HR generates single-use QR	Generate QR Code

Attendance Business Rules: - BR-HR-06: $\text{delay_minutes} = \text{check_in} - \text{shift_start_time}$ (if positive) - BR-HR-07: $\text{working_hours} = \text{check_out} - \text{check_in}$ (in hours) - BR-HR-08: Status transitions: absent → checked_in → present - BR-HR-09: QR codes are single-use (enforced by used_qr_codes PK)

Feature Group C: Leave Management

CRUD	Tables	Output	Input	Actor	Description	Feature
C	leave_requests	Leave request (status=pending_manager)	type, start_date, end_date, reason	Employee	Employee requests time off	Submit Leave Request
U	leave_requests	Status → pending_hr OR rejected	request_id, approve/reject	Hall Manager	Hall Manager reviews request	Manager Approve Leave

CRUD	Tables	Output	Input	Actor	Description	Feature
U	leave_requests	Status → pending_owner OR rejected	request_id, approve/reject	HR Manager	HR reviews manager-approved request	HR Approve Leave
U	leave_requests	Status → approved OR rejected	request_id, approve/reject	Owner	Final approval	Owner Approve Leave
R	leave_requests	Leave list with balances	-	Employee	Employee views own leave history	View My Leaves
R/U	leave_requests	Leave dashboard	Filters	HR Manager	HR views/manages all leave requests	Manage All Leaves

:Leave Approval Workflow

Employee → pending_manager → Hall Manager → pending_hr → HR Manager → pending_owner → Owner → approve

Leave Business Rules: - BR-HR-10: Leave types: Annual, Sick, Emergency, Unpaid, Maternity/Paternity - BR-HR-11: Annual leave deducted from annual_leave_balance (default 24 days/year) - BR-HR-12: Emergency leaves may bypass manager approval - BR-HR-13: Rejected requests must include rejection_reason - BR-HR-14: Approved annual leaves reduce annual_leave_balance by days count

Feature Group D: Absence Management (Advanced)

CRUD	Tables	Actor	Description	Feature
C	hr_absence_requests	Employee	Employee submits pre-planned absence	Submit Absence Request
U	hr_absence_requests	Hall Manager	Manager approves/rejects absence	Manager Review Absence
U	hr_absence_requests	HR Manager	HR final decision	HR Review Absence
C	hr_absence_records	HR Manager	System/HR records unexcused absence	Record Unauthorized Absence
U	hr_absence_records, hr_absence_audit	HR Manager	HR changes default 2-day penalty	Override Absence Penalty
C	hr_violations_warnings	System	System triggers warning for consecutive absences	Auto-Warning Trigger
U	hr_absence_settings	HR Manager	Configure penalty defaults	Manage Absence Settings

Absence Business Rules: - BR-HR-15: Default penalty for unauthorized absence = 2 days salary deduction - BR-HR-16: HR can override to: deduct_2, deduct_1, casual_leave, unpaid_leave, cancelled -

BR-HR-17: 3 consecutive unauthorized absences → first_warning - BR-HR-18: Continued absence after warning → final_warning - BR-HR-19: Continued absence after final warning → termination - BR-HR-20: All absence changes logged in hr_absence_audit

Feature Group E: Overtime & Missions

CRUD	Tables	Actor	Description	Feature
C	overtime	Employee / HR	Record overtime hours	Submit Overtime
U	overtime	Hall Manager	Manager approves overtime	Approve Overtime
C	missions	Employee	Employee requests to leave for errand	Submit Mission/Errand
U	missions	Hall Manager	Manager approves mission	Approve Mission
U	missions, used_qr_codes	Employee	Employee scans QR leaving for mission	QR Mission Out
U	missions, used_qr_codes	Employee	Employee scans QR returning from mission	QR Mission Return
C	friday_work_requests	Hall Manager	Department requests employees work on Friday	Friday Work Request
U	friday_work_requests	HR Manager	HR approves Friday work	Approve Friday Work

Business Rules: - BR-HR-21: Overtime rate default = 1.5x base hourly rate - BR-HR-22: Overtime must be approved before payroll calculation - BR-HR-23: Missions track actual_time_out and actual_time_in via QR - BR-HR-24: Friday work requests include employee UUID array

Feature Group F: Complaints & Disciplinary

CRUD	Tables	Actor	Description	Feature
C	complaints	Employee	Employee files complaint (can be anonymous)	Submit Complaint
U	complaints	HR Manager	HR resolves complaint	Resolve Complaint
C	disciplinary_actions	HR Manager	HR issues disciplinary action	Issue Disciplinary Action
C	employee_warnings	HR Manager	HR issues formal warning	Issue Warning
R	employee_warnings	HR Manager / Employee	View employee warnings history	View Warnings

Business Rules: - BR-HR-25: Anonymous complaints hide employee identity (is_anonymous=true) - BR-HR-26: Warning types escalate: verbal → written → final → suspension → termination - BR-HR-27: Warnings can be appealed (status → appealed) - BR-HR-28: Warnings can include deduction_amount and deduction_days

Feature Group G: Performance & Training

CRUD	Tables	Actor	Description	Feature
C	performance_reviews	Hall Manager / HR	Manager creates review for employee	Create Performance Review
U	performance_reviews	Reviewer	Submit completed review	Submit Review
U	performance_reviews	Employee	Employee acknowledges review	Acknowledge Review
C	training_courses	HR Manager	Plan training course	Create Training Course
C	training_enrollments	HR Manager	Enroll employee in course	Enroll Employee
U	training_enrollments	HR Manager	Mark course as completed	Record Completion
C	asset_assignments	HR Manager	Assign company asset to employee	Assign Asset
U	asset_assignments	HR Manager	Record asset return	Return Asset

Business Rules: - BR-HR-29: Performance scores are 1-5 for each dimension (attendance, quality, teamwork, initiative, communication) - BR-HR-30: Review workflow: draft → submitted → reviewed → acknowledged - BR-HR-31: Training types: internal, external, online - BR-HR-32: Asset types: laptop, phone, tablet, vehicle, key, uniform, tool, other

Feature Group H: Offboarding

CRUD	Tables	Actor	Description	Feature
C	offboarding	HR Manager	Start employee separation process	Initiate Offboarding
U	offboarding (it_cleared)	IT Admin	IT confirms equipment returned	IT Clearance
U	offboarding (hr_cleared)	HR Manager	HR confirms HR obligations cleared	HR Clearance
U	offboarding (finance_cleared)	Accountant	Finance confirms financial obligations cleared	Finance Clearance
U	offboarding, users (status→inactive)	HR Manager	All clearances done, complete process	Complete Offboarding

Business Rules: - BR-HR-33: All three clearances (IT, HR, Finance) required before completion - BR-HR-34: Offboarding deactivates user account (status → inactive)

Feature Group I: Recruitment / ATS

CRUD	Tables	Actor	Description	Feature
-	(handled via ats_applications context)	HR Manager	Create job posting	Post Job
C	ats_applications	External / HR	Candidate applies with CV	Submit Application

CRUD	Tables	Actor	Description	Feature
U	ats_applications (ai_score, ai_verdict)	AI Engine	AI analyzes CV and scores candidate	AI CV Scoring
U	ats_applications	HR Manager	HR reviews AI-scored application	Review Application
U	ats_applications (status→interview)	HR Manager	Move candidate to interview stage	Schedule Interview
U/C	ats_applications (status→hired), users	HR Manager	Convert to employee	Hire Candidate

Business Rules: - BR-HR-35: AI verdict: accepted, rejected, or review - BR-HR-36: Application workflow: pending → screening → interview → offered → hired / rejected - BR-HR-37: AI matches skills from CV against job requirements

PART 5 & 6 — PROCESSES

Process HR-P01: Leave Request Approval Cycle

الخاصية	التفاصيل والبيان
Process ID	HR-P01
Start Actor	Employee
End Actor	Employee (receives approval/rejection notification)
المحفز	Employee submits leave request
Main Flow	Employee submits leave request → 2. Status = pending_manager → 3. Hall Manager reviews → 4. If approved: status = pending_hr → 5. HR Manager reviews → 6. If approved: status = pending_owner → 7. Owner reviews → 8. If approved: status = approved, annual_leave_balance decreased → 9. Notification sent to Employee
Alternative Flows	3a. Manager rejects → status = rejected, notification to Employee. 5a. HR rejects → same. 7a. Owner rejects → same
الجدول المعنية	leave_requests, users (annual_leave_balance), notifications
قواعد العمل	BR-HR-10 through BR-HR-14
الأخطاء المحتملة	Insufficient leave balance, Invalid date range, Leave conflict
Notifications	Each approval/rejection generates notification to employee

Process HR-P02: Attendance Check-In/Out

الخاصية	التفاصيل والبيان
Process ID	HR-P02

الخاصية	التفاصيل والبيان
Start Actor	Employee
End Actor	System (attendance recorded)
المحفز	Employee scans QR code at factory gate
Main Flow	Employee scans QR → 2. System validates QR hash → 3. If check-in: Create attendance record with check_in, .1 calculate delay_minutes → 4. If check-out: Update attendance with check_out, calculate working_hours → 5. If late: Create late_deductions record
الجدول المعنوية	used_qr_codes, attendance, late_deductions

Process HR-P03: Unauthorized Absence Handling

الخاصية	التفاصيل والبيان
Process ID	HR-P03
Start Actor	System (detects absence)
End Actor	Employee (receives penalty/warning)
المحفز	Employee does not check in by shift end
Main Flow	System detects no attendance for employee on workday → 2. HR records unauthorized absence → 3. Default 2- .1 day salary penalty applied → 4. System checks consecutive absence count → 5. If ≥ 3 consecutive: Generate first_warning → 6. If continued: final_warning → 7. If continued: termination recommendation
الجدول المعنوية	hr_absence_records, hr_violations_warnings, hr_absence_audit
قواعد العمل	BR-HR-15 through BR-HR-20

Process HR-P04: Offboarding Clearance

الخاصية	التفاصيل والبيان
Process ID	HR-P04
Start Actor	HR Manager
End Actor	HR Manager (process completed)
المحفز	Employee resignation/termination

الخاصية	التفاصيل والبيان
Main Flow	HR initiates offboarding → 2. IT receives clearance request → 3. IT confirms equipment returned .1 (it_cleared=true) → 4. Finance confirms no pending dues (finance_cleared=true) → 5. HR confirms HR obligations clear (hr_cleared=true) → 6. All cleared → status=completed → 7. User account deactivated
الجدول المعنية	offboarding, users

Process HR-P05: ATS Recruitment with AI

الخاصية	التفاصيل والبيان
Process ID	HR-P05
Start Actor	Candidate (submits CV) or HR Manager (enters CV)
End Actor	HR Manager (hires or rejects)
Main Flow	CV submitted → 2. AI Engine analyzes cv_text → 3. AI calculates ai_score, sets ai_verdict → 4. HR reviews AI .1 analysis → 5. If promising: status → screening → interview → offered → hired → 6. Create user account
الجدول المعنية	ats_applications, users

الجزء 8 — المخططات الانسيابية لمسارات العمل

Leave Approval Workflow

```
flowchart TD
  A[👤 Employee] --> B[Submit Leave Request]
  B --> C[Status: pending_manager]
  C --> D{Hall Manager Review}
  D -->|Approve| E[Status: pending_hr]
  D -->|Reject| F[Status: rejected]
  E --> G{HR Manager Review}
  G -->|Approve| H[Status: pending_owner]
  G -->|Reject| F
  H --> I{Owner Review}
  I -->|Approve| J[Status: approved]
  I -->|Reject| F
  J --> K[Deduct from annual_leave_balance]
  K --> L[📧 Notify Employee]
  F --> L
  L --> M[👤 Employee]
```

Attendance Flow

```

flowchart TD
    A[👤 Employee] --> B[Scan QR at Gate]
    B --> C{First Scan Today?}
    C -->|Yes - Check In| D[Create Attendance Record]
    C -->|No - Check Out| E[Update Attendance Record]
    D --> F{Late?}
    F -->|Yes| G[Calculate delay_minutes]
    G --> H[Create late_deductions]
    F -->|No| I[Status: present]
    E --> J[Calculate working_hours]
    J --> K[Status: present]
    H --> I
  
```

Absence Penalty Escalation

```

flowchart TD
    A[System Detects Absence] --> B[HR Records Absence]
    B --> C[Apply 2-Day Penalty Default]
    C --> D{HR Override?}
    D -->|Yes| E[Apply HR Decision]
    D -->|No| F[Keep Default Penalty]
    E --> G{Check Consecutive Days}
    F --> G
    G -->|< 3 days| H[Record Only]
    G -->|>= 3 days| I[Issue First Warning]
    I --> J{Continued Absence?}
    J -->|Yes| K[Issue Final Warning]
    K --> L{Still Absent?}
    L -->|Yes| M[Recommend Termination]
    L -->|No| N[Employee Returns]
  
```

Offboarding Flow

```

flowchart TD
    A[👤 HR Manager] --> B[Initiate Offboarding]
    B --> C[IT Clearance Request]
    B --> D[Finance Clearance Request]
    B --> E[HR Clearance Request]
    C --> F{IT Cleared?}
    D --> G{Finance Cleared?}
    E --> H{HR Cleared?}
    F -->|✓| I[it_cleared = true]
    G -->|✓| J[finance_cleared = true]
    H -->|✓| K[hr_cleared = true]
    I --> L{All Three Cleared?}
    J --> L
    K --> L
    L -->|Yes| M[Status: completed]
    M --> N[Deactivate User Account]
    N --> O[👤 HR Manager - Confirmed]
  
```

ATS Recruitment Flow

flowchart TD

```
A[📄 Candidate CV] --> B[Submit Application]
B --> C[🤖 AI Engine Analyzes CV]
C --> D[AI Score + Verdict]
D --> E{AI Verdict}
E -->|Accepted| F[Status: screening]
E -->|Review| G[HR Manual Review]
E -->|Rejected| H[Status: rejected]
G -->|Pass| F
G -->|Fail| H
F --> I[Schedule Interview]
I --> J{Interview Result}
J -->|Pass| K[Status: offered]
J -->|Fail| H
K --> L{Accept Offer?}
L -->|Yes| M[Status: hired]
L -->|No| H
M --> N[Create Employee Record]
N --> O[👤 HR Manager]
```

PART 14 & 15 — SECTION ERD

```

erDiagram
    USERS ||--o{ ATTENDANCE : "tracks"
    USERS ||--o{ LEAVE_REQUESTS : "submits"
    USERS ||--o{ OVERTIME : "records"
    USERS ||--o{ MISSIONS : "requests"
    USERS ||--o{ COMPLAINTS : "files"
    USERS ||--o{ DISCIPLINARY_ACTIONS : "receives"
    USERS ||--o{ OFFBOARDING : "undergoes"
    USERS ||--o{ EMPLOYEE_WARNINGS : "receives"
    USERS ||--o{ PERFORMANCE_REVIEWS : "reviewed_in"
    USERS ||--o{ TRAINING_ENROLLMENTS : "enrolled_in"
    USERS ||--o{ ASSET_ASSIGNMENTS : "assigned"
    USERS ||--o{ HR_ABSENCE_REQUESTS : "submits"
    USERS ||--o{ HR_ABSENCE_RECORDS : "has"
    USERS ||--o{ HR_VIOLATIONS_WARNINGS : "receives"
    USERS ||--o{ FRIDAY_WORK_REQUESTS : "requests"
    TRAINING_COURSES ||--o{ TRAINING_ENROLLMENTS : "has"
    USERS ||--o{ ATS_APPLICATIONS : "reviews"

    USERS {
        uuid id PK
        text employee_id UK
        text full_name
        text email UK
        text role
        text department
        text position
        numeric base_salary
        text shift
        text status
        numeric annual_leave_balance
        date hire_date
        boolean insurance_active
    }

    ATTENDANCE {
        uuid id PK
        uuid employee_id FK
        text employee_name
        text department
        date date
        timestampz check_in
        timestampz check_out
        text shift
        integer delay_minutes
        numeric working_hours
        text status
        text qr_check_in
        text qr_check_out
    }

    LEAVE_REQUESTS {
        uuid id PK
        uuid employee_id FK
        text type
        date start_date

```

```
    date end_date
    numeric days
    text reason
    text status
    text rejection_reason
    text approved_by
}

OVERTIME {
    uuid id PK
    uuid employee_id FK
    date date
    numeric hours
    text reason
    text status
    numeric rate
    text approved_by
}

MISSIONS {
    uuid id PK
    uuid employee_id FK
    date mission_date
    text reason
    time time_out
    time time_in
    text status
    text qr_out
    text qr_in
    numeric overtime_hours
}

COMPLAINTS {
    uuid id PK
    uuid employee_id FK
    text subject
    text description
    boolean is_anonymous
    text status
}

DISCIPLINARY_ACTIONS {
    uuid id PK
    uuid employee_id FK
    text type
    text reason
    uuid issued_by FK
}

OFFBOARDING {
    uuid id PK
    uuid employee_id FK
    date separation_date
    text reason
    boolean it_cleared
    boolean hr_cleared
    boolean finance_cleared
    text status
}
```

```
LATE_DEDUCTIONS {
    uuid id PK
    text employee_name
    date date
    integer delay_minutes
    text deduction_type
    numeric amount
}

HR_ABSENCE_REQUESTS {
    uuid id PK
    uuid employee_id
    text type
    date start_date
    date end_date
    boolean emergency_flag
    text status
    text manager_notes
    text hr_notes
}

HR_ABSENCE_RECORDS {
    uuid id PK
    uuid employee_id
    date date
    text type
    numeric penalty_days
    text hr_decision
    text status
}

HR_VIOLATIONS_WARNINGS {
    uuid id PK
    uuid employee_id
    text warning_level
    text reason
    integer consecutive_days
    text status
}

HR_ABSENCE_AUDIT {
    uuid id PK
    uuid record_id
    text table_name
    text action
    text old_value
    text new_value
    text changed_by
}

PERFORMANCE_REVIEWS {
    uuid id PK
    uuid employee_id FK
    uuid reviewer_id FK
    text review_period
    integer overall_rating
    integer attendance_score
    integer quality_score
    text status
}
```

```
TRAINING_COURSES {
    uuid id PK
    text title
    text trainer
    text course_type
    text department
    date start_date
    date end_date
    text status
}

TRAINING_ENROLLMENTS {
    uuid id PK
    uuid course_id FK
    uuid employee_id FK
    text status
    numeric score
    text certificate_url
}

ASSET_ASSIGNMENTS {
    uuid id PK
    text asset_name
    text asset_type
    text serial_number
    uuid assigned_to FK
    uuid assigned_by FK
    date assigned_date
    date return_date
    text status
}

EMPLOYEE_WARNINGS {
    uuid id PK
    uuid employee_id FK
    text warning_type
    text reason
    date incident_date
    uuid issued_by FK
    numeric deduction_amount
    text status
}

FRIDAY_WORK_REQUESTS {
    uuid id PK
    text department
    uuid requested_by FK
    uuid_arr employees
    date request_date
    text status
}

ATS_APPLICATIONS {
    uuid id PK
    text job_title
    text department
    text candidate_name
    text candidate_email
    text cv_text
}
```



```

numeric ai_score
text ai_verdict
text status
}

```

PART 16 — CRUD ANALYSIS

Tables	Actor	D	U	R	C	Feature
users	HR Manager				✓	Create Employee
users	HR Manager			✓		View Employees
users	HR Manager		✓			Update Employee
users	Owner	✓				Delete Employee
attendance, used_qr_codes	Employee			✓	✓	QR Check-In
attendance, used_qr_codes	Employee		✓	✓		QR Check-Out
attendance	Employee			✓		View My Attendance
attendance	HR Manager			✓		View All Attendance
leave_requests	Employee				✓	Submit Leave
leave_requests	Hall Manager		✓	✓		Approve Leave (Manager)
leave_requests	HR Manager		✓	✓		Approve Leave (HR)
leave_requests	Owner		✓	✓		Approve Leave (Owner)
overtime	Employee				✓	Submit Overtime
overtime	Hall Manager		✓	✓		Approve Overtime
missions	Employee				✓	Submit Mission
complaints	Employee				✓	Submit Complaint
complaints	HR Manager		✓	✓		Resolve Complaint
employee_warnings	HR Manager				✓	Issue Warning
performance_reviews	Hall Manager/HR				✓	Create Performance Review
training_courses	HR Manager				✓	Create Training
training_enrollments	HR Manager				✓	Enroll in Training
asset_assignments	HR Manager				✓	Assign Asset
asset_assignments	HR Manager		✓			Return Asset
offboarding	HR Manager				✓	Initiate Offboarding

Tables	Actor	D	U	R	C	Feature
offboarding	HR/IT/Finance		✓			Complete Offboarding
ats_applications	HR/Candidate				✓	Submit ATS Application
ats_applications	HR Manager		✓	✓		Review ATS Application
hr_absence_records	HR Manager				✓	Record Absence
hr_absence_records	HR Manager		✓			Override Absence Penalty
friday_work_requests	Hall Manager				✓	Friday Work Request

الجزء 17 — مصفوفة الصلاحيات التفصيلية للقسم

Owner	HR Mgr	Hall Mgr	Employee	Feature
✓	✓	✓	✓	View Own Profile
✓	✓	Team	X	View All Employees
✓	✓	X	X	Create Employee
✓	✓	X	X	Edit Employee
✓	X	X	X	Delete Employee
✓	✓	✓	✓	View Own Attendance
✓	✓	Team	X	View All Attendance
✓	✓	✓	✓	Submit Leave
X	X	✓	X	Approve Leave (L1)
X	✓	X	X	Approve Leave (L2)
✓	X	X	X	Approve Leave (L3)
✓	✓	✓	✓	Submit Overtime
✓	✓	✓	X	Approve Overtime
✓	✓	✓	✓	Submit Complaint
✓	✓	X	X	Resolve Complaint
✓	✓	X	X	Issue Warning
✓	✓	X	X	Manage Recruitment
✓	✓	X	X	Process Offboarding
✓	✓	X	X	Manage Training
✓	✓	Create	X	Performance Reviews

Owner	HR Mgr	Hall Mgr	Employee	Feature
✓	✓	X	X	Absence Management
✓	✓	✓	X	Friday Work Requests
✓	✓	X	X	Export Reports

الجزء 18 — قواعد العمل الحاكمة (Business Rules) SUMMARY

Impact	Rule	ID
Prevents duplicate employee codes	employee_id must be unique	BR-HR-01
Prevents duplicate accounts	Email and username must be unique	BR-HR-02
Privacy and security	Employees can only view their own data	BR-HR-03
Auto-calculate lateness	Delay = check_in - shift_start (if positive)	BR-HR-06
Auto-calculate work duration	working_hours = check_out - check_in	BR-HR-07
Constrained by DB CHECK	Leave types: Annual/Sick/Emergency/Unpaid/Maternity	BR-HR-10
Balance tracking	Annual leave deducted from balance (default 24 days)	BR-HR-11
Accountability	Rejected requests must include rejection_reason	BR-HR-13
Standard penalty	Default unauthorized absence penalty = 2 days salary	BR-HR-15
Escalation policy	consecutive absences → first_warning +3	BR-HR-17
Employee protection	Anonymous complaints hide employee identity	BR-HR-25
Progressive discipline	Warning escalation: verbal→written→final→suspension→termination	BR-HR-26
Standardized evaluation	Performance scores 1-5 per dimension	BR-HR-29
Complete exit process	All 3 clearances required for offboarding completion	BR-HR-33
Automated screening	AI CV scoring: accepted/rejected/review	BR-HR-35

الجزء 22 — إشعارات وتنبيهات القسم

Type	Receiver	Sender	Event
Notification	Hall Manager	Employee	Leave Request Submitted
Notification	HR Manager	Hall Manager	Leave Approved by Manager
Notification	Owner	HR Manager	Leave Approved by HR

Type	Receiver	Sender	Event
Notification	Employee	Owner / HR / Manager	Leave Final Decision
Notification	Hall Manager	Employee	Overtime Submitted
Notification	Employee	HR Manager	Warning Issued
Notification	Employee	System	Absence Penalty Applied
Notification	IT/Finance	HR Manager	Offboarding Initiated
Notification	Employee	HR Manager	Training Enrollment
Notification	Employee	Reviewer	Performance Review Created
Notification	HR Manager	System/External	ATS New Application
Notification	Hall Manager	HR Manager	Friday Work Approved

PART 23 — ERROR FLOWS

Recovery	Database Effect	Impact	Error Scenario
Employee must choose Unpaid leave	No record created	Request blocked	Leave request with insufficient balance
Check existing record first	No change	Insert fails	Duplicate attendance for same date
Generate new QR	No attendance update	Check-in rejected	QR code already used
Employee adjusts dates	No record created	Request should be rejected	Leave date conflict (overlapping)
Complete remaining clearances	Status stays pending	Cannot complete	Offboarding without all clearances
Use valid type from enum	CHECK constraint violation	Insert fails	Invalid leave type
HR must provide reason	Override recorded without context	Audit incomplete	Absence penalty override without reason

الجزء 4 — القسم 3: الرواتب، الأجور، والتعويضات 💰

4.1 الغرض من الوحدة وأهميتها التشغيلية

Automates monthly salary calculation integrating data from Attendance, Overtime, Leaves, Loans, and .Late Deductions modules. Manages employee loans, medical reimbursements, and expense claims

Why: Manual payroll calculation for hundreds of factory workers across shifts is error-prone and time-consuming

Problem Solved: Automated net salary = base_salary + overtime_pay + bonuses + performance_bonus - penalties - late_deductions - absence_deductions - loan_installment

Users: HR Manager (calculate/process), Owner (approve funding), Employee (view own), Accountant (payment processing)

4.2 الجهات الفاعلة في هذه الوحدة

Role in This Section	Actor
Calculates payroll, manages adjustments, processes loans	HR Manager
Approves payroll funding, approves loans, approves medical disbursements	Owner
Views own salary, submits loan/medical/expense requests	Employee
Processes final payments, records in GL	Accountant

4.3 الميزات التفصيلية والوظائف

Business Rules	CRUD	Tables	Output	Input	Actor	Feature
BR-PAY-01: Net = Base + OT + Bonuses - Penalties - Deductions	C	payroll, attendance, overtime, loans, late_deductions	Payroll records for all employees	Month (YYYY-MM)	HR Manager	Calculate Monthly Payroll
	R	payroll	Summary with totals	Month filter	HR Manager	View Payroll Dashboard
BR-PAY-02: Owner must approve before payment	U	payroll	Status → funds_released	Month	Owner	Release Funds
BR-PAY-03: Funds must be released first	U	payroll	Status → paid, paid_date set	Month	HR Manager	Mark as Paid
BR-PAY-04: Employee sees only own records	R	payroll	Own payroll history	-	Employee	View My Salary

Business Rules	CRUD	Tables	Output	Input	Actor	Feature
	C/R/U	salary_adjustments	Adjustment record	Employee, amount, type	HR Manager	Salary Adjustments
BR-PAY-05: Monthly_deduction = amount / installments	C	loans	Loan request (status=pending)	Amount, installments, reason	Employee	Submit Loan Request
BR-PAY-06: Owner approval required	U	loans	Status → active	loan_id	Owner	Approve Loan
BR-PAY-07: Employee confirms money received	U	loans	employee_receipt_status → confirmed	loan_id	Employee	Confirm Loan Receipt
	U	loans	hr_final_approval = true	loan_id	HR Manager	HR Final Loan Approval
BR-PAY-08: Deferred months skip deduction	U	loans	Month added to []deferred_months	loan_id, month	HR Manager	Defer Loan Month
BR-PAY-09: Auto-complete when fully paid	U	loans	Status → completed	remaining = 0	System	Complete Loan
	C	medical_requests	Request (status=pending)	Description, document, amount	Employee	Submit Medical Request
BR-PAY-10: Owner approval required	U	medical_requests	Status → approved_by_owner	request_id	Owner	Approve Medical
	U	medical_requests	Status → disbursed	request_id	HR Manager	Disburse Medical
	C	expenses	Expense (status=pending)	Type, amount, date, description	Employee	Submit Expense Claim
	U	expenses	Status → approved	expense_id	HR Manager	Approve Expense
BR-PAY-11: Deduction based on delay_minutes threshold	C	late_deductions	Deduction record	delay_minutes	System	Late Deduction Calc

Processes

Process PAY-P01: Monthly Payroll Calculation

flowchart TD

```

A[👤 HR Manager] --> B[Select Month]
B --> C[Fetch All Active Employees]
C --> D[For Each Employee]
D --> E[Get base_salary from users]
E --> F[Calculate overtime_pay from approved overtime]
F --> G[Get bonuses + performance_bonus]
G --> H[Calculate late_deductions from attendance delays]
H --> I[Calculate absence_deductions from absences]
I --> J[Get active loan monthly_deduction]
J --> K{Deferred Month?}
K -->|Yes| L[Skip Loan Deduction]
K -->|No| M[Include Loan Deduction]
L --> N[Net = Base + OT + Bonuses - Penalties - Late - Absence - Loan]
M --> N
N --> O[Insert Payroll Record]
O --> P[(payroll DB)]
P --> Q[Status: processing]
Q --> R[👤 HR Manager - Review]

```

Start Actor: HR Manager | **End Actor:** HR Manager (reviews generated payroll)

Process PAY-P02: Loan Lifecycle

flowchart TD

```

A[👤 Employee] --> B[Submit Loan Request]
B --> C[Status: pending]
C --> D{Owner Approval}
D -->|Approve| E[Status: active]
D -->|Reject| F[Status: rejected]
E --> G[👤 Employee Confirms Receipt]
G --> H[HR Final Approval]
H --> I[Monthly Deduction Begins]
I --> J{Each Month}
J --> K{Deferred?}
K -->|No| L[Deduct from Payroll]
K -->|Yes| M[Skip This Month]
L --> N[remaining_amount -= monthly_deduction]
N --> O{remaining = 0?}
O -->|Yes| P[Status: completed]
O -->|No| J
F --> Q[👤 Employee Notified]
P --> Q

```

Start Actor: Employee | **End Actor:** Employee (loan completed or rejected)

Process PAY-P03: Medical Request

flowchart TD

```
A[👤 Employee] --> B[Submit Medical Request + Document]
B --> C[Status: pending]
C --> D{Owner Review}
D -->|Approve| E[Status: approved_by_owner]
D -->|Reject| F[Status: rejected]
E --> G[HR Processes Disbursement]
G --> H[Status: disbursed]
H --> I[👤 Employee Receives Payment]
```

Start Actor: Employee | **End Actor:** Employee (receives disbursement)

Section ERD

```

erDiagram
    USERS ||--o{ PAYROLL : "has"
    USERS ||--o{ LOANS : "requests"
    USERS ||--o{ MEDICAL_REQUESTS : "submits"
    USERS ||--o{ EXPENSES : "claims"
    USERS ||--o{ LATE_DEDUCTIONS : "incurs"

    PAYROLL {
        uuid id PK
        uuid employee_id FK
        text month
        numeric base_salary
        numeric overtime_pay
        numeric bonuses
        numeric performance_bonus
        numeric penalties
        numeric late_deductions
        numeric absence_deductions
        numeric net_salary
        text status
        date paid_date
    }

    LOANS {
        uuid id PK
        uuid employee_id FK
        numeric amount
        integer installments
        numeric monthly_deduction
        numeric remaining_amount
        text status
        text reason
        text_arr deferred_months
        text approved_by
        text employee_receipt_status
        boolean hr_final_approval
    }

    MEDICAL_REQUESTS {
        uuid id PK
        uuid employee_id FK
        text description
        text document_url
        numeric amount
        text status
        text owner_comments
        text hr_comments
        text rejection_reason
    }

    EXPENSES {
        uuid id PK
        uuid employee_id FK
        text type
        numeric amount
        date date
  
```

```

    text description
    text status
}

LATE_DEDUCTIONS {
    uuid id PK
    text employee_name
    date date
    integer delay_minutes
    text deduction_type
    numeric amount
}

```

CRUD & Permissions

Accountant	Owner	HR Mgr	Employee	Feature
✓	✓	✓	✓	View Own Salary
✓	✓	✓	X	View All Payroll
X	✓	✓	X	Calculate Payroll
X	✓	X	X	Release Funds
✓	✓	✓	X	Mark Paid
X	X	✓	✓	Submit Loan
X	✓	X	X	Approve Loan
X	X	✓	✓	Submit Medical
X	✓	X	X	Approve Medical
X	✓	✓	X	Disburse Medical
X	X	✓	✓	Submit Expense
X	✓	✓	X	Approve Expense

Business Rules

Rule	ID
$\text{net_salary} = \text{base_salary} + \text{overtime_pay} + \text{bonuses} + \text{performance_bonus} - \text{penalties} - \text{late_deductions} - \text{absence_deductions}$	BR-PAY-01
'Payroll must be in 'funds_released' status before marking as 'paid'	BR-PAY-02

Rule	ID
Only Owner can release funds	BR-PAY-03
Employee can only view own payroll records	BR-PAY-04
$\text{monthly_deduction} = \text{amount} / \text{installments}$	BR-PAY-05
Loans require Owner approval	BR-PAY-06
Employee must confirm receipt before HR final approval	BR-PAY-07
Deferred months (TEXT[]) skip payroll deduction for that month	BR-PAY-08
Loan auto-completes when remaining_amount reaches 0	BR-PAY-09
Medical requests require Owner approval before HR can disburse	BR-PAY-10
Payroll status flow: processing → funds_released → paid	BR-PAY-11

الجزء 4 — القسم 4: المالية، الحسابات، والخزينة

4.1 الغرض من الوحدة وأهميتها التشغيلية

Comprehensive double-entry accounting system managing all financial operations of the factory: treasury, banking, invoicing, check lifecycle, journal entries, cost centers, fixed assets, taxes, and financial closings.

Why: A manufacturing operation requires rigorous financial tracking across multiple revenue streams, supplier payments, payroll, and maintenance costs

Problem Solved: Integrated GL with full invoice lifecycle, check management, custody tracking, tax filing, and period closings

Users: CFO, Chief Accountant, Accountant, Owner

4.2 الجهات الفاعلة في هذه الوحدة

Responsibilities	Actor
Strategic financial oversight, Approve large transactions, Financial reporting	CFO
Journal entry approval, Financial closing, Tax filing	Chief Accountant
Data entry (invoices, payments, checks), Daily operations	Accountant
Final approval for major financial decisions	Owner
Purchase invoice creation	Procurement Manager
Sales invoice creation	Sales Manager

4.3 الميزات التفصيلية والوظائف

CRUD	Tables	Actor	Feature
C/R/U	finance_safes	Accountant	Manage Safes (Treasury)
C/R/U	finance_bank_accounts	Accountant	Manage Bank Accounts
C/R/U	finance_clients	Sales/Accountant	Manage Finance Clients
C/R/U	finance_suppliers	Procurement/Accountant	Manage Finance Suppliers
C/R/U	finance_invoices	Accountant	Create Invoice (Sales/Purchase/Return)
C/R	finance_collections	Accountant	Record Collection/Payment
C/R/U	finance_checks	Accountant	Manage Checks (Incoming/Outgoing)
C/R/U	finance_custodies	Accountant	Issue Custody/Advance
U	finance_custodies	Employee/Accountant	Settle Custody
C/R	finance_journal_entries, finance_journal_lines	Accountant	Create Journal Entry
U	finance_journal_entries	Chief Accountant	Approve Journal Entry
U	finance_journal_entries	Chief Accountant	Post Journal Entry
C/R/U	finance_cost_centers	CFO	Manage Cost Centers
C/R/U	finance_fixed_assets	Accountant	Manage Fixed Assets
C/R/U	finance_expense_categories	CFO	Manage Expense Categories
C/R/U	finance_taxes	Chief Accountant	Tax Calculation & Filing
C/R	finance_closings	Chief Accountant	Financial Closing (Daily/Monthly/Yearly)
C/R	general_ledger	System/Accountant	Record GL Transaction

CRUD	Tables	Actor	Feature
C/R/U	chart_of_accounts	CFO	Manage Chart of Accounts
C/R/U	currencies	CFO	Manage Currencies
C/R	exchange_rates	Accountant	Update Exchange Rates
C/R/U	department_budgets	CFO	Department Budgets
R/U	doc_sequences	System	Auto Document Numbering

Processes

Process FIN-P01: Invoice Lifecycle

flowchart TD

```

A[👤 Accountant] --> B[Create Invoice]
B --> C{Invoice Type}
C -->|Sales| D[Link to finance_client]
C -->|Purchase| E[Link to finance_supplier]
D --> F[Calculate: subtotal - discount + tax + additions = total]
E --> F
F --> G[Status: unpaid]
G --> H[Record Payment/Collection]
H --> I{Full Payment?}
I -->|Yes| J[Status: paid]
I -->|No| K[Status: partial]
K --> L[remaining = total - paid_amount]
L --> H
J --> M[Auto GL Entry]
M --> N[👤 Accountant - Complete]

```

Process FIN-P02: Check Lifecycle (Incoming)

flowchart TD

```

A[📄 Receive Check from Client] --> B[Status: pending]
B --> C[Deposit to Bank]
C --> D[Status: under_collection]
D --> E{Bank Result}
E -->|Cleared| F[Status: collected]
E -->|Returned| G[Status: bounced]
F --> H[Update Bank Balance]
G --> I[Update Client Balance]
H --> J[GL Entry: Debit Bank, Credit AR]
I --> K[Notify Sales Manager]

```

Process FIN-P03: Journal Entry Flow

flowchart TD

```

A[👤 Accountant] --> B[Create Journal Entry]
B --> C[Add Journal Lines]
C --> D{Debit = Credit?}
D -->|No| E[❌ Error: Unbalanced]
D -->|Yes| F[Status: draft]
F --> G[Submit for Approval]
G --> H{Chief Accountant Review}
H -->|Approve| I[Status: approved]
H -->|Reject| J[Status: cancelled]
I --> K[Post to GL]
K --> L[Status: posted]
L --> M[Update Account Balances]

```

Process FIN-P04: Financial Closing

flowchart TD

```

A[👤 Chief Accountant] --> B[Select Closing Type]
B --> C{Type}
C -->|Daily| D[Reconcile Today's Transactions]
C -->|Monthly| E[Reconcile Month's Transactions]
C -->|Yearly| F[Full Year Reconciliation]
D --> G[Create Closing Record]
E --> G
F --> G
G --> H[is_locked = true]
H --> I[No Edits Allowed for Closed Period]
I --> J[👤 Chief Accountant - Confirmed]

```

Section ERD

```

erDiagram
    FINANCE_SAFES ||--o{ FINANCE_COLLECTIONS : "receives"
    FINANCE_BANK_ACCOUNTS ||--o{ FINANCE_COLLECTIONS : "receives"
    FINANCE_CLIENTS ||--o{ FINANCE_INVOICES : "billed"
    FINANCE_SUPPLIERS ||--o{ FINANCE_INVOICES : "bills"
    FINANCE_INVOICES ||--o{ FINANCE_COLLECTIONS : "paid_via"
    FINANCE_INVOICES ||--o{ FINANCE_CHECKS : "paid_by"
    FINANCE_CLIENTS ||--o{ FINANCE_CHECKS : "issues"
    FINANCE_SUPPLIERS ||--o{ FINANCE_CHECKS : "receives"
    FINANCE_JOURNAL_ENTRIES ||--|{ FINANCE_JOURNAL_LINES : "contains"
    FINANCE_COST_CENTERS ||--o{ FINANCE_COST_CENTERS : "parent"
    CHART_OF_ACCOUNTS ||--o{ CHART_OF_ACCOUNTS : "parent"

    FINANCE_SAFES {
        uuid id PK
        text name
        numeric balance
        boolean is_active
    }

    FINANCE_BANK_ACCOUNTS {
        uuid id PK
        text bank_name
        text account_number
        numeric balance
        boolean is_active
    }

    FINANCE_CLIENTS {
        uuid id PK
        text name
        text phone
        text email
        numeric credit_limit
        numeric balance
    }

    FINANCE_SUPPLIERS {
        uuid id PK
        text name
        text phone
        numeric balance
    }

    FINANCE_INVOICES {
        uuid id PK
        text invoice_number
        text invoice_type
        uuid client_id FK
        uuid supplier_id FK
        numeric subtotal
        numeric discount
        numeric tax_amount
        numeric total
        numeric paid_amount
        numeric remaining
  
```

```

    text status
}

FINANCE_COLLECTIONS {
    uuid id PK
    uuid invoice_id FK
    text collection_type
    numeric amount
    uuid safe_id FK
    uuid bank_id FK
    text reference_number
}

FINANCE_CHECKS {
    uuid id PK
    text check_number
    text check_type
    numeric amount
    text bank_name
    date due_date
    text status
    uuid client_id FK
    uuid supplier_id FK
    uuid invoice_id FK
}

FINANCE_CUSTODIES {
    uuid id PK
    uuid employee_id
    text employee_name
    numeric amount
    numeric settled_amount
    numeric remaining
    text status
}

FINANCE_JOURNAL_ENTRIES {
    uuid id PK
    text entry_number
    date entry_date
    text description
    numeric total_debit
    numeric total_credit
    text status
    uuid approved_by
}

FINANCE_JOURNAL_LINES {
    uuid id PK
    uuid entry_id FK
    text account_name
    numeric debit
    numeric credit
    text cost_center
}

FINANCE_COST_CENTERS {
    uuid id PK
    text name
    text code

```



```

    uuid parent_id FK
    boolean is_active
}

FINANCE_FIXED_ASSETS {
    uuid id PK
    text name
    text asset_code
    text category
    date purchase_date
    numeric purchase_cost
    numeric depreciation_rate
    numeric accumulated_depreciation
    numeric current_value
    text status
}

FINANCE_TAXES {
    uuid id PK
    text tax_type
    text period
    numeric taxable_amount
    numeric tax_amount
    text status
}

FINANCE_CLOSINGS {
    uuid id PK
    text closing_type
    text period
    boolean is_locked
}

CHART_OF_ACCOUNTS {
    uuid id PK
    text code UK
    text name_en
    text name_ar
    text account_type
    uuid parent_id FK
    numeric balance
}

GENERAL_LEDGER {
    uuid id PK
    timestampz transaction_date
    text account_type
    text account_name
    numeric debit
    numeric credit
    text reference_type
    uuid reference_id
}

DEPARTMENT_BUDGETS {
    uuid id PK
    text department
    text fiscal_year
    text budget_type
    numeric allocated_amount
}

```

```
numeric spent_amount
numeric remaining_amount
text status
}

DOC_SEQUENCES {
  uuid id PK
  text doc_type UK
  text prefix
  integer last_number
}
```

Business Rules

Rule	ID
Invoice total = subtotal - discount + tax_amount + additions	BR-FIN-01
remaining = total - paid_amount	BR-FIN-02
Invoice status: unpaid → partial → paid (based on remaining)	BR-FIN-03
Check lifecycle: pending → under_collection → collected/bounced/cancelled	BR-FIN-04
Journal entry must balance: total_debit = total_credit	BR-FIN-05
Journal workflow: draft → approved → posted / cancelled	BR-FIN-06
Posted journal entries cannot be edited	BR-FIN-07
Financial closing locks period — no edits allowed	BR-FIN-08
Cost centers support hierarchy (parent_id self-reference)	BR-FIN-09
Chart of accounts supports hierarchy (parent_id self-reference)	BR-FIN-10
Fixed asset current_value = purchase_cost - accumulated_depreciation	BR-FIN-11
Custody remaining = amount - settled_amount	BR-FIN-12
Document numbers auto-increment via doc_sequences	BR-FIN-13
Budget remaining = allocated_amount - spent_amount	BR-FIN-14
Each (department, fiscal_year, budget_type) is unique	BR-FIN-15

Permissions Matrix

Owner	CFO	Chief Acct	Accountant	Feature
✓	✓	✓	✓	Manage Safes

Owner	CFO	Chief Acct	Accountant	Feature
✓	✓	✓	✓	Manage Bank Accounts
✓	✓	✓	✓	Create Invoices
✓	✓	✓	✓	Record Payments
✓	✓	✓	✓	Manage Checks
✓	✓	✓	✓	Create Journal Entry
✓	✓	✓	X	Approve Journal Entry
✓	✓	✓	X	Post Journal Entry
✓	✓	✓	X	Financial Closing
✓	✓	✓	X	Manage Chart of Accounts
✓	✓	X	X	Department Budgets
✓	✓	✓	X	Tax Filing
✓	✓	✓	✓	View Financial Reports

=====

الجزء 4 — الأقسام 5 إلى 8: سلسلة التوريد والعمليات الصناعية

القسم 5: المشتريات وإدارة الموردين

Purpose 5.1

Manages the complete procurement lifecycle from departmental purchase requests through supplier quotations, purchase order creation, goods receipt, and settlement. Includes AI-powered supplier performance evaluation.

Actors 5.2

Responsibilities	Actor
Submit purchase requests	Any Department Head
Check stock availability, dispense items	Warehouse Manager
Approve requests, create POs, manage suppliers	Procurement Manager

Responsibilities	Actor
Request quotations, source suppliers	Procurement Specialist
Inspect incoming materials	Quality Manager
Approve major purchases	Owner

Purchase Request Workflow 5.3

Department → pending_warehouse → Warehouse checks stock →
 IF available: dispensed (from warehouse)
 IF not available: pending_manager → Procurement Manager approves →
 quotation_requested → purchased → pending_finance → received

Features 5.4

CRUD	Tables	Actor	Feature
C	purchase_requests	Dept Head	Submit Purchase Request
R/U	purchase_requests, inventory_items	Warehouse Mgr	Warehouse Stock Check
U/C	purchase_requests, inventory_transactions	Warehouse Mgr	Dispense from Stock
U	purchase_requests	Procurement Mgr	Approve Purchase Request
C	purchase_orders	Procurement Specialist	Create Purchase Order
U	purchase_orders	Procurement Mgr / Owner	Approve Purchase Order
U	purchase_orders, inventory_items	Warehouse Mgr	Receive Goods
C/U	qc_inspections, purchase_orders	QC Inspector	QC Incoming Material
C/R/U/D	suppliers	Procurement Mgr	Manage Suppliers
R	suppliers	Supplier	Supplier Portal Login
R	supplier_orders	Supplier	View Supplier Orders
C	supplier_transactions	Accountant	Record Supplier Transaction
C/R	supplier_performance	AI/Procurement	AI Supplier Performance

Process Flow

flowchart TD

```

A[👤 Department Head] --> B[Submit Purchase Request]
B --> C[Status: pending_warehouse]
C --> D{Warehouse Check}
D -->|In Stock| E[Dispense from Warehouse]
D -->|Out of Stock| F[Status: pending_manager]
E --> G[Status: dispensed]
F --> H{Procurement Manager}
H -->|Approve| I[Status: approved]
H -->|Reject| J[Status: rejected]
I --> K[Request Quotations from Suppliers]
K --> L[Status: quotation_requested]
L --> M[Create Purchase Order]
M --> N[Status: purchased]
N --> O[Goods Arrive at Warehouse]
O --> P[QC Inspection]
P --> Q{QC Result}
Q -->|Pass| R[Status: received]
Q -->|Fail| S[Return to Supplier]
R --> T[Update Inventory]
T --> U[Status: pending_finance]
U --> V[Finance Settlement]
V --> W[👤 Department Head - Complete]
  
```

Section ERD

erDiagram

```

PURCHASE_REQUESTS ||--o{ PURCHASE_ORDERS : "fulfilled_by"
PURCHASE_REQUESTS }o--|| INVENTORY_ITEMS : "for_item"
SUPPLIERS ||--o{ SUPPLIER_ORDERS : "receives"
SUPPLIERS ||--o{ SUPPLIER_TRANSACTIONS : "paid_via"
SUPPLIERS ||--o{ SUPPLIER_PERFORMANCE : "evaluated"
SUPPLIERS ||--o{ PURCHASE_ORDERS : "supplies"

```

```

PURCHASE_REQUESTS {
    uuid id PK
    uuid item_id FK
    text item_name
    integer requested_quantity
    text unit
    date delivery_date
    text status
    text requested_by
    text department
    text approved_by
}

```

```

PURCHASE_ORDERS {
    uuid id PK
    text po_number
    uuid supplier_id FK
    text supplier_name
    date order_date
    date delivery_date
    jsonb items
    numeric total
    text status
    text quality_status
    uuid approved_by FK
}

```

```

SUPPLIERS {
    uuid id PK
    text company_name
    text contact_person
    text email UK
    text phone
    text password_hash
}

```

```

SUPPLIER_ORDERS {
    uuid id PK
    uuid supplier_id FK
    text order_details
    numeric total_amount
    text status
}

```

```

SUPPLIER_TRANSACTIONS {
    uuid id PK
    uuid supplier_id FK
    numeric amount
}

```

```

text payment_method
text check_status
date transaction_date
}

SUPPLIER_PERFORMANCE {
  uuid id PK
  uuid supplier_id FK
  date period_start
  date period_end
  integer total_orders
  integer on_time_deliveries
  integer late_deliveries
  numeric quality_score
  numeric overall_score
  text ai_recommendation
}

```

Business Rules

Rule	ID
PR workflow: pending_warehouse → dispensed OR pending_manager → approved → quotation_requested → purchased → pending_finance → received	BR-PROC-01
Warehouse checks stock before escalating to procurement	BR-PROC-02
PO must reference a supplier	BR-PROC-03
PO quality_status must be 'approved' before closing	BR-PROC-04
Supplier performance AI score considers: on-time delivery, quality, price competitiveness, return rate	BR-PROC-05
Suppliers can login to portal with email + password_hash	BR-PROC-06



القسم 6: إدارة المخازن والمستودعات

Purpose 6.1

Real-time inventory tracking with stock movements, low-stock alerts, and material traceability across all warehouses.

Actors 6.2

Responsibilities	Actor
Full CRUD on inventory, Process movements	Warehouse Manager
Request materials (out), Return excess	Production Manager
Add incoming stock (in)	Procurement Manager

Features 6.3

CRUD	Tables	Actor	Feature
C	inventory_items	Warehouse Mgr	Add Inventory Item
U	inventory_items	Warehouse Mgr	Update Item
C	inventory_transactions, stock_movements	Warehouse Mgr	Stock In (Receive)
C	inventory_transactions, stock_movements	Warehouse Mgr	Stock Out (Dispense)
C	stock_movements	Warehouse Mgr	Stock Transfer
C	stock_movements	Warehouse Mgr	Stock Adjustment
R	inventory_items (quantity < min_quantity)	System	Low Stock Alert
R	inventory_items, stock_movements	Warehouse Mgr	View Stock Report

Process Flow

```
flowchart TD
    A[📦 Stock Movement Event] --> B{Movement Type}
    B -->|In| C[Increase quantity]
    B -->|Out| D[Decrease quantity]
    B -->|Transfer| E[Move between warehouses]
    B -->|Adjustment| F[Correct quantity]
    B -->|Return| G[Return to stock]
    C --> H[Update inventory_items.quantity]
    D --> H
    E --> H
    F --> H
    G --> H
    H --> I{quantity < min_quantity?}
    I -->|Yes| J[⚠️ Low Stock Alert]
    I -->|No| K[Normal]
    J --> L[Notify Warehouse Manager]
```


Section ERD

```

erDiagram
    INVENTORY_ITEMS ||--o{ INVENTORY_TRANSACTIONS : "tracked_by"
    INVENTORY_ITEMS ||--o{ STOCK_MOVEMENTS : "moved"

    INVENTORY_ITEMS {
        uuid id PK
        text name
        text category
        integer quantity
        integer min_quantity
        numeric last_purchase_price
        text supplier_name
    }

    INVENTORY_TRANSACTIONS {
        uuid id PK
        uuid item_id FK
        text item_name
        text transaction_type
        integer quantity
        text requested_by
        text processed_by
        timestampz date
        text notes
    }

    STOCK_MOVEMENTS {
        uuid id PK
        uuid item_id
        text item_name
        text warehouse
        text movement_type
        text source_module
        uuid source_id
        numeric quantity
        numeric unit_cost
        numeric total_cost
        text batch_number
        text created_by_name
    }

```

Business Rules

Rule	ID
Categories: Maintenance, Workshop, Supplies, Chemicals, Fixed Assets	BR-INV-01
Low stock alert when quantity < min_quantity	BR-INV-02
Transaction types: in, out	BR-INV-03
Movement types: in, out, transfer, adjustment, return	BR-INV-04

Rule	ID
Stock out cannot reduce quantity below 0	BR-INV-05
Movements linked to source_module (procurement, production, sales)	BR-INV-06



القسم 7: الإنتاج، التصنيع، وتتبع الدفعات

Purpose 7.1

Manages manufacturing operations from Bill of Materials through production stages to finished batches .with full material traceability

Actors 7.2

Responsibilities	Actor
Create/manage production orders, BOM, stages	Production Manager
Production planning and scheduling	Planning Manager
QC at production stages	Quality Manager
Provide raw materials	Warehouse Manager

Features 7.3

CRUD	Tables	Feature
C/R/U/D	bom, bom_items	Manage BOM
C	production_orders	Create Production Order
C/R/U	production_stage_templates	Define Stage Templates
C/R/U	production_stage_logs	Track Production Stages
C/R	production_consumption	Record Material Consumption
C	production_batches	Create Batch
R/U	production_batches, qc_inspections	Track Batch Quality

Process Flow

flowchart TD

```
A[👤 Production Manager] --> B[Create Production Order]
B --> C[Select Product + BOM]
C --> D[Request Materials from Warehouse]
D --> E[Record Consumption]
E --> F[Stage 1: Raw Material Prep]
F --> G{QC Required?}
G -->|Yes| H[QC Inspection]
G -->|No| I[Next Stage]
H -->|Pass| I
H -->|Fail| J[Stage Failed - Rework]
I --> K[Stage 2: Processing]
K --> L[Stage N: Final]
L --> M[Create Batch]
M --> N[Batch QC Inspection]
N -->|Pass| O[Status: finished]
N -->|Fail| P[Status: qc_failed]
O --> Q[Move to Finished Goods Warehouse]
Q --> R[👤 Warehouse Manager]
```

Start Actor: Production Manager | **End Actor:** Warehouse Manager (receives finished goods)

Section ERD

```

erDiagram
    PRODUCTS ||--o{ BOM : "has"
    BOM ||--|{ BOM_ITEMS : "contains"
    BOM_ITEMS }o--|| INVENTORY_ITEMS : "uses"
    PRODUCTION_ORDERS ||--o{ PRODUCTION_STAGE_LOGS : "has"
    PRODUCTION_ORDERS ||--o{ PRODUCTION_CONSUMPTION : "consumes"
    PRODUCTION_ORDERS ||--o{ PRODUCTION_BATCHES : "produces"
    PRODUCTION_STAGE_TEMPLATES ||--o{ PRODUCTION_STAGE_LOGS : "defines"
    PRODUCTION_BATCHES }o--o| QC_INSPECTIONS : "inspected_by"
    PRODUCTS ||--o{ PRODUCTION_BATCHES : "produced_as"

    BOM {
        uuid id PK
        uuid product_id FK
        text name
        integer version
        text status
    }

    BOM_ITEMS {
        uuid id PK
        uuid bom_id FK
        text material_name
        uuid material_id FK
        numeric quantity
        text unit
        numeric waste_percent
        numeric cost_per_unit
    }

    PRODUCTION_STAGE_LOGS {
        uuid id PK
        uuid production_order_id FK
        uuid stage_template_id FK
        text stage_name
        text status
        timestampz started_at
        timestampz completed_at
        numeric quantity_in
        numeric quantity_out
        numeric waste_quantity
        text qc_result
    }

    PRODUCTION_CONSUMPTION {
        uuid id PK
        uuid production_order_id FK
        uuid material_id FK
        numeric planned_qty
        numeric actual_qty
        numeric waste_qty
        uuid batch_id
    }

    PRODUCTION_BATCHES {
        uuid id PK

```

```
text batch_number UK
uuid production_order_id FK
uuid product_id FK
numeric quantity
text status
uuid qc_inspection_id FK
date manufactured_date
date expiry_date
text warehouse_location
}
```

Business Rules

Rule	ID
BOM versions allow multiple recipes per product	BR-PROD-01
Stage workflow: pending → in_progress → completed/failed/skipped	BR-PROD-02
Stages with is_qc_required=true must pass QC before proceeding	BR-PROD-03
Material consumption tracks planned vs actual vs waste	BR-PROD-04
Batch lifecycle: in_production → qc_pending → qc_passed/qc_failed → finished → shipped	BR-PROD-05
Traceability: Batch → Production Order → BOM → Materials → Supplier	BR-PROD-06
waste_quantity = quantity_in - quantity_out (per stage)	BR-PROD-07

القسم 8: توكيد ومراقبة الجودة

Purpose 8.1

Quality gate for both incoming raw materials and finished products, ensuring factory output meets .standards

Actors 8.2

Responsibilities	Actor
Perform inspections, Record results	QC Inspector
Review inspections, Approve conditional items	Quality Manager
Submit batches for QC	Production Manager
Submit incoming materials for QC	Procurement Manager

Features 8.3

CRUD	Tables	Feature
C	qc_inspections	Create Incoming Material Inspection
C	qc_inspections	Create Finished Product Inspection
U	qc_inspections	Record Inspection Result
U	qc_inspections	Pass/Fail Decision
U	qc_inspections	Conditional Approval
R	qc_inspections	View QC Dashboard

Process Flow

flowchart TD

```

A[📦 Material/Product Arrives] --> B{Type}
B -->|Incoming Material| C[QC Inspector Inspects]
B -->|Finished Product| D[QC Inspector Inspects Batch]
C --> E{Result}
D --> E
E -->|Passed| F[Status: passed]
E -->|Failed| G[Status: failed]
E -->|Conditional| H[Status: conditional_approval]
F --> I[Accept into Inventory / Ship]
G --> J[Return / Rework / Scrap]
H --> K[Quality Manager Review]
K --> L{Decision}
L -->|Accept| I
L -->|Reject| J

```

Section ERD

erDiagram

```

QC_INSPECTIONS {
    uuid id PK
    text reference_type
    uuid reference_id
    text item_name
    text inspector_name
    text status
    text notes
    timestampz inspection_date
}

```

Business Rules

Rule	ID
reference_type: incoming_material or finished_product	BR-QC-01
Status: pending → passed / failed / conditional_approval	BR-QC-02
Failed incoming materials must be returned to supplier	BR-QC-03
Failed batches must be reworked or scrapped	BR-QC-04
QC inspections linked to batches via production_batches.qc_inspection_id	BR-QC-05

Cross-Section Relationships (5-6-7-8)

```
flowchart LR
  PROC[🛒 Procurement] -->|Purchase Order| QC1[🔍 QC: Incoming]
  QC1 -->|Passed| INV[📦 Inventory]
  INV -->|Raw Materials| PROD[🏭 Production]
  PROD -->|Finished Batch| QC2[🔍 QC: Finished]
  QC2 -->|Passed| INV2[📦 Finished Goods]
  INV2 -->|Ready to Ship| SALES[📊 Sales]
```

الجزء 4 — الأقسام 9 إلى 12: المبيعات، الأسطول، الهندسة، والصيانة

القسم 9: المبيعات، العملاء، والكتالوج العام 📊

Purpose 9.1

End-to-end sales management from public product catalog through quote requests, customer registration, sales order processing, and delivery. The public-facing catalog allows guests to browse products and request quotes without authentication.

Actors 9.2

Responsibilities	Actor
Manage clients, process sales orders, manage products	Sales Manager
Browse catalog, register, request quotes	Customer/Guest
Approve large orders, view sales dashboard	Owner
Delivery of confirmed orders	Logistics Manager

Process Flows 9.3

Sales Order Lifecycle

```
flowchart TD
    A[👤 Customer] --> B[Browse Public Catalog]
    B --> C[Request Quote]
    C --> D[👤 Sales Manager Reviews]
    D --> E{Existing Client?}
    E -->|No| F[Customer Registration Request]
    F --> G{Sales Approves Registration?}
    G -->|Yes| H[Create Client Record]
    G -->|No| I[Reject]
    E -->|Yes| J[Create Sales Order]
    H --> J
    J --> K[Status: draft]
    K --> L[Confirm Order]
    L --> M[Status: confirmed]
    M --> N[Production / Stock Check]
    N --> O[Status: delivered]
    O --> P[Create Invoice]
    P --> Q[Status: invoiced]
    Q --> R[Payment Received]
    R --> S[Status: paid]
    S --> T[👤 Customer - Complete]
```

Start Actor: Customer | **End Actor:** Customer (receives product)

Quote Request Flow

flowchart TD

```
A[👤 Guest/Customer] --> B[Fill Quote Form]
B --> C[Select Product + Quantity + Specs]
C --> D[Submit Quote Request]
D --> E[Status: pending]
E --> F[👤 Sales Manager Reviews]
F --> G{Decision}
G -->|Quote| H[Status: quoted - Send Price]
G -->|Reject| I[Status: rejected]
H --> J{Customer Accepts?}
J -->|Yes| K[Convert to Sales Order]
K --> L[Status: converted_to_so]
J -->|No| M[No Action]
```

Section ERD

```

erDiagram
    PRODUCT_CATEGORIES ||--o{ PRODUCTS : "contains"
    PRODUCTS ||--o{ PRODUCT_IMAGES : "has"
    PRODUCTS ||--o{ PRODUCT_DOCUMENTS : "has"
    PRODUCTS ||--o{ QUOTE_REQUESTS : "quoted_for"
    CLIENTS ||--o{ SALES_ORDERS : "places"
    CLIENTS ||--o{ QUOTE_REQUESTS : "requests"
    PRODUCTS ||--o{ CUSTOMER_REGISTRATION_REQUESTS : "interested_in"
    CLIENTS ||--o{ CUSTOMER_FEEDBACK : "gives"

    PRODUCT_CATEGORIES {
        uuid id PK
        text name_ar
        text name_en
        text slug UK
        text icon
    }

    PRODUCTS {
        uuid id PK
        text name_ar
        text name_en
        text slug UK
        text code
        text sku
        uuid category_id FK
        text material
        text color
        text grade
        jsonb technical_specs
        text unit
        integer moq
        boolean is_public
        boolean is_featured
        numeric cost
        numeric selling_price
        integer current_stock
        integer views
    }

    PRODUCT_IMAGES {
        uuid id PK
        uuid product_id FK
        text image_url
        boolean is_main
        integer display_order
    }

    PRODUCT_DOCUMENTS {
        uuid id PK
        uuid product_id FK
        text title_ar
        text file_url
        text doc_type
        boolean is_public
    }

```

```
SALES_ORDERS {
  uuid id PK
  text so_number
  uuid client_id FK
  text client_name
  date order_date
  jsonb items
  numeric total
  numeric paid_amount
  numeric remaining
  text invoice_type
  text status
  text delivery_status
}

CLIENTS {
  uuid id PK
  text company_name
  text contact_person
  text phone
  text email
  text address
  text status
}

QUOTE_REQUESTS {
  uuid id PK
  text quote_number UK
  uuid product_id FK
  uuid client_id FK
  text company_name
  numeric quantity
  text required_specs
  date required_delivery_date
  text status
}

CUSTOMER_REGISTRATION_REQUESTS {
  uuid id PK
  text company_name
  text contact_person
  text phone
  uuid requested_product_id FK
  text status
}

CUSTOMER_FEEDBACK {
  uuid id PK
  text customer_name
  uuid order_id
  integer rating
  text feedback_type
  text message
  text status
  text response
}
```

Business Rules

Rule	ID
SO workflow: draft → confirmed → delivered → invoiced → paid / cancelled	BR-SALES-01
Invoice types: cash, credit, tax	BR-SALES-02
remaining = total - paid_amount	BR-SALES-03
'Public catalog shows only products where is_public=true AND status='active	BR-SALES-04
Quote workflow: pending → reviewed → quoted → rejected / converted_to_so	BR-SALES-05
Customer registration requires Sales Manager approval	BR-SALES-06
Product views counter incremented on public page view	BR-SALES-07
Feedback rating: 1-5 stars	BR-SALES-08

القسم 10: الحركة، النقل، وإدارة الأسطول

Purpose 10.1

Comprehensive fleet management including internal/external drivers, company vehicles with odometer .tracking, trip scheduling, incident management, and fleet maintenance

Actors 10.2

Responsibilities	Actor
Full fleet management, trip scheduling	Logistics Manager
Trip execution, odometer recording	Internal Driver
External transport services	External Driver
View fleet dashboard, costs	Owner

Process Flows 10.3

Trip Lifecycle

```

flowchart TD
    A[👤 Logistics Manager] --> B[Create Trip]
    B --> C{Driver Type}
    C -->|Internal| D[Assign Internal Driver + Vehicle]
    C -->|External| E[Assign External Driver]
    D --> F[Status: مجدولة - Scheduled]
    E --> F
    F --> G[Loading]
    G --> H[Status: جاري التحميل]
    H --> I[Departure]
    I --> J[Status: في الطريق]
    J --> K[Arrival]
    K --> L[Status: تم التوصيل]
    L --> M[Record Distance + Fuel]
    M --> N[Calculate Transport Cost]
    N --> O[👤 Logistics Manager - Complete]
  
```

Fleet Alert System

```

flowchart TD
    A[🔔 System Daily Check] --> B{Check Alerts}
    B --> C[Driver License < 30 days]
    B --> D[Vehicle License < 30 days]
    B --> E[Insurance < 30 days]
    B --> F[Odometer near Maintenance Due]
    C --> G[🚨 Alert: License Expiry]
    D --> G
    E --> G
    F --> H[🚨 Alert: Maintenance Due]
    G --> I[Notify Logistics Manager]
    H --> I
  
```

Section ERD

```

erDiagram
    FLEET_INTERNAL_DRIVERS ||--o{ FLEET_TRIPS : "drives"
    FLEET_EXTERNAL_DRIVERS ||--o{ FLEET_TRIPS : "drives"
    FLEET_VEHICLES ||--o{ FLEET_TRIPS : "used_in"
    FLEET_VEHICLES ||--o{ FLEET_MAINTENANCE : "maintained"
    FLEET_VEHICLES ||--o{ FLEET_INCIDENTS : "involved_in"
    FLEET_TRIPS ||--o{ FLEET_INCIDENTS : "has"
    FLEET_INTERNAL_DRIVERS ||--o{ FLEET_VEHICLES : "assigned_to"
    CARS ||--o{ CAR_TRIPS : "has"
    CARS ||--o{ CAR_MAINTENANCE_LOGS : "maintained"

    FLEET_INTERNAL_DRIVERS {
        uuid id PK
        uuid user_id FK
        text driver_code UK
        text driver_name
        text license_number
        date license_expiry
        text status
        integer trips_count
        numeric total_km
        numeric rating
    }

    FLEET_EXTERNAL_DRIVERS {
        uuid id PK
        text driver_name
        text company_name
        text car_number
        numeric capacity
        numeric transport_rate
        text status
    }

    FLEET_VEHICLES {
        uuid id PK
        text car_number UK
        text plate_number
        text model
        integer year
        uuid current_driver_id FK
        text status
        numeric odometer
        numeric fuel_consumption
        date insurance_expiry
        date license_expiry
    }

    FLEET_TRIPS {
        uuid id PK
        text trip_number UK
        text client_name
        uuid sales_order_id
        text driver_type
        uuid internal_driver_id FK
        uuid external_driver_id FK

```

```

        uuid vehicle_id FK
        text status
        numeric distance
        numeric transport_cost
        numeric fuel_qty
    }

    FLEET_INCIDENTS {
        uuid id PK
        uuid trip_id FK
        uuid vehicle_id FK
        text incident_type
        text details
        numeric cost
    }

    FLEET_MAINTENANCE {
        uuid id PK
        uuid vehicle_id FK
        text maintenance_type
        text details
        numeric cost
        text status
    }

    CARS {
        uuid id PK
        text plate_number UK
        text car_model
        uuid driver_id FK
        numeric current_odometer
        date license_expiry_date
        text status
    }

    CAR_TRIPS {
        uuid id PK
        uuid car_id FK
        uuid driver_id FK
        date trip_date
        text destination
        numeric odometer_start
        numeric odometer_end
        numeric distance_covered
        numeric fuel_consumed_liters
        text status
    }

    CAR_MAINTENANCE_LOGS {
        uuid id PK
        uuid car_id FK
        text log_type
        numeric cost
        numeric odometer_at_log
    }

```

Business Rules

Rule	ID
Trip status (Arabic):جدولة → جاري التحميل → في الطريق → تم التوصيل / ملغاة	BR-FLEET-01
Incident types:مخالفة (violation), حادث (accident)	BR-FLEET-02
Fleet alerts: License expiry < 30 days, Insurance expiry < 30 days, Maintenance due	BR-FLEET-03
distance_covered = odometer_end - odometer_start (GENERATED column)	BR-FLEET-04
Car maintenance log types: Fuel, Maintenance, Tires, Oil Change	BR-FLEET-05
Trips linked to sales_orders (delivery) and purchase_orders (pickup)	BR-FLEET-06

القسم 11: الهندسة والمكتب الفني

Purpose 11.1

Manage engineering projects (civil, electrical, mechanical, production line) and technical drawings with .multi-department approval workflow

Process Flow 11.2

```
flowchart TD
    A[👤 Engineering Manager] --> B[Create Project]
    B --> C[Status: planning]
    C --> D[Design Phase]
    D --> E[Upload Drawings]
    E --> F[Status: designing]
    F --> G{Drawing Approval}
    G -->|Approved| H[Status: in_progress]
    G -->|Rejected| I[Revise Drawing]
    I --> E
    H --> J[Construction/Implementation]
    J --> K[Status: supervision]
    K --> L[Project Complete]
    L --> M[Status: completed]
    M --> N[Workflow Approvals by Departments]
    N --> O[👤 Engineering Manager - Complete]
```


Section ERD

```

erDiagram
    USERS ||--o{ ENGINEERING_PROJECTS : "manages"
    ENGINEERING_PROJECTS ||--o{ ENGINEERING_DRAWINGS : "has"
    USERS ||--o{ ENGINEERING_DRAWINGS : "uploads"

    ENGINEERING_PROJECTS {
        uuid id PK
        uuid employee_id FK
        text engineer_name
        text title
        text project_type
        text description
        text technical_specs
        integer progress
        text status
        jsonb workflow_approvals
    }

    ENGINEERING_DRAWINGS {
        uuid id PK
        uuid project_id FK
        uuid uploaded_by FK
        text title
        text file_url
        text status
        text approved_by
        text comments
    }

```

Business Rules

Rule	ID
Project types: civil, electrical, mechanical, production_line, other	BR-ENG-01
Project workflow: planning → designing → in_progress → supervision → completed	BR-ENG-02
Drawing workflow: pending → approved / rejected	BR-ENG-03
workflow_approvals (JSONB) tracks per-department sign-offs	BR-ENG-04
Progress tracked as percentage (0-100)	BR-ENG-05

القسم 12: الصيانة والمعدات والماكينات

Purpose 12.1

Machine maintenance tracking, external maintenance company management, equipment registry with .rental tracking, and facilities management

Process Flows 12.2

Maintenance Visit Flow

flowchart TD

```

A[ 🛠 Issue Reported ] --> B[ Create Maintenance Request ]
B --> C{ Internal or External? }
C --> | Internal Technician | D[ Assign Technician ]
C --> | External Company | E[ Create Maintenance Visit ]
D --> F[ Repair + Log ]
E --> G[ Company Sends Technician ]
G --> H[ Work Performed ]
H --> I[ Record: labor_cost + parts_cost ]
I --> J[ Before/After Photos ]
J --> K[ Status: completed ]
K --> L{ Payment }
L --> M[ Create Maintenance Payment ]
M --> N{ Payment Method }
N --> | Treasury | O[ Deduct from Safe ]
N --> | Bank | P[ Bank Transfer ]
O --> Q[ Status: paid ]
P --> Q
F --> R[ Update machine status ]
Q --> R
R --> S[ 👤 Maintenance Manager ]
  
```

Equipment Rental Flow

flowchart TD

```
A[👤 Maintenance Mgr] --> B[Need Equipment]
B --> C{Own or Rent?}
C -->|Own| D[Check Equipment Registry]
C -->|Rent| E[Create Rental Contract]
E --> F[Status: draft]
F --> G{Approval}
G -->|Approved| H[Status: active]
H --> I[Equipment in Use]
I --> J{Contract Expiring?}
J -->|< 30 days| K[Status: expiring - Alert]
J -->|Expired| L[Status: expired]
K --> M{Renew or Return?}
L --> M
M -->|Return| N[Status: returned]
M -->|Renew| O[Extend Contract]
```

Section ERD

```

erDiagram
    MACHINES ||--o{ MAINTENANCE_LOGS : "has"
    MAINTENANCE_COMPANIES ||--o{ MAINTENANCE_VISITS : "performs"
    MAINTENANCE_VISITS ||--o{ MAINTENANCE_PAYMENTS : "paid_via"
    EQUIPMENT_CATEGORIES ||--o{ EQUIPMENT : "categorizes"
    EQUIPMENT ||--o{ EQUIPMENT_RENTALS : "rented"
    SUPPLIERS ||--o{ EQUIPMENT : "owns"
    SUPPLIERS ||--o{ EQUIPMENT_RENTALS : "rents_from"

    MACHINES {
        uuid id PK
        text name
        text code UK
        text location
        text status
        date last_maintenance_date
        date next_maintenance_date
    }

    MAINTENANCE_LOGS {
        uuid id PK
        uuid machine_id FK
        text technician_name
        text issue_description
        text action_taken
        text parts_used
        numeric cost
    }

    MAINTENANCE_COMPANIES {
        uuid id PK
        text name
        text contact_person
        text phone
        date contract_start
        date contract_end
        numeric rating
        text status
        integer total_visits
        numeric total_spending
    }

    MAINTENANCE_VISITS {
        uuid id PK
        text visit_number UK
        uuid company_id FK
        date visit_date
        text problem_description
        text work_description
        numeric labor_cost
        numeric parts_cost
        numeric total_cost
        text status
        text_arr before_photos
        text_arr after_photos
    }

```

```

MAINTENANCE_PAYMENTS {
  uuid id PK
  uuid visit_id FK
  uuid company_id FK
  numeric amount
  text payment_method
  text status
}

EQUIPMENT {
  uuid id PK
  text name
  text code UK
  uuid category_id FK
  text serial_number
  text status
  uuid owner_supplier_id FK
  numeric daily_rate
  numeric monthly_rate
  numeric purchase_cost
}

EQUIPMENT_RENTALS {
  uuid id PK
  uuid equipment_id FK
  text rental_number UK
  uuid supplier_id FK
  date start_date
  date end_date
  text rate_type
  numeric rate_amount
  numeric total_cost
  text status
}

```

Business Rules

Rule	ID
Machine status: operational, down, under_maintenance	BR-MAINT-01
Visit total_cost = labor_cost + parts_cost	BR-MAINT-02
Visit payment flow: pending → approved → paid / cancelled	BR-MAINT-03
Payment methods: treasury (safe) or bank	BR-MAINT-04
Maintenance company rating: 0-5 scale	BR-MAINT-05
Company status: active, inactive, blacklisted	BR-MAINT-06
Equipment status: available, rented, maintenance, retired	BR-MAINT-07
Rental status: draft → active → expiring → expired / returned / cancelled	BR-MAINT-08
Rate types: daily, weekly, monthly	BR-MAINT-09

الجزء 4 — الأقسام 13 إلى 15 + التحليل الشامل للنظام (الأجزاء 30-24)



القسم 13: التمريض والعيادة الطبية للمصنع

Purpose 13.1

Manages factory clinic operations including employee health visits, work injury documentation (OSHA-style), rest permits, pharmacy inventory, and medication dispensing

Actors 13.2

Responsibilities	Actor
Record clinic visits, injuries, dispense medication, issue rest permits	Nurse
Oversee clinic operations, manage pharmacy stock, approve medical leave	Nursing Management
View medical reports, integrate with absence/leave system	HR Manager
View medical dashboards, approve medical reimbursements	Owner

Process Flows 13.3

Clinic Visit Flow

```

flowchart TD
    A[👤 Employee] --> B[Visit Factory Clinic]
    B --> C[👩 Nurse Records Visit]
    C --> D[Record Vitals: BP, Temp, HR, Blood Sugar]
    D --> E[Complaint + Diagnosis]
    E --> F{Treatment}
    F -->|Medication| G[Dispense from Pharmacy]
    F -->|Rest| H[Issue Rest Permit]
    F -->|Hospital| I[Hospital Transfer Referral]
    G --> J[Record in clinic_dispense_logs]
    J --> K[Update medication stock]
    H --> L{Permit Type}
    L -->|Clinic Rest| M[Rest in Clinic]
    L -->|Rest of Day| N[Gate Pass Authorized]
    L -->|Sick Leave Short| O[1-3 Day Leave]
    L -->|Sick Leave Extended| P[Extended Medical Leave]
    I --> Q[Record referral_details]
    M --> R{Disposition}
    N --> R
    O --> R
    P --> R
    Q --> R
    R --> S[👤 Employee Returns/Leaves]
  
```

Work Injury Documentation

```

flowchart TD
    A[🚨 Work Injury Occurs] --> B[👩 Nurse Records Injury]
    B --> C[Record: location, severity, injury_type]
    C --> D[Record: root_cause, immediate_action]
    D --> E{Severity}
    E -->|Minor/Moderate| F[First Aid + Return to Work]
    E -->|Severe/Critical| G[Hospital Transfer]
    G --> H[Record: hospital_name, hospitalized=true]
    H --> I[Track lost_work_days]
    F --> J[Supervisor Notified]
    I --> J
    J --> K{Investigation Needed?}
    K -->|Yes| L[Status: osh_investigation]
    K -->|No| M[Status: recovered]
    L --> N[Record investigation_notes]
    N --> M
  
```

Section ERD

```

erDiagram
    USERS ||--o{ CLINIC_VISITS : "visits"
    USERS ||--o{ CLINIC_INJURIES : "injured"
    USERS ||--o{ CLINIC_REST_PERMITS : "receives"
    CLINIC_MEDICATIONS ||--o{ CLINIC_DISPENSE_LOGS : "dispensed"
    CLINIC_VISITS ||--o{ CLINIC_DISPENSE_LOGS : "triggers"
    USERS ||--o{ CLINIC_DISPENSE_LOGS : "receives_meds"

    CLINIC_VISITS {
        uuid id PK
        uuid employee_id FK
        text employee_name
        text department
        text visit_type
        text blood_pressure
        numeric temperature
        numeric heart_rate
        numeric blood_sugar
        text complaint
        text diagnosis
        text treatment
        text medicine_dispensed
        text disposition
        text attended_by
    }

    CLINIC_INJURIES {
        uuid id PK
        timestamp incident_date
        uuid employee_id FK
        text department
        text location
        text severity
        text injury_type
        text description
        text root_cause
        text immediate_action
        boolean hospitalized
        text hospital_name
        numeric lost_work_days
        text status
        text investigation_notes
    }

    CLINIC_REST_PERMITS {
        uuid id PK
        uuid employee_id FK
        text permit_type
        timestamp start_time
        timestamp end_time
        text diagnosis
        boolean gate_pass_authorized
        text status
        timestamp returned_at
        text issued_by
    }

```



```
CLINIC_MEDICATIONS {
  uuid id PK
  text name
  text generic_name
  text category
  text unit
  numeric current_stock
  numeric min_threshold
  date expiry_date
  text batch_no
  text location
}

CLINIC_DISPENSE_LOGS {
  uuid id PK
  uuid medication_id FK
  text medication_name
  uuid employee_id FK
  numeric quantity
  text reason
  uuid visit_id FK
  text dispensed_by
}
```

Business Rules

Rule	ID
Visit types: checkup, acute, work_injury, follow_up, first_aid	BR-NURSE-01
Dispositions: return_to_work, clinic_rest, rest_permit, hospital_transfer	BR-NURSE-02
Injury severity: minor, moderate, severe, critical	BR-NURSE-03
Injury types: cut, burn, fracture, chemical, eye, shock, fall, other	BR-NURSE-04
Permit types: clinic_rest, rest_of_day, sick_leave_short, sick_leave_extended	BR-NURSE-05
Medication dispensing decreases current_stock	BR-NURSE-06
Low medication alert when current_stock < min_threshold	BR-NURSE-07
Medication categories: analgesic, antiseptic, wound_care, gastrointestinal, respiratory, burn_care, emergency	BR-NURSE-08
Injury status: under_treatment → rest → hospitalized → recovered / osh_investigation	BR-NURSE-09



القسم 14: الشؤون القانونية والامتثال

Purpose 14.1

.Track legal issues, compliance matters, and case management

Section ERD 14.2

```
erDiagram
    USERS ||--o{ LEGAL_ISSUES : "reports"

    LEGAL_ISSUES {
        uuid id PK
        text title
        text description
        uuid reported_by FK
        text reported_name
        text status
        text assigned_to
    }
```

Business Rules

Rule	ID
'Default status: 'open	BR-LEGAL-01
Cases assigned to legal advisor/assigned_to	BR-LEGAL-02
Status flow: open → in_progress → resolved / closed	BR-LEGAL-03

Suggestion for Future Enhancement: This module is minimal. Recommended additions: - [TIP!] Case categories (employment law, contract disputes, regulatory compliance) - Document attachments for legal documents - Deadline tracking for court dates - Linked employees and incidents - Legal cost tracking - External lawyer management



القسم 15: تكنولوجيا المعلومات، النظم، والإدارة العامة

Purpose 15.1

Central hub for IT support, system administration, collaboration tools (chat, tasks, calendar), document management, approval workflows, AI-powered reports, and KPI dashboards

Subsystems 15.2

IT Support 15.2.1

```
erDiagram
    USERS ||--o{ IT_TICKETS : "submits"

    IT_TICKETS {
        uuid id PK
        uuid employee_id FK
        text department
        text issue_type
        text description
        text priority
        text status
        text assigned_to
    }
```

Notifications & Announcements 15.2.2

```
erDiagram
    USERS ||--o{ NOTIFICATIONS : "receives"
    USERS ||--o| NOTIFICATION_PREFERENCES : "configures"

    NOTIFICATIONS {
        uuid id PK
        uuid user_id FK
        text type
        text title
        text message
        boolean read
    }

    ANNOUNCEMENTS {
        uuid id PK
        text title
        text message
        text priority
        text department
    }

    NOTIFICATION_PREFERENCES {
        uuid id PK
        uuid user_id FK
        boolean email_enabled
        boolean push_enabled
        boolean leave_notifications
        boolean attendance_notifications
        boolean payroll_notifications
        boolean task_notifications
        boolean chat_notifications
    }
```

Collaboration (Chat, Tasks, Calendar) 15.2.3

```

erDiagram
    CHAT_CHANNELS ||--|{ CHAT_MESSAGES : "contains"
    USERS ||--o{ CHAT_MESSAGES : "sends"
    USERS ||--o{ TASKS : "assigned"
    USERS ||--o{ CALENDAR_EVENTS : "organizes"

    CHAT_CHANNELS {
        uuid id PK
        text name
        text channel_type
        uuid_arr members
        uuid created_by FK
    }

    CHAT_MESSAGES {
        uuid id PK
        uuid channel_id FK
        uuid sender_id FK
        text message
        text message_type
        text file_url
    }

    TASKS {
        uuid id PK
        text title
        text module
        uuid assigned_to FK
        uuid assigned_by FK
        text priority
        text status
        date due_date
    }

    CALENDAR_EVENTS {
        uuid id PK
        text title
        text event_type
        timestampz start_time
        timestampz end_time
        boolean all_day
        uuid organizer_id FK
        uuid_arr attendees
        boolean is_recurring
    }

```

Document Management 15.2.4

```
erDiagram
    DOCUMENT_CATEGORIES ||--o{ DOCUMENTS : "contains"
    DOCUMENT_CATEGORIES ||--o{ DOCUMENT_CATEGORIES : "parent"

    DOCUMENTS {
        uuid id PK
        text title
        uuid category_id FK
        text file_url
        text file_type
        text module
        uuid uploaded_by FK
        integer version
    }

    DOCUMENT_CATEGORIES {
        uuid id PK
        text name
        text name_ar
        uuid parent_id FK
    }
```

Approval Workflows 15.2.5

```
erDiagram
    APPROVAL_WORKFLOWS ||--|{ APPROVAL_STEPS : "has"
    APPROVAL_WORKFLOWS ||--o{ APPROVAL_REQUESTS : "processes"
    APPROVAL_REQUESTS ||--|{ APPROVAL_ACTIONS : "tracks"

    APPROVAL_WORKFLOWS {
        uuid id PK
        text name
        text module
        boolean is_active
    }

    APPROVAL_STEPS {
        uuid id PK
        uuid workflow_id FK
        integer step_order
        text approver_role
        uuid approver_id FK
        text approval_type
        integer auto_approve_after_hours
    }

    APPROVAL_REQUESTS {
        uuid id PK
        uuid workflow_id FK
        integer current_step
        text entity_type
        uuid entity_id
        uuid requested_by FK
        text status
    }

    APPROVAL_ACTIONS {
        uuid id PK
        uuid request_id FK
        integer step_order
        text action
        uuid acted_by FK
        text comments
    }
```

AI Reports & KPIs 15.2.6

```
erDiagram
    USERS ||--o{ AI_REPORTS : "generates"
    KPI_DEFINITIONS ||--o{ KPI_VALUES : "measured_by"

    AI_REPORTS {
        uuid id PK
        text report_type
        text period
        text title
        jsonb content
        text summary
        uuid generated_by FK
        integer report_year
        text language
    }

    KPI_DEFINITIONS {
        uuid id PK
        text name
        text module
        text formula
        text unit
        numeric target_value
        numeric warning_threshold
        numeric danger_threshold
    }

    KPI_VALUES {
        uuid id PK
        uuid kpi_id FK
        text period
        numeric actual_value
        text department
    }
```


الجزء 24 — شبكة العلاقات التبادلية بين الوحدات الـ 15

Inter-Module Data Flow

```

flowchart TD
    subgraph External
        CUST[👤 Customer/Guest]
        SUPP[👤 Supplier]
    end

    subgraph Core
        AUTH[🔒 Auth & Security]
        HR[👥 HR & People]
        PAY[💰 Payroll]
        FIN[🏦 Finance]
    end

    subgraph Supply
        PROC[📋 Procurement]
        INV[📦 Inventory]
    end

    subgraph Manufacturing
        PROD[🏭 Production]
        QC[🔍 Quality]
    end

    subgraph Customer_Facing
        SALES[📊 Sales & CRM]
        FLEET[🚚 Fleet]
    end

    subgraph Support
        ENG[⚙️ Engineering]
        MAINT[🔧 Maintenance]
        NURSE[👩 Nursing]
        LEGAL[⚖️ Legal]
        IT[💻 IT & Admin]
    end

    AUTH -->|Users & Roles| HR
    AUTH -->|Permissions| IT
    HR -->|Employee Data| PAY
    HR -->|Attendance/OT/Deductions| PAY
    PAY -->|Salary Payments| FIN

    CUST -->|Orders| SALES
    SALES -->|Sales Orders| PROD
    SALES -->|Delivery Requests| FLEET
    SALES -->|Sales Invoices| FIN
  
```

```
SUPP -->|Supply Materials| PROC
PROC -->|Purchase Orders| INV
PROC -->|Purchase Invoices| FIN
PROC -->|Incoming Material| QC

INV -->|Raw Materials| PROD
PROD -->|Finished Batches| QC
QC -->|Approved Goods| INV

FLEET -->|Driver Employee Data| HR
FLEET -->|Transport Costs| FIN

MAINT -->|Spare Parts| INV
MAINT -->|Payments| FIN
ENG -->|Production Lines| PROD
NURSE -->|Rest Permits| HR
LEGAL -->|Employee Cases| HR

IT -->|Notifications| AUTH
IT -->|AI Reports| FIN
IT -->|Approvals| HR
```

Key Cross-Module Relationships

Data Shared	Relationship	To Module	From Module
	Salary base for payroll calculation	users.base_salary	HR → Payroll
	Auto deductions/additions	delay_minutes, overtime_hours	Attendance → Payroll
	Stock increase	Purchase completed	Procurement → Inventory
	QC inspection trigger	Incoming material	Procurement → Quality
	Stock decrease	Material consumption	Inventory → Production
	Final QC inspection	Finished batch	Production → Quality
	Revenue recognition	Sales order invoiced	Sales → Finance
	Trip scheduling	Delivery order	Sales → Fleet
	Expense recording	Purchase invoice	Procurement → Finance
	Cost tracking	Transport costs	Fleet → Finance
	Expense recording	Maintenance payments	Maintenance → Finance
	Stock decrease	Spare parts used	Maintenance → Inventory
	Leave/absence integration	Rest permits, injuries	Nursing → HR
	Cross-cutting concern	Notifications, Approvals	IT → All

الجزء 25 — رحلات البيانات وتكامل العمليات الكبرى

Journey 1: Sales Order to Delivery

flowchart LR

```

A[👤 Customer] -->|Quote Request| B[🏢 Sales]
B -->|Approved Quote| C[Sales Order Created]
C -->|Check Stock| D{In Stock?}
D -->|Yes| E[📦 Reserve from Inventory]
D -->|No| F[🏭 Create Production Order]
F -->|Consume Materials| G[📦 Inventory Decrease]
G -->|Manufacture| H[🏭 Production Stages]
H -->|Finished Batch| I[🔍 QC Inspection]
I -->|Passed| J[📦 Add to Finished Goods]
E --> K[🚚 Schedule Trip]
J --> K
K -->|Deliver| L[👤 Customer Receives]
L -->|Invoice| M[💰 Finance: Create Invoice]
M -->|Payment| N[💰 Finance: Record Payment]

```

Journey 2: Purchase to Production

flowchart LR

```

A[🏭 Production Needs Materials] -->|Purchase Request| B[📋 Procurement]
B -->|Check Warehouse| C{In Stock?}
C -->|Yes| D[📦 Dispense from Warehouse]
C -->|No| E[Create Purchase Order]
E -->|Send to Supplier| F[👤 Supplier Ships]
F -->|Goods Arrive| G[🔍 QC: Incoming Inspection]
G -->|Passed| H[📦 Add to Inventory]
G -->|Failed| I[Return to Supplier]
H --> D
D -->|Consume| J[🏭 Production: Material Consumption]
J -->|Process| K[🏭 Manufacturing Stages]
K -->|Batch Ready| L[🔍 QC: Finished Product]
L -->|Passed| M[📦 Finished Goods Inventory]

```

Journey 3: Employee Lifecycle

flowchart LR

```

A[📄 Recruitment/ATS] -->|AI CV Score| B{Hired?}
B -->|Yes| C[👤 Create Employee]
C -->|Auto Permissions| D[🔑 Role + Permissions]
D -->|Daily| E[📅 Attendance + Leave + Overtime]
E -->|Monthly| F[💰 Payroll Calculation]
F -->|Payment| G[🏦 Finance: Salary Payment]
E -->|If Injured| H[🏥 Clinic Visit]
E -->|Performance| I[📊 Performance Review]
C -->|Assigned| J[💻 Assets + Equipment]
C -->|Eventually| K[🚪 Offboarding]
K -->|Clearances| L[IT + HR + Finance Clear]
L -->|Complete| M[Account Deactivated]

```

الجزء 26 — مخطط المعمارية الشامل للنظام (Master ERD)

```

erDiagram
    COMPANIES ||--o{ BRANCHES : "has"
    BRANCHES ||--o{ USERS : "employs"
    USERS ||--o{ ATTENDANCE : "tracks"
    USERS ||--o{ LEAVE_REQUESTS : "submits"
    USERS ||--o{ PAYROLL : "paid"
    USERS ||--o{ LOANS : "borrows"

    SUPPLIERS ||--o{ PURCHASE_ORDERS : "supplies"
    SUPPLIERS ||--o{ SUPPLIER_PERFORMANCE : "evaluated"

    PURCHASE_REQUESTS }o--|| INVENTORY_ITEMS : "for"
    INVENTORY_ITEMS ||--o{ INVENTORY_TRANSACTIONS : "moved"
    INVENTORY_ITEMS ||--o{ BOM_ITEMS : "used_in"

    PRODUCTS ||--o{ BOM : "has_recipe"
    BOM ||--||{ BOM_ITEMS : "contains"
    PRODUCTION_ORDERS ||--o{ PRODUCTION_STAGE_LOGS : "stages"
    PRODUCTION_ORDERS ||--o{ PRODUCTION_BATCHES : "produces"
    PRODUCTION_BATCHES }o--o| QC_INSPECTIONS : "inspected"

    CLIENTS ||--o{ SALES_ORDERS : "places"
    PRODUCTS ||--o{ QUOTE_REQUESTS : "quoted"
    PRODUCT_CATEGORIES ||--o{ PRODUCTS : "categorizes"

    FLEET_VEHICLES ||--o{ FLEET_TRIPS : "used_in"
    FLEET_INTERNAL_DRIVERS ||--o{ FLEET_TRIPS : "drives"

    MACHINES ||--o{ MAINTENANCE_LOGS : "maintained"
    MAINTENANCE_COMPANIES ||--o{ MAINTENANCE_VISITS : "performs"
    EQUIPMENT_CATEGORIES ||--o{ EQUIPMENT : "groups"
    EQUIPMENT ||--o{ EQUIPMENT_RENTALS : "rented"

    USERS ||--o{ CLINIC_VISITS : "visits"
    CLINIC_MEDICATIONS ||--o{ CLINIC_DISPENSE_LOGS : "dispensed"

    FINANCE_INVOICES ||--o{ FINANCE_COLLECTIONS : "paid"
    FINANCE_JOURNAL_ENTRIES ||--||{ FINANCE_JOURNAL_LINES : "contains"
    CHART_OF_ACCOUNTS ||--o{ CHART_OF_ACCOUNTS : "parent"

    USERS ||--o{ ENGINEERING_PROJECTS : "manages"
    ENGINEERING_PROJECTS ||--o{ ENGINEERING_DRAWINGS : "has"

    APPROVAL_WORKFLOWS ||--||{ APPROVAL_STEPS : "has"
    APPROVAL_WORKFLOWS ||--o{ APPROVAL_REQUESTS : "processes"

    TRAINING_COURSES ||--o{ TRAINING_ENROLLMENTS : "enrolls"
    USERS ||--o{ PERFORMANCE_REVIEWS : "reviewed"

```

الجزء 27 — المسار العام لعمليات المصنع المتكاملة

```

flowchart TD
    subgraph "🔑 Authentication"
        LOGIN[Login] --> PERMS[Permission Check]
    end

    subgraph "👤 HR Operations"
        PERMS --> ATTEND[Attendance]
        PERMS --> LEAVES[Leave Management]
        PERMS --> RECRUIT[Recruitment/ATS]
    end

    subgraph "💰 Compensation"
        ATTEND --> PAYROLL[Payroll Calc]
        LEAVES --> PAYROLL
        PAYROLL --> PAYMENT[Salary Payment]
    end

    subgraph "🛒 Procurement"
        PR[Purchase Request] --> WCHECK{Stock Check}
        WCHECK -->|Available| DISP[Dispense]
        WCHECK -->|Unavailable| PO[Purchase Order]
        PO --> RECV[Goods Receipt]
        RECV --> QCI[QC: Incoming]
    end

    subgraph "📦 Inventory"
        QCI -->|Pass| STOCK_IN[Stock In]
        DISP --> STOCK_OUT[Stock Out]
    end

    subgraph "🏭 Production"
        STOCK_OUT --> CONSUME[Material Consumption]
        CONSUME --> STAGES[Manufacturing Stages]
        STAGES --> BATCH[Create Batch]
    end

    subgraph "🔍 Quality"
        BATCH --> QCF[QC: Finished Product]
        QCF -->|Pass| FG[Finished Goods]
    end

    subgraph "📊 Sales"
        QUOTE[Quote Request] --> SO[Sales Order]
        SO --> FG
    end

    subgraph "🚚 Fleet"
        SO --> TRIP[Schedule Trip]
        TRIP --> DELIVER[Delivery]
    end

    subgraph "💵 Finance"
        DELIVER --> INV_S[Sales Invoice]
        PO --> INV_P[Purchase Invoice]
        PAYMENT --> GL[General Ledger]
    end
  
```

```
INV_S --> GL
INV_P --> GL
end
```

الجزء 30 — تقرير التدقيق النهائي والتوصيات وخارطة المستقبل

Validation Summary

Count	Status	Category
+137	Complete	Tables Documented
Section ERDs + 1 Master 15	Complete	ERD Diagrams Created
Processes +25	Complete	Process Flows
Rules +90	Complete	Business Rules
All 15 Modules	Complete	CRUD Analysis
All Modules	Complete	Permissions Matrix
All FK Documented	Complete	Relationships
Data Journeys 3	Complete	Cross-Module Flows

Issues & Recommendations

Recommendation	Impact	Issue	#
Implement proper RLS policies based on user_id and role	Security	(USING (true)) All RLS policies are OPEN	1
Add case categories, documents, deadlines, costs	Feature Gap	Legal module is minimal (1 table)	2
Confirm table exists or create it — it's referenced by FK	Missing DDL	No production_orders CREATE TABLE found in read SQL files	3
Use a separate departments table with FK instead of varchar	Data Integrity	and users.role CHECK users.department constraints were DROPPED	4
Consider using Supabase Auth or adding bcrypt/argon2 verification	Security	Custom auth (not Supabase Auth) stores users password_hash in	5
Add employee_id UUID FK reference	Data Integrity	uses employee_name TEXT, not late_deductions FK	6

Recommendation	Impact	Issue	#
Use English status codes with Arabic translations in UI	Consistency 	uses Arabic status values <code>fleet_vehicles</code>	7
Add indexes on all FK columns for join performance	 Performance	No explicit <code>indexes</code> on many FK columns	8
Table exists but needs full feature integration	 Completeness	Missing <code>spare_parts_requests</code> in detailed analysis	9
Create or confirm table in production DB	Missing  DDL	No <code>driver_costs</code> table found (referenced in codebase)	10

Suggestions for Future Addition (المش موجود — اقتراحات للإضافة لاحقاً)

Reason	Module	Suggested Feature	#
Allow employees to check-in/out, view salary, submit requests via mobile	HR	Employee Self-Service Portal (mobile)	1
Automated shift planning with swap management	HR	Shift Scheduling System	2
MRP (Material Requirements Planning) integration	Production	Production Planning & Scheduling	3
Allow customers to track orders, view invoices, pay online	Sales	Customer Portal (full)	4
Allow suppliers to submit quotations, update delivery status	Procurement	Supplier Portal (enhanced)	5
Live OEE metrics, machine utilization, waste tracking	Production	Real-time Production Dashboard	6
Auto-generate POs when stock hits reorder point	Inventory	Automated Inventory Reordering	7
Issue invoices in USD, EUR, SAR alongside EGP	Finance	Multi-Currency Invoicing	8
Track costs per engineering project	Engineering	Project Costing	9
AI-based failure prediction using maintenance history	Maintenance	Predictive Maintenance	10
Barcode scanning for stock movements	Inventory	Barcode/RFID Warehouse	11
Case management, documents, lawyer fees, court dates	Legal	Advanced Legal Module	12
Track employee borrowing patterns	Payroll	Employee Loan History (credit score)	13
Dashboard tracking factory-wide safety metrics	Nursing/HSE	Safety Incident Board	14
REST API for third-party integrations	IT	API Gateway	15

=====

Department Guide — دليل الأقسام

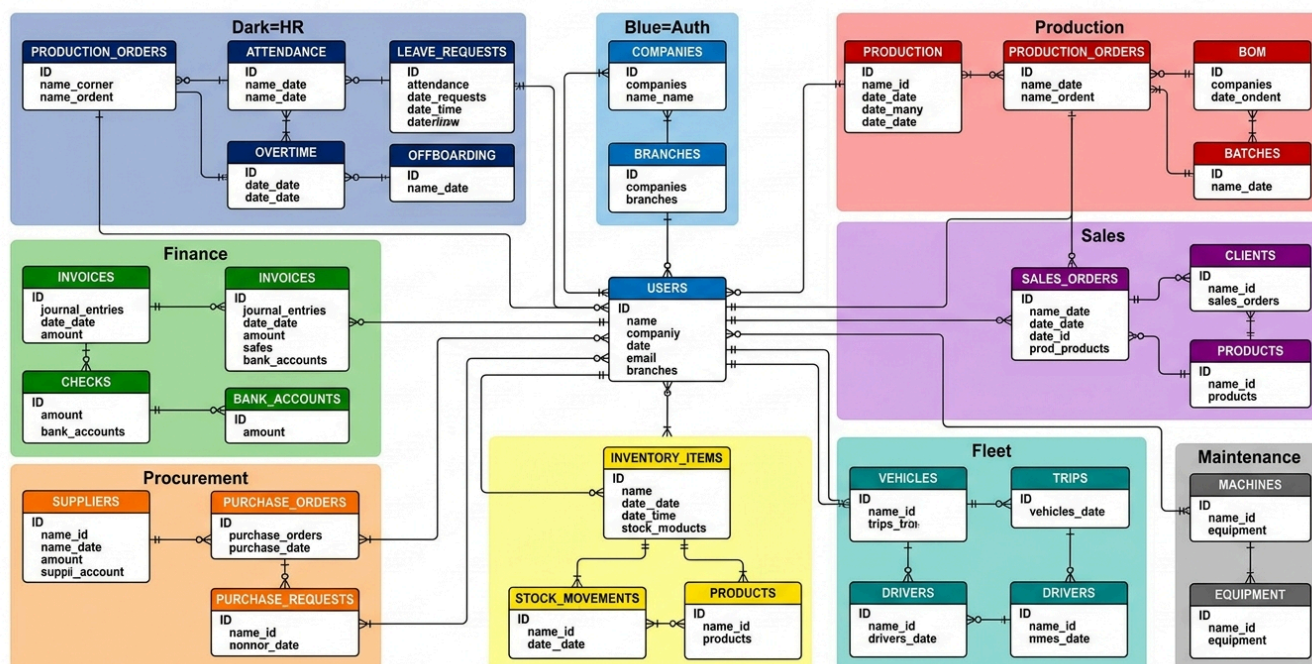
Ninja Factory ERP — All 16 Departments

كل قسم بالشاشات والدورات بتاعته — ليه الصفحات دي في القسم ده

ERD Images

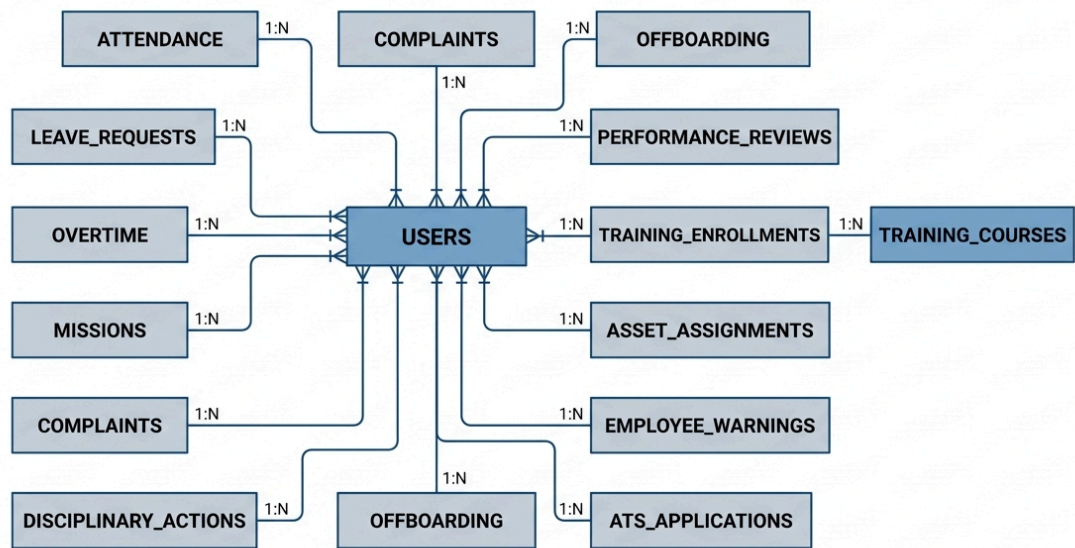
Master System ERD

Ninja Factory ERP — Master ERD



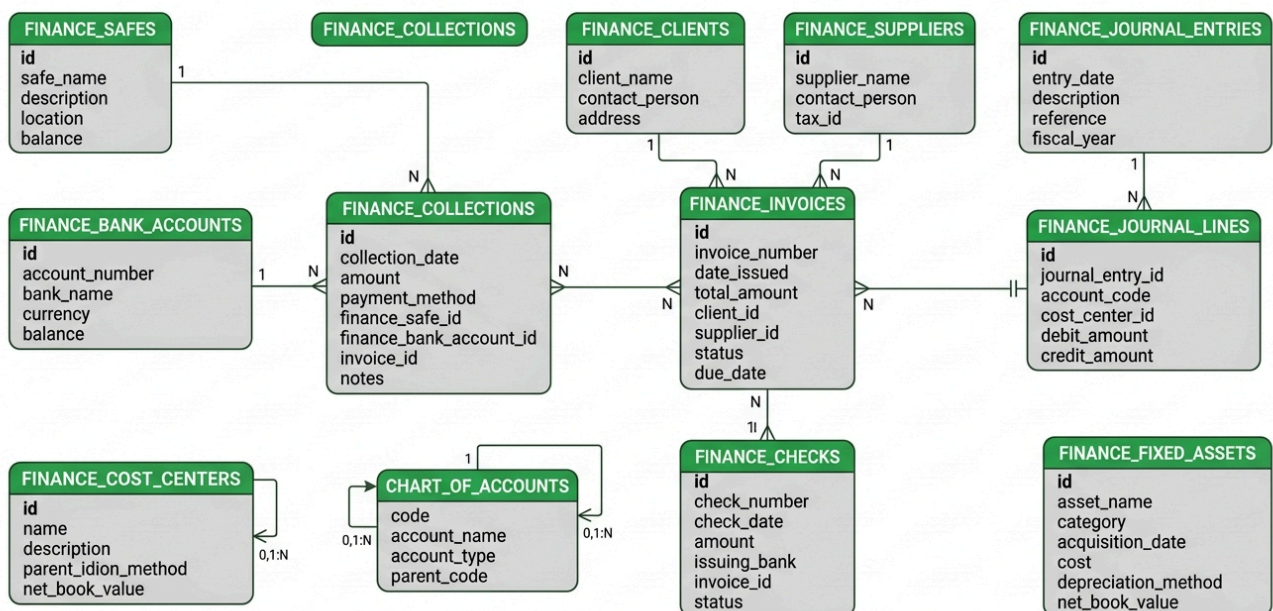
HR & People Management ERD

HR & People Management — ERD



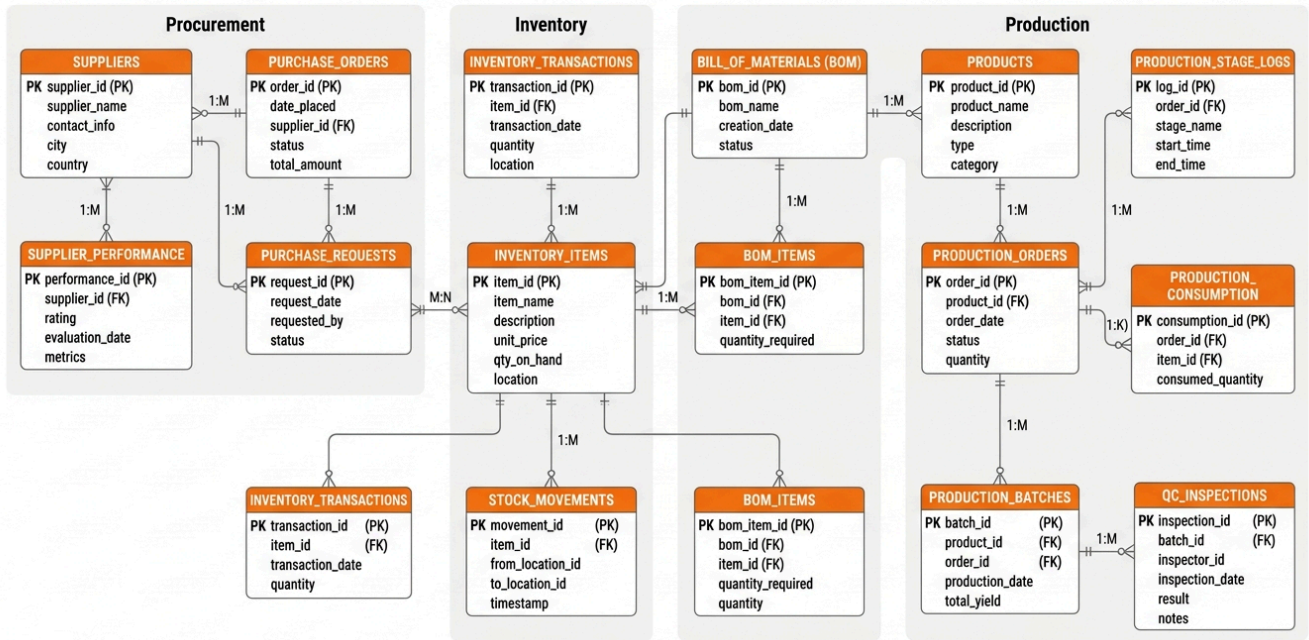
Finance & Accounting ERD

Finance & Accounting — ERD



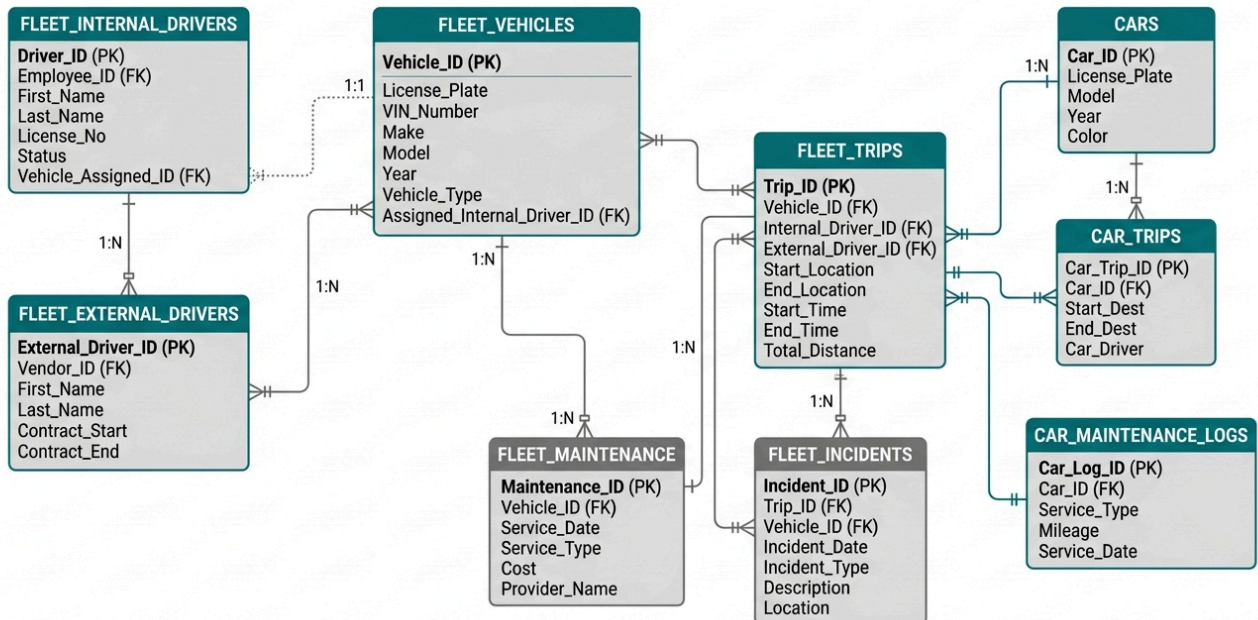
Supply Chain ERD (Procurement → Inventory → Production → QC)

SUPPLY CHAIN — ERD (Procurement → Inventory → Production → QC)

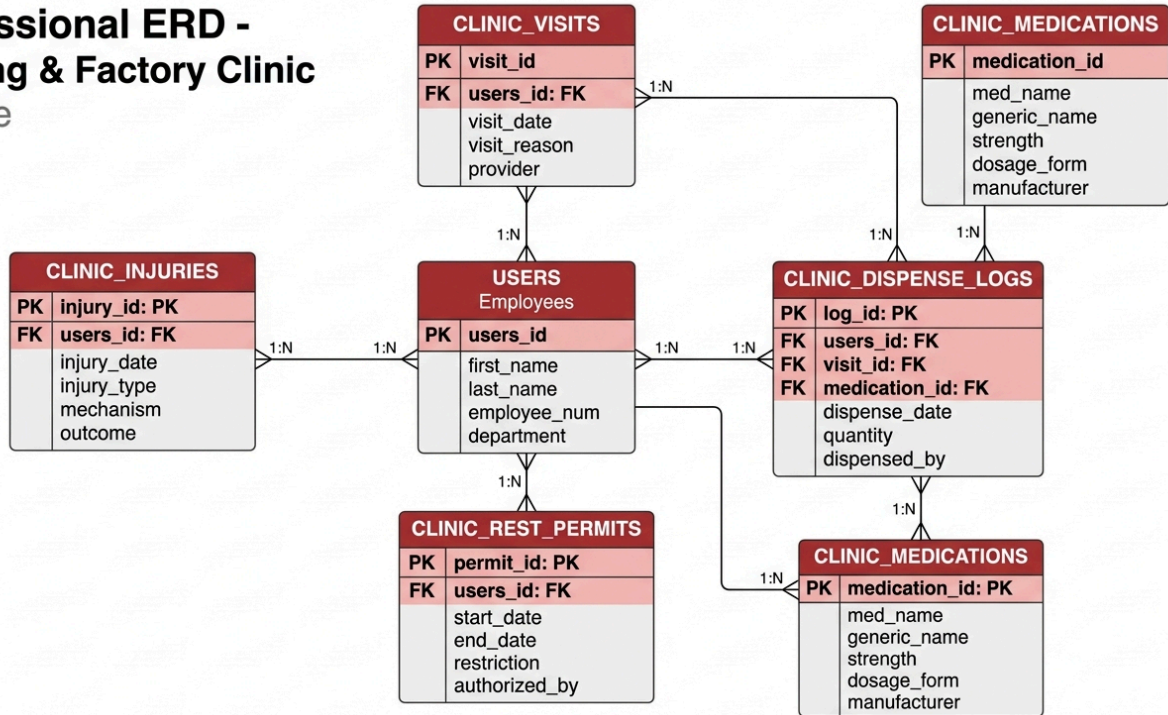


Fleet & Logistics ERD

Fleet & Logistics — ERD



Nursing & Clinic ERD

Professional ERD -
Nursing & Factory Clinic
Module

1. قسم الموارد البشرية — HR Department

الغرض: إدارة شاملة لدورة حياة الموظف من التعيين حتى الاستقالة

من يعمل هنا: HR Manager, HR Staff

الشاشات ولماذا

#	الشاشة	لماذا موجودة
1	employees	إدارة بيانات جميع الموظفين — إضافة، تعديل، حذف، بحث
2	attendance	متابعة حضور وانصراف جميع الموظفين يومياً
3	leaves	إدارة طلبات الإجازات — مراجعة وموافقة/رفض
4	absence-leave	نظام الغياب المتقدم — تسجيل الغياب غير المبرر وتطبيق الجزاءات
5	shifts	إدارة الورديات — صباحي، مسائي، ليلي، إداري
6	overtime	إدارة ساعات العمل الإضافي والموافقة عليها
7	all-delays	عرض جميع حالات التأخير مع حساب الخصومات
8	all-missions	إدارة المأموريات الخارجية لجميع الموظفين

#	الشاشة	لماذا موجودة
9	employee-warnings	إصدار الإنذارات والجزاءات التأديبية
10	asset-assignment	تسليم وتتبع العهد (للابتوب، موبايل، مفاتيح، زي)
11	payroll	حساب وصرف المرتبات الشهرية
12	payroll-funding	تمويل المرتبات (موافقة صاحب المصنع)
13	hr-adjustments	تعديلات المرتبات (بدلات، حوافز، خصومات)
14	recruitment	إدارة التوظيف — إعلانات الوظائف
15	hr-ats	نظام تتبع المتقدمين مع تقييم AI للسيرة الذاتية
16	loans	إدارة سلف الموظفين
17	expenses	إدارة مصروفات الموظفين
18	complaints	إدارة شكاوى الموظفين (سرية أو معلنة)
19	medical-requests	طلبات العلاج الطبي والصرف
20	offboarding	إجراءات إنهاء الخدمة والمخالفات
21	training	إدارة الدورات التدريبية
22	performance-reviews	تقييم أداء الموظفين الدوري
23	hr-qr-generator	توليد أكواد QR للحضور
24	friday-work	طلبات العمل يوم الجمعة
25	documents	إدارة مستندات الموظفين

دورات القسم (Cycles)

الدورة 1: دورة الإجازة

وافق/يرفض → المالك يوافق/يرفض → خصم من رصيد الإجازات HR → موظف يقدم طلب → مدير الصالة يوافق/يرفض

الدورة 2: دورة الغياب والجزاءات

يمكن بعدل → 3 أيام متتالية → إنذار أول → إنذار نهائي → فصل HR → غياب بدون إذن → خصم يومين تلقائي

الدورة 3: دورة التوظيف

فرز → مقابلة → عرض → تعيين → إنشاء حساب → AI تقييم → CVs إعلان وظيفة → استلام

الدورة 4: دورة إنهاء الخدمة

إغلاق الحساب → HR مخالصة مالية → مخالصة IT → بدء الإجراء → مخالصة

2. قسم المالية — Finance Department

الغرض: إدارة الحسابات والخزينة والمراجعة المالية

من يعمل هنا: CFO, Chief Accountant, Accountant

الشاشات

#	الشاشة	لماذا موجودة
1	chart-of-accounts	شجرة الحسابات — تصنيف جميع الحسابات المالية
2	financial-reports	التقارير المالية — ميزان المراجعة، قائمة الدخل
3	petty-cash	إدارة الخزينة والبنوك والعهد
4	driver-payments	مدفوعات السائقين والنقل

دورات القسم

الدورة 1: دورة الفاتورة

إنشاء فاتورة → غير مسددة → تحصيل جزئي → تحصيل كامل → مسددة → قيد يومية

الدورة 2: دورة الشيكات

استلام شيك → قيد انتظار → إيداع بالبنك → تحت التحصيل → محصل / مرتد

الدورة 3: دورة قيود اليومية

إنشاء قيد (مسودة) → مراجعة → اعتماد → ترحيل → تحديث الأرصدة

الدورة 4: دورة الإقفال

إقفال يومي → إقفال شهري → إقفال سنوي → قفل الفترة (لا تعديل)

3. قسم الإنتاج — Production Department

الغرض: إدارة عمليات التصنيع من المواد الخام حتى المنتج النهائي
من يعمل هنا: Production Manager, Planning Manager, Line Workers

الشاشات

#	الشاشة	لماذا موجودة
1	production	إدارة أوامر الإنتاج
2	erp-products	إدارة المنتجات ووصفات التصنيع (BOM)

دورات القسم

الدورة 1: دورة أمر الإنتاج



الدورة 2: دورة التتبع (Traceability)



4. قسم المخازن — Warehouse Department

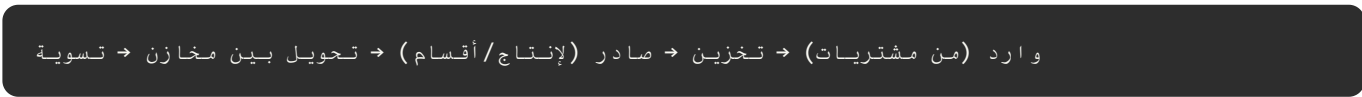
الغرض: إدارة المخزون وحركات الأصناف
من يعمل هنا: Warehouse Manager, Storekeepers

الشاشات

#	الشاشة	لماذا موجودة
1	inventory	إدارة جميع أصناف المخزون (خامات، قطع غيار، مستلزمات)
2	purchase-requests	طلبات الشراء من الأقسام — الفحص والصرف

دورات القسم

الدورة 1: دورة حركة المخزون



الدورة 2: دورة طلب الشراء من المخزن

قسم يطلب صنف → المخزن يفحص → متوفر: صرف مباشر → غير متوفر: تحويل للمشتريات

5. قسم المشتريات — Procurement Department

الغرض: شراء المواد الخام والمستلزمات وإدارة علاقات الموردين

من يعمل هنا: Procurement Manager, Procurement Specialist

الشاشات

#	الشاشة	لماذا موجودة
1	purchase-requests	مراجعة طلبات الشراء من الأقسام
2	erp-suppliers	إدارة بيانات الموردين
3	supplier-portal	بوابة المورد الخارجي لتتبع الطلبات
4	supplier-performance	تقييم أداء الموردين بالذكاء الاصطناعي

دورات القسم

الدورة 1: دورة الشراء

أسعار → إنشاء أمر شراء → إرسال للمورد → استلام البضاعة → فحص الجودة → إدخال المخزن → تسوية مالية

6. قسم الجودة — Quality Department

الغرض: ضمان جودة المواد الواردة والمنتجات النهائية

من يعمل هنا: Quality Manager, QC Inspector

الشاشات

#	الشاشة	لماذا موجودة
1	quality	إدارة فحوصات الجودة

دورات القسم

الدورة 1: فحص المواد الواردة

ناجح: إدخال مخزن → فشل: إرجاع للمورد → QC مادة خام تصل → فحص

الدورة 2: فحص المنتج النهائي

ناجح: منتج نهائي → فشل: إعادة تصنيع / خردة → QC دفعة إنتاج جاهزة → فحص

7. قسم المبيعات — Sales Department

الغرض: إدارة المبيعات والعلاء وكتالوج المنتجات

من يعمل هنا: Sales Manager, Sales Staff

الشاشات

#	الشاشة	لماذا موجودة
1	erp-sales	إدارة أوامر البيع
2	erp-products	إدارة كتالوج المنتجات (عام ومخفي)
3	customer-requests	طلبات تسجيل العملاء الجدد وعروض الأسعار

دورات القسم

الدورة 1: دورة البيع

إنشاء أمر بيع → إنتاج/مرف من المخزن → توصيل → فاتورة → تحصيل → Sales عميل يطلب عرض سعر → مراجعة

الدورة 2: تسجيل عميل جديد

يراجع → موافقة → إنشاء حساب عميل Sales → زائر يملأ نموذج تسجيل

8. قسم الهندسة — Engineering Department

الغرض: إدارة المشاريع الهندسية والرسومات الفنية

من يعمل هنا: Engineering Manager, Engineer, Technical Office

الشاشات

#	الشاشة	لماذا موجودة
1	engineering	إدارة المشاريع الهندسية والرسومات

دورات القسم

الدورة 1: دورة المشروع الهندسي

تخطيط → تصميم → رفع رسومات → اعتماد الرسومات → تنفيذ → إشراف → انتهاء → اعتمادات الأقسام

9. قسم الصيانة — Maintenance Department

الغرض: صيانة الماكينات والمعدات وإدارة شركات الصيانة الخارجية

من يعمل هنا: Maintenance Manager, Technicians

الشاشات

#	الشاشة	لماذا موجودة
1	maintenance	إدارة الماكينات وسجلات الصيانة
2	equipment	إدارة المعدات والإيجارات

دورات القسم

الدورة 1: دورة الصيانة الداخلية

عطل/موعد صيانة → تعيين فني → إصلاح → تسجيل السجل → تحديث حالة الماكينة

الدورة 2: دورة الصيانة الخارجية

طلب صيانة → تعيين شركة خارجية → زيارة فنية → تنفيذ الأعمال → تسجيل التكاليف → فاتورة → دفع

الدورة 3: دورة إيجار المعدات

حاجة لمعدة → إنشاء عقد إيجار → تفعيل → استخدام → إنتهاء/تجديد → إرجاع

10. قسم الأسطول واللوجستيات — Fleet & Logistics

الغرض: إدارة السيارات والسائقين والرحلات

من يعمل هنا: Logistics Manager, Internal Drivers

الشاشات

#	الشاشة	لماذا موجودة
1	fleet	إدارة السيارات والسائقين والرحلات

دورات القسم

الدورة 1: دورة الرحلة

جدولة رحلة → تعيين سائق + سيارة → تحميل → انطلاق → في الطريق → توصيل → تسجيل المسافة والوقود

الدورة 2: دورة صيانة الأسطول

تنبيه صيانة (عداد الكيلومترات) → طلب صيانة → تنفيذ → تسجيل التكلفة → تحديث حالة السيارة

11. قسم التمريض والعيادة — Nursing & Clinic

الغرض: إدارة العيادة الطبية للمصنع

من يعمل هنا: Nurse, Nursing Management

الشاشات

#	الشاشة	لماذا موجودة
1	nursing-page	تسجيل زيارات العيادة والإصابات والأدوية
2	nursing-medical-approvals	اعتمادات طبية

دورات القسم

الدورة 1: دورة الزيارة الطبية

عيادة → تسجيل العلامات الحيوية → تشخيص → علاج/صرف دواء → قرار (رجوع للعمل / راحة / تحويل مستشفى)

الدورة 2: دورة إصابة العمل

تسجيل الإصابة → إسعافات أولية → تقييم الخطورة → تحويل مستشفى (إذا لزم) → تحقيق السلامة → متابعة

الدورة 3: دورة صيدلية العيادة

استلام أدوية → تخزين → صرف للموظفين → تتبع المخزون → تنبيه نقص → طلب إعادة تموين

12. قسم تكنولوجيا المعلومات — IT Department

الغرض: دعم فني وإدارة النظام والصلاحيات

من يعمل هنا: IT Admin, IT Support

الشاشات

#	الشاشة	لماذا موجودة
1	screen-permissions	إدارة صلاحيات الشاشات والأفعال لكل دور
2	system-settings	إعدادات النظام العامة
3	audit-log	سجل المراجعة — تتبع جميع العمليات
4	login-history	سجل تسجيل الدخول — متابعة أمنية
5	activity-log-page	سجل النشاطات التفصيلي
6	notification-settings	إعدادات الإشعارات
7	ai-mind	المحادثة مع الذكاء الاصطناعي

دورات القسم

الدورة 1: دورة تذكرة الدعم الفني

موظف يفتح تذكرة → تعيين فني → جاري الحل → تم الحل

الدورة 2: دورة إدارة الصلاحيات

مستخدم جديد → تحديد الدور → تطبيق قالب الصلاحيات → تعديل فردي إذا لزم → بث التغييرات فوري

13. قسم التعبئة والتغليف — Packaging Department

الغرض: تعبئة وتغليف المنتجات النهائية

من يعمل هنا: Packaging Workers, Hall Manager

الشاشات

يستخدم الشاشات المشتركة: `scan-checkout` , `scan-checkin` , `my-leaves` , `my-attendance` , `dashboard`

دورات القسم

استلام منتج نهائي من الإنتاج → تعبئة → تغليف → تخزين في مخزن المنتجات النهائية

14. قسم التخطيط — Planning Department

الغرض: تخطيط الإنتاج والجدولة

من يعمل هنا: Planning Manager

الشاشات

يستخدم شاشات الإنتاج + التقارير + KPIs

دورات القسم

تحليل طلبات البيع → تخطيط الإنتاج → جدولة الواردات → متابعة التنفيذ → تعديل الخطة

15. قسم الصحة والسلامة — Health & Safety Department

الغرض: السلامة المهنية وتتبع الإصابات

من يعمل هنا: HSE Officers

الشاشات

يستخدم شاشات التمريض (إصابات العمل) + التقارير

دورات القسم

تقييم المخاطر → وضع إجراءات وقائية → تسجيل الحوادث → تحقيق → إجراءات تصحيحية

16. الأقسام الإدارية — Administration & Secretariat

الغرض: الشؤون الإدارية والسكرتارية والقانونية

من يعمل هنا: Admin Staff, Legal Advisor

الشاشات

#	الشاشة	لماذا موجودة
1	legal-affairs	تتبع القضايا القانونية
2	announcements	إدارة الإعلانات والتعميمات
3	document-management	إدارة المستندات الرسمية

دورات القسم

دورة القضية القانونية:

تسجيل قضية → تعيين محامي/مسؤول → متابعة → حل → إغلاق

الشاشات المشتركة لجميع الموظفين (Self-Service)

كل موظف في أي قسم يقدر يوصل للشاشات دي:

#	الشاشة	الغرض
1	dashboard	لوحة المعلومات الشخصية
2	hr-personal	بيانات الموظف الشخصية
3	my-attendance	سجل الحضور الشخصي
4	scan-checkin	تسجيل الدخول بـ QR
5	scan-checkout	تسجيل الخروج بـ QR
6	my-leaves	طلبات الإجازات

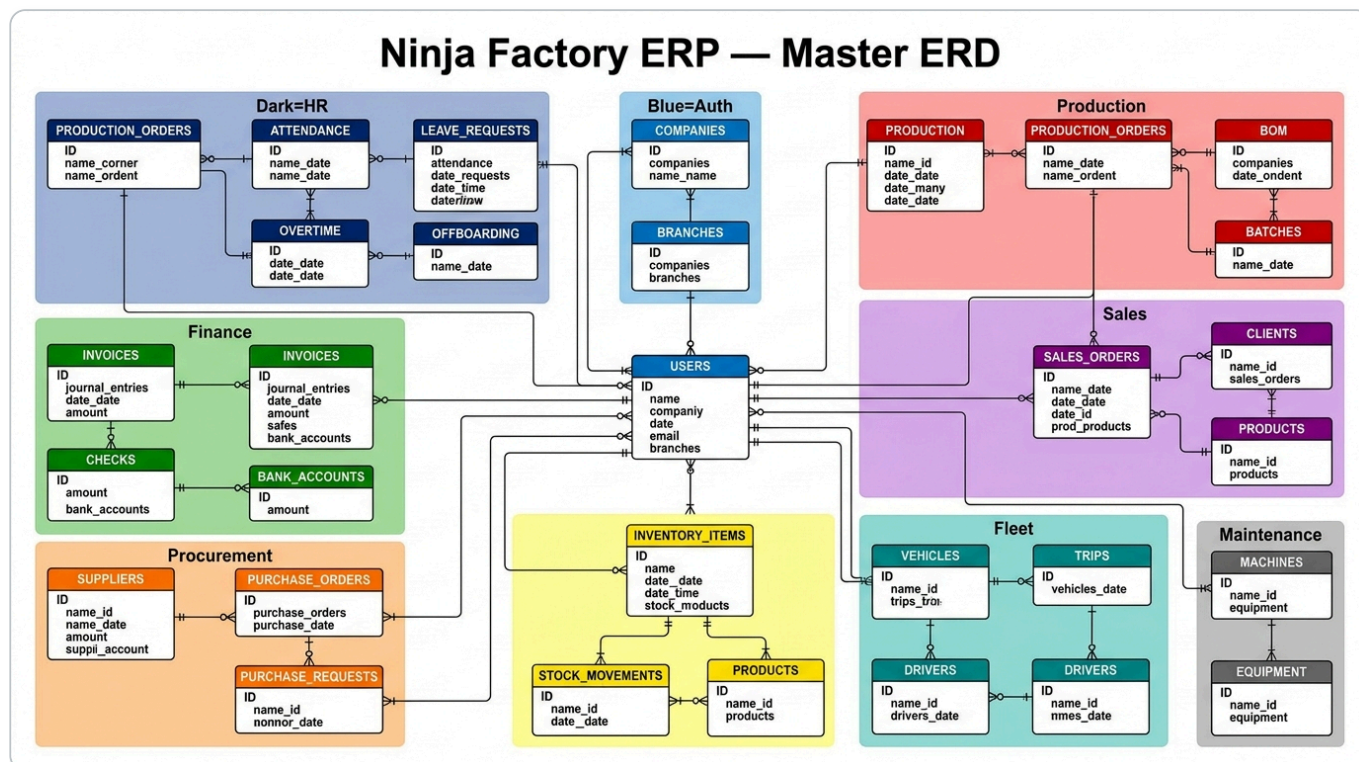
#	الشاشة	الغرض
7	my-salary	كشف المرتب
8	my-overtime	ساعات العمل الإضافي
9	my-loans	السلف
10	my-medical	طلبات العلاج
11	my-delays	حالات التأخير
12	my-missions	المأموريات
13	my-expenses	المصروفات
14	complaints	تقديم شكوى
15	announcements	عرض الإعلانات
16	internal-chat	المحادثات الداخلية
17	shift-swap	تبادل الورديات
18	calendar	التقويم
19	task-management	إدارة المهام
20	org-directory	دليل المنظمة

معرض مخططات قواعد البيانات (ERD Architectural 10)

(Diagrams)

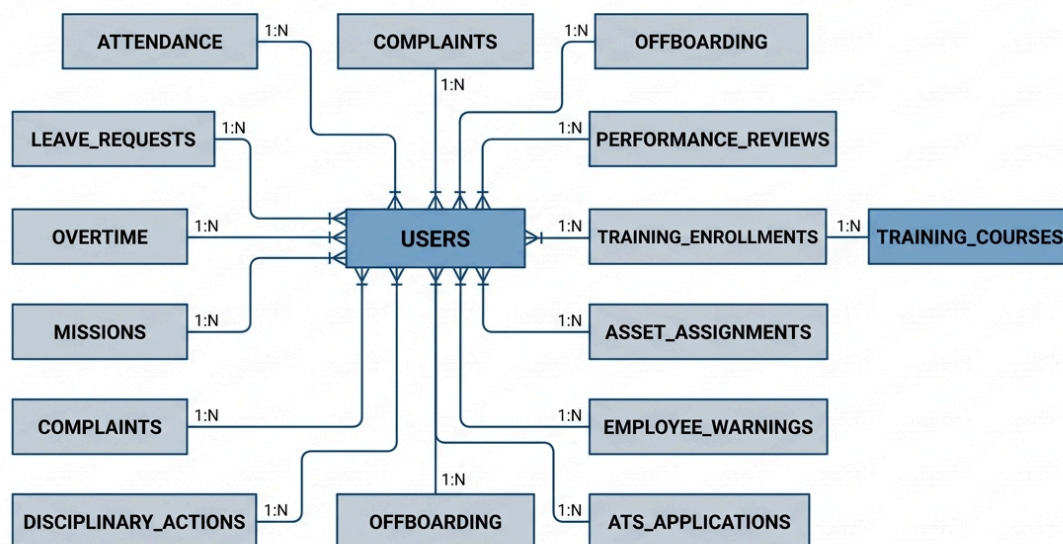
مخططات الكيانات والعلاقات التفصيلية لكافة قطاعات مصنع نينجا للبولىمر والبلاستيك

1. المخطط العام الشامل للنظام المتكامل (Master ERP Architecture ERD)

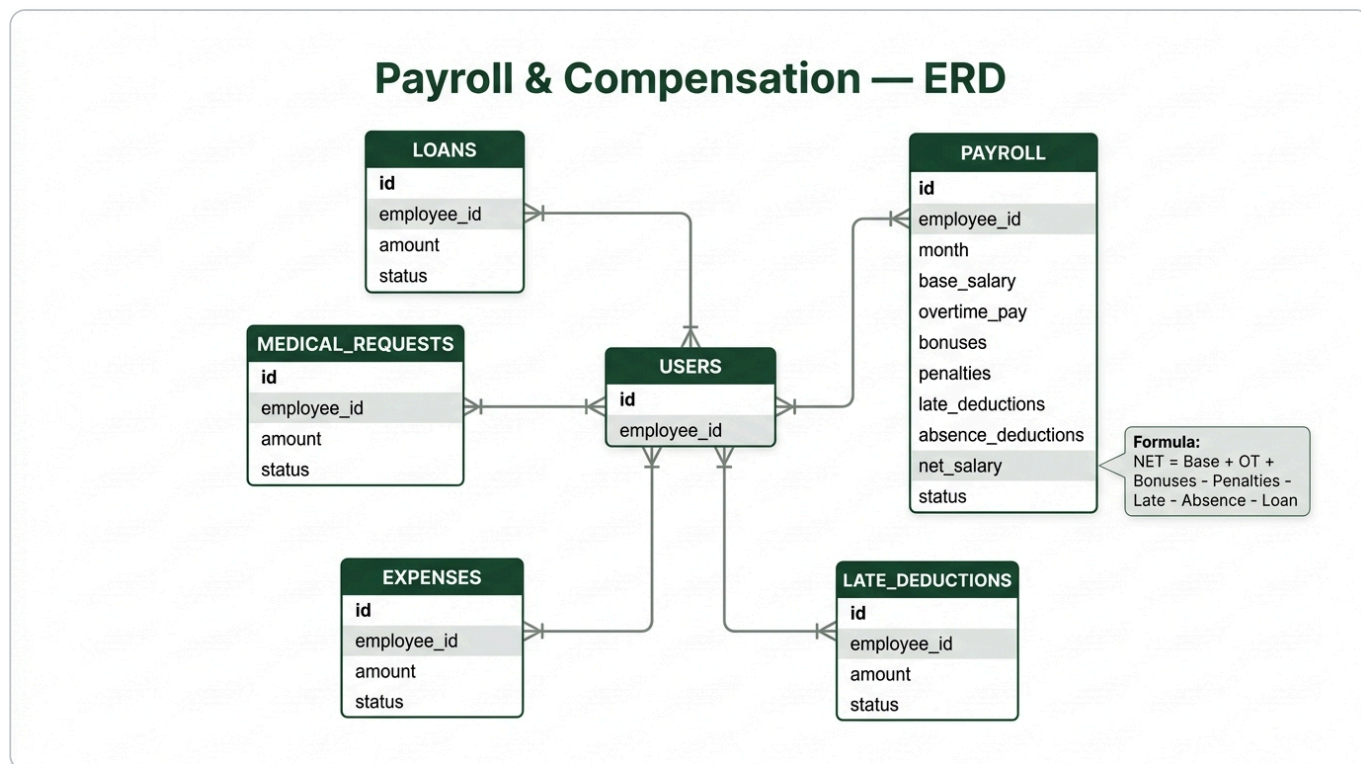


2. مخطط الموارد البشرية وشؤون العاملين (HR & Workforce Management ERD)

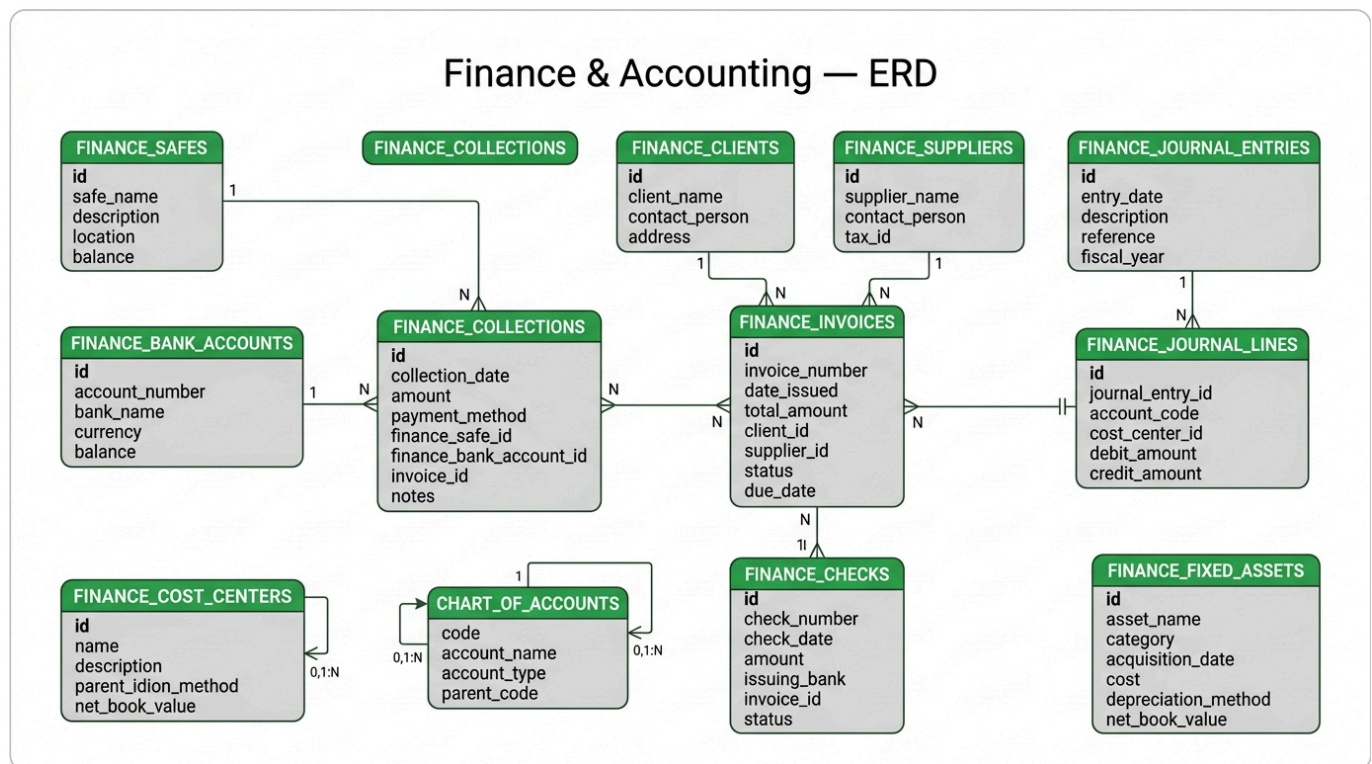
HR & People Management — ERD



3. مخطط المرتبات والأجور والبدلات والسلف (Payroll & Compensation ERD)

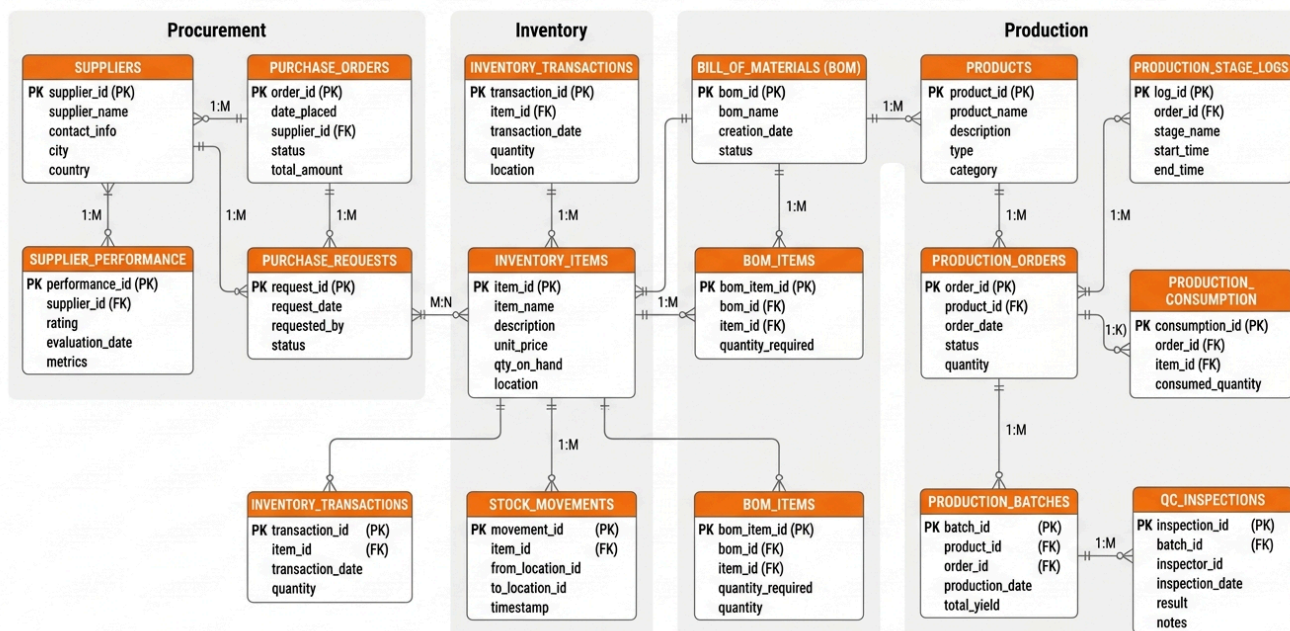


4. مخطط النظام المالي والمحاسبي المزدوج (Finance & Accounting ERD)



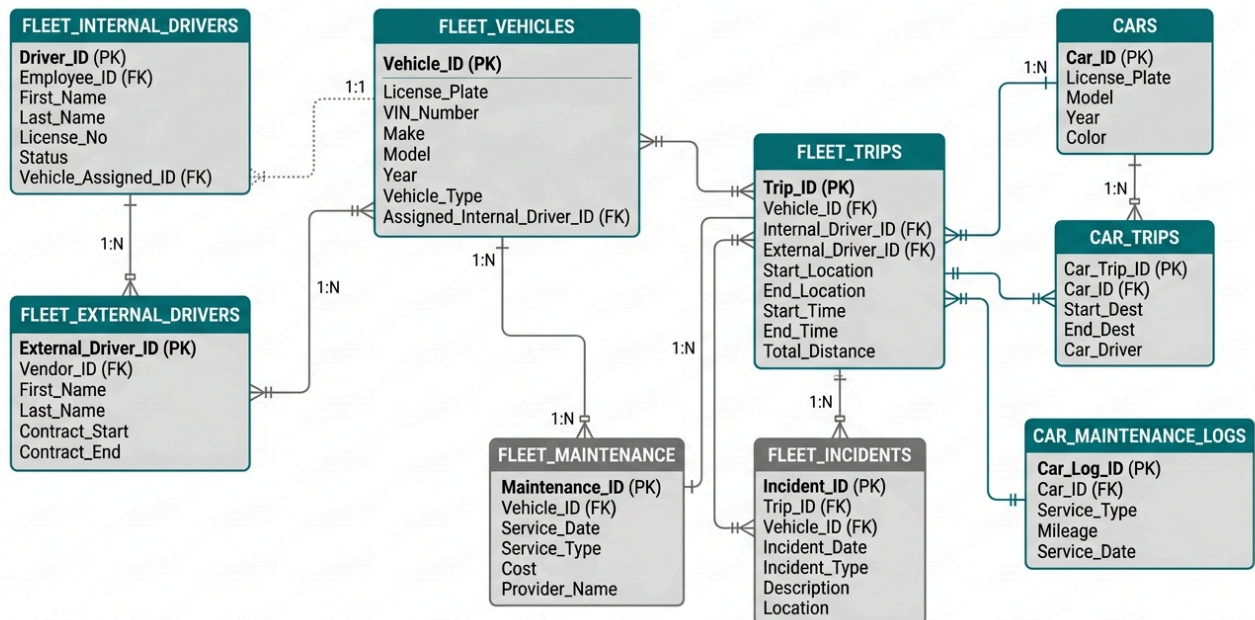
5. مخطط سلسلة التوريد (المشتريات ← المخازن ← الإنتاج ← الجودة)

SUPPLY CHAIN — ERD (Procurement → Inventory → Production → QC)

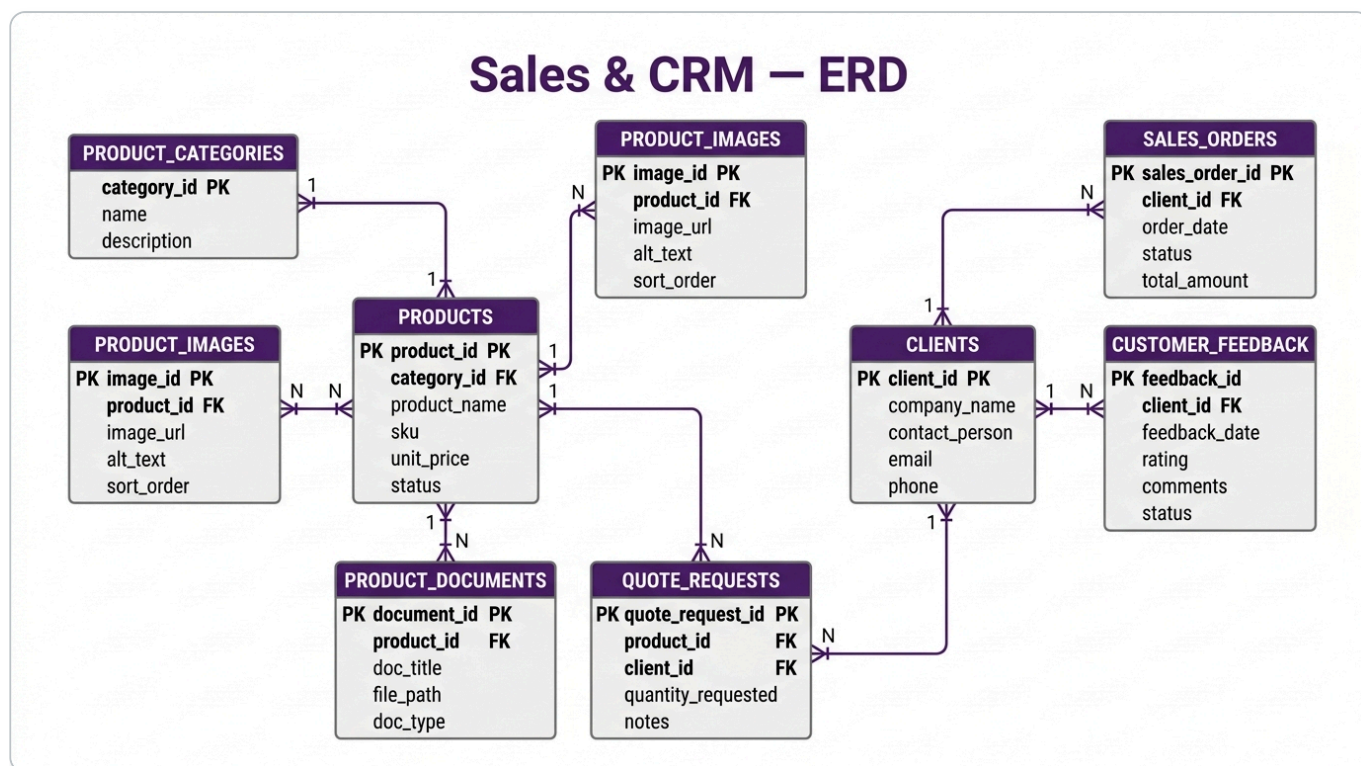


6. مخطط الحركة والنقل وإدارة الأسطول (Fleet & Logistics ERD)

Fleet & Logistics — ERD

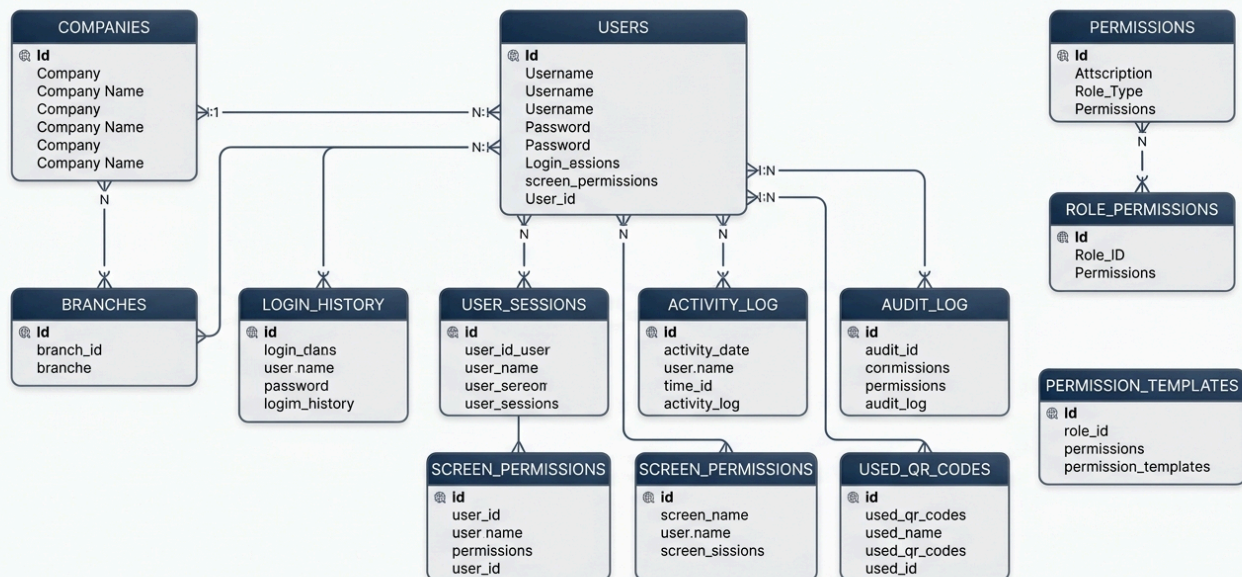


7. مخطط المبيعات وإدارة العملاء والكتالوج (Sales & CRM ERD)



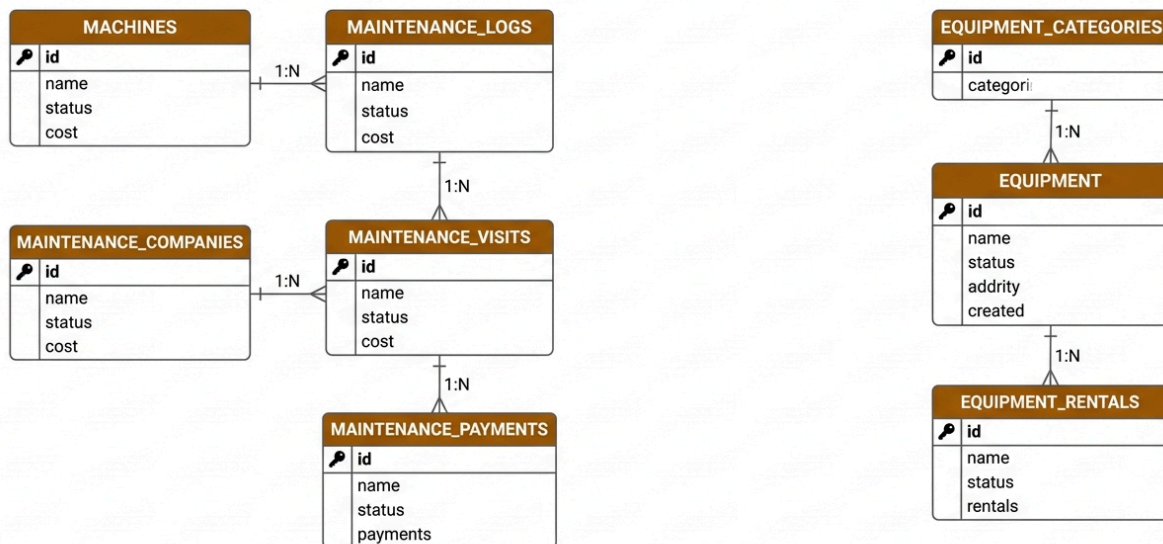
8. مخطط الأمان والتحقق وإدارة الجلسات والصلاحيات (Auth & Security ERD)

Authentication & Security — ERD



9. مخطط الصيانة والماكينات والمعدات الصناعية (Maintenance & Assets ERD)

Maintenance & Equipment — ERD



10. مخطط التمرريض والعيادة الطبية للمصنع (Nursing & Clinic ERD)

Professional ERD - Nursing & Factory Clinic Module

